GEK-83755

DRIVE SYSTEM INSTRUCTIONS

DATA-TRONIC™ GAS TURBINE CONTROL OPERATION & MAINTENANCE MANUAL

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

DATA-TRONIC™ GAS TURBINE CONTROL OPERATION & MAINTENANCE MANUAL

GEK-83755

ISSUE DATE: July 1981

The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, USA.

DIRECTO-MATIC® is a registered trademark of General Electric Company.
DATA-TRONIC™ and SPEEDTRONIC™ are trademarks of General Electric Company.

TABLE OF CONTENTS

DATA-TRONIC™ SYSTEM OVERVIEW 1

Data Acquisition System 1
Data Collection 1
Communication with Remote Station Communication Controller (SCC) 1
Commands from the Station Communication Controller 1
Station Operator Console (SOC) 1
Station Communications controller 1
Supervisory Control and Data Acquisition (SCADA) Interface 4
Data Analyzer and Processor (DAP) Interface 4
Plant data Logger Interface 4

DATA ACQUISITION SYSTEM DESCRIPTION 4

Data Acquisition System Small Electronics Module 4
Internal Connections 4
External Connections 4
Circuit Boards 10
DS3800HMPA Microprocessor 10
DS3800HXPA Microprocessor Expander 10
DS3800NADC Analog to Digital Converter 10
DS3800HMHA Master Highway Controller 11
DS3800HRMA Dual Ported RAM 11
DS3800HRRB Relay Output 11
Analog Input Rack (AIR) 11
Point Cards 11
Amplifier Cards 11
Circuit Boards 11
IC3600AMINI 11
IC3600QMID2 11
IC3600QMID3 11
IC3600APSJ Isolated Power Supply 11
IC3600AEXA Expander Card 11
ICS3600AJBA Jumper Card 11
DS3820PS3 Power Supply 13
SPEEDTRONIC™ Control Interface 13
Digital Inputs to the Data Acquisition System from the SPEEDTRONIC™ Control 13
Digital Outputs to the SPEEDTRONIC™ Control from the Data Acquisition System 13
Contact Outputs from the Data Acquisition System to the SPEEDTRONIC™ Control 13
Analog Inputs to the Data Acquisition System from the SPEEDTRONIC™ Control 13
DS3820LIMA Isolation Module 13
DS3820DDPC Data Acquisition System Display Option 13
Teleprinter Option 18
Cables 18

STATION OPERATORS CONSOLE (SOC) DESCRIPTION 18

Station Communications Controller Medium Electronic Module (SCC MEM) 18
Internal Connections 18
External Connections 18
Connections to Top of Medium Electronics Module 18
Connections to the Front of Circuit Boards 18
Circuit Boards 24
DS3800HMPA Microprocessor 24
DS3800HXPA Microprocessor Expander 24
DS3800HMHA Master Highway Controller 24
DS3800HRMA Dual Ported RAM 24
DS3800HRRB Relay Output Card 24
DS3800HCMA Dual Communication Controller 24
DS3800HSCC Non Isolated Input 26
CRT's 28
Status CRT 28
Alarm CRT 30
Power Supplies 30
DS3820PS3 Power Supply 30
PS-28A 26
Teleprinter Option 31
DS3820LIMA Isolation Module 32
Limited Distance Data Set (LDDS) 32
Supervisory Control and Data Acquisition (SCADA) Link Option 32
Data Analyzer and Processor (DAP) Link Option 32
Logger Interface Option 32
Cables 34
Redundant Option 34

INSTALLATION NOTES 37

Data Acquisition System Installation 37
Party Line Connectors 37
Party Line Terminators 37
Station Operator's Console Installation 41
Party Line Connector 41
Party Line Terminator 41

OPERATOR INSTRUCTIONS 41

Device Summary 41
Unit Select Switch 41
Master Control Switch 41
Load Control Switch 41
Base Indication Light 41
Peak Indicating Light 41
Watts Switch 41
Gas Monitor Switch 41
Override Indicating Light 43
Monitor Indicating Light 43
Fast Load Start Pushbutton 43
52G Trip Pushbutton 43
52G Open Indicating Light 43
52G Closed Indicating Light 43
Alarm Indicating Light 43
Silence Pushbutton 43
Acknowledge Pushbutton 43
Clear Pushbutton 43
Review Pushbutton 43
Channel Select Switch 43
Control Location Switch 43
Ready Indicating Light 43

Sequence in Progress Indicating Light 43
Flame Indicating Light 43
Complete Sequence Indicating Light 43
Temperature Control Indicating Light 43
Fire Zone 1 Indicating Light 43
Fire Zone 2 Indicating Light 43
Status CRT 44
Teleprinter 44
Beeper "A" 44
Beeper "B" 44
Channel Selection 44
Unit Selection 44
Control Location 44
CAB or Local 44
Station or Central 44
Supervisory Control and Data Acquisition (SCADA) 44
Status CRT 44
Header Line 45
Control Location 45
Gen Brk 52G 45
Megawatts 45
Megavars 45
Kilovolts 45
EXH Median 45
Temp Spread 45
Gas P2 45
Gas Fuel Flow 45
Speed-RPM 45
Alarm CRT 45
Header Line 46
Alarm Messages 46
Turbine Control 48
Start 48
Fast Load Start 49
Stop 49
Base Load Select 49
Peak Load Select 49
Raise Watts 50
Lower Watts 50
Gas Monitor 50
Gas Override 50
Generator Control 51

DATA-TRONIC™ SYSTEM OVERVIEW

The DATA-TRONIC™ control system allows a remote operator to control up to eight gas turbines (units) from a remote location (See Figure 1). The communications between the units and the remote operator is with a serial party line. The serial party line allows simultaneous two-way communication between the remote operator and the units with only two twisted pairs of wires. There are options available to accommodate interface to a maintenance computer, customer's load dispatch computer or plant logger.

Data Acquisition System (DAS)

The Data Acquisition System (DAS) is a microprocessor based system that performs the following functions:

1. Collects large amounts of data from several sources.
2. Communicates with the remote station communication controller (SCC) via the party line.
3. Receives commands from the Station Communication Controller (SCC) (See Figure 2) via the party line and gives the command to the SPEEDTRONIC™ control or generator control in the form of a contact closure.

Data Collection. The Data Acquisition System gathers data from three sources. Those three sources are the SPEEDTRONIC control, analog input rack (AIR), and the Integrated Temperature System (ITS). The data from the SPEEDTRONIC control is both analog and digital data and includes alarms. The data from the Analog Input Rack is all analog and consists of generator Resistance Temperature Detector's (RTD) and several other analog data signals. The data from the Integrated Temperature System is both analog and digital. This data is transferred from the integrated Temperature System to the Data Acquisition System on an RS-232 serial link.

Communication with Remote Station Communication Controller (SCC). The Station Communication Controller and the Data Acquisition System are in a master/slave relationship with the Station Communication Controller serving as the master and the Data Acquisition System acting as the slave. The Station Communication Controller initiates all communications and the Data Acquisition System answers back. Each Data Acquisition System is polled by the Station Communication Controller once per second for data. The Station Communication Controller also can send commands to the Data Acquisition System. Communication is accomplished by use of an RS-422 SDLC link at 9600 baud.

Commands from the Station Communication Controller. Commands such as start, stop, raise/lower watts may be sent from the Station Communication Controller to the Data Acquisition System. The Data Acquisition System will act on those commands by closing the appropriate output contact(s).

Station Operator Console (SOC)

The Station Operator Console is the console from which an operator may control up to eight units. Located on the console is a unit selector switch which allows the operator to select the unit he desires to control and also the switches, indicating lights and pushbuttons which allow the operator to control the turbine. Also located on the console are two CRT displays. One CRT displays status information for all eight units. The second CRT displays alarm information for all eight units. The optional teleprinter is located on the console to allow the operator to set time and request limited logs.

Station Communications Controller (SCC). The Station Communication Controller is a microprocessor-based system that is mounted within the Station Operator Console. The Station Communications Controller communicates with up to eight Data Acquisition Systems. The switches and pushbuttons on the Station Operator Console are wired to the Station Communication Controller. The Station Communication Controller scans these inputs and sends commands to the selected Data Acquisition System based on these inputs.

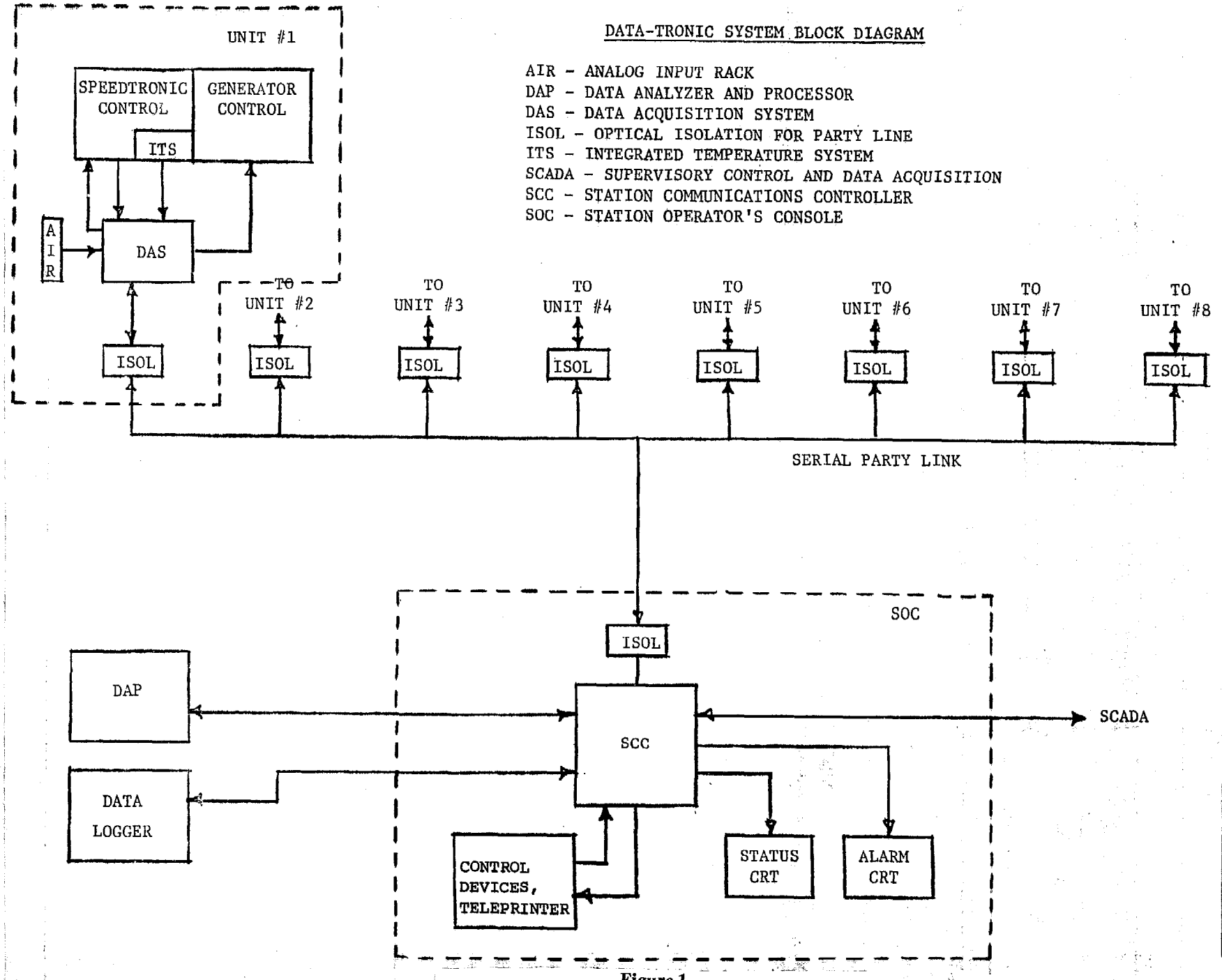


Figure 1
DATA-TRONIC™ Gas Turbine Control System

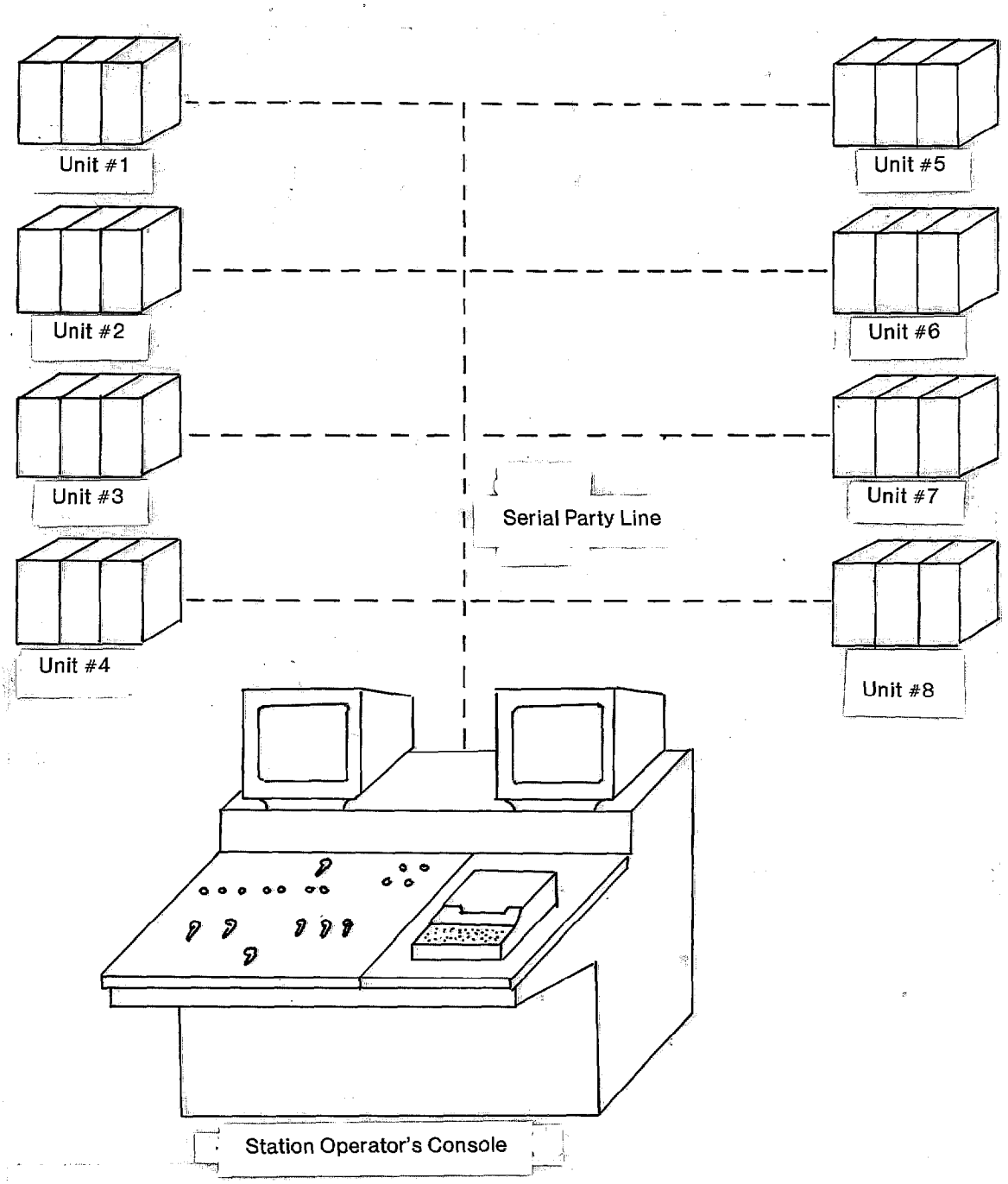


Figure 2
DATA-TRONIC™ Gas Turbine Control System

Supervisory Control and Data Acquisition (SCADA) Interface. Interface to the customer's load dispatch computer or SCADA is an available option. SCADA can control individual units if the station operator has selected that unit for SCADA control. SCADA may also request data for all eight units. This is an RS-232 serial interface between SCADA and the Station Communication Controller.

Data Analyzer and Processor (DAP) Interface. Interface to a maintenance oriented minicomputer or DAP is an available option. The Data Analyzer and Processor has access to all data within the entire system. The Data Analyzer and Processor gathers this data via a serial RS-232 link to the Station Communications Controller. The Data Analyzer and Processor maintains historical data and uses it to print trend logs.

Plant Data Logger Interface. Interface to a plant data logger is an available option. The data logger can interface with more than one Station Communication Controller making it truly a plant logger. The data logger gathers most of its status and alarm data via a serial link to the Station Communication Controller. The data logger can also accept direct data inputs for digital and analog signals. The data logger has an alarm printer to print out system alarms. A separate printer records events and prints out logs on demand.

DATA ACQUISITION SYSTEM DESCRIPTION

Data Acquisition System Small Electronics Module

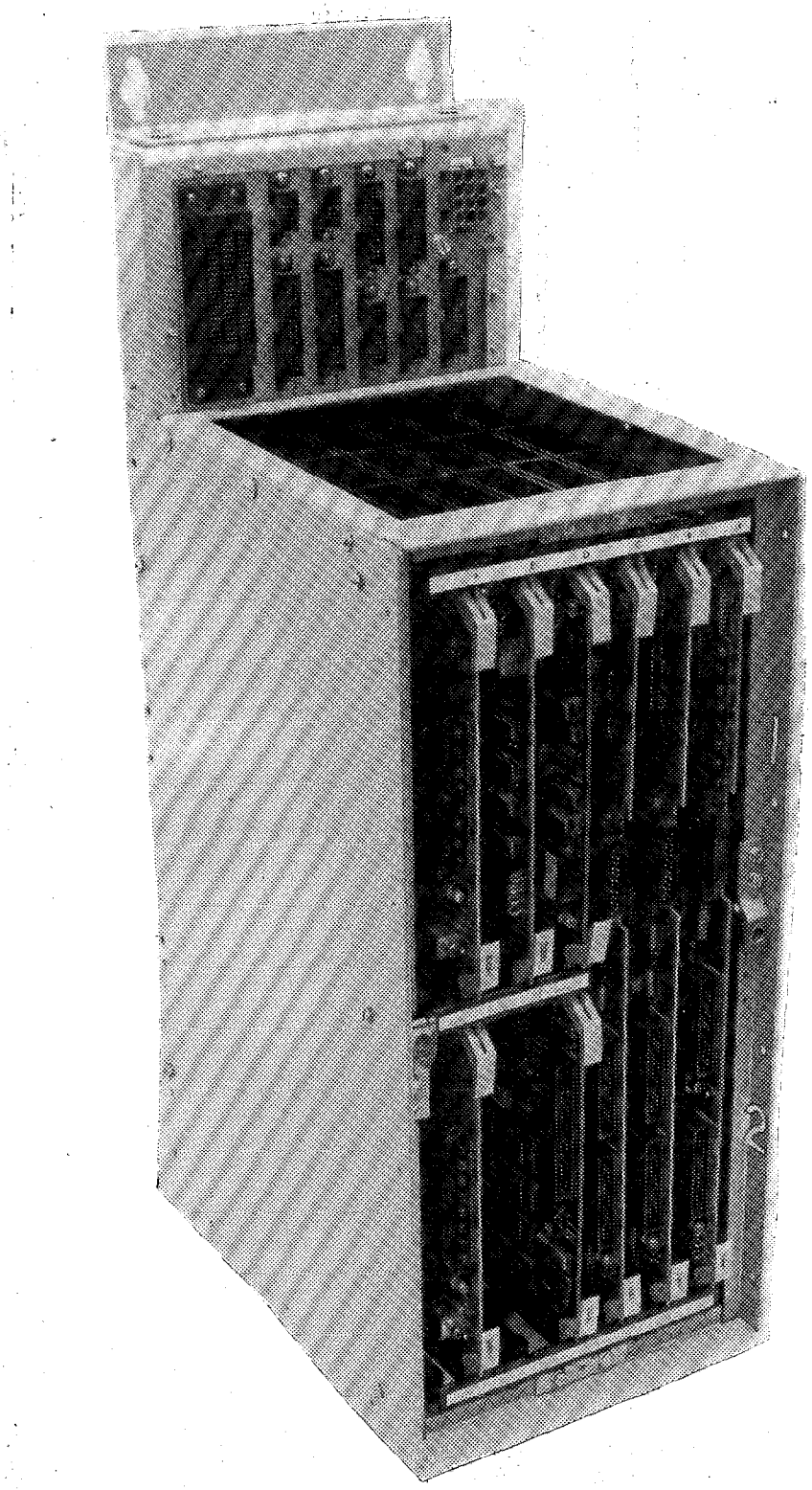
The Data Acquisition System Small Electronics Module (SEM) is the electronics module which houses the main processor board and other related boards. The Data Acquisition System Small Electronics Module is shown in Figure 3. Circuit boards plug into the Small Electronics Module. An identification strip indicates which board plugs into which board slot. (See Figure 4).

Internal Connections. All connections between circuit boards are made on the backplane at the rear of the Small Electronics Module. The printed wire backplane provides all power supply bus connections. All other connections are wire wrap connections between wire posts. (See Figure 5).

External Connections. External connections to the signal electronics module are either made at the top of the small electronics module or to the circuit boards directly.

1. Connections to the top of Small Electronics Module receptacles are provided at the top of the Small Electronics Module for plug-in cables. Example: Receptacle JA is the connection point where the power suply cable plugs in (See Figure 6).

2. Some circuit boards have receptacles mounted on them. These are also known as card-front connections. Example: RS-232 connectors on the DS3800HMPA board (See Figure 7).



(DSF50458)

Figure 3
Data Acquisition System Small Electronics Module

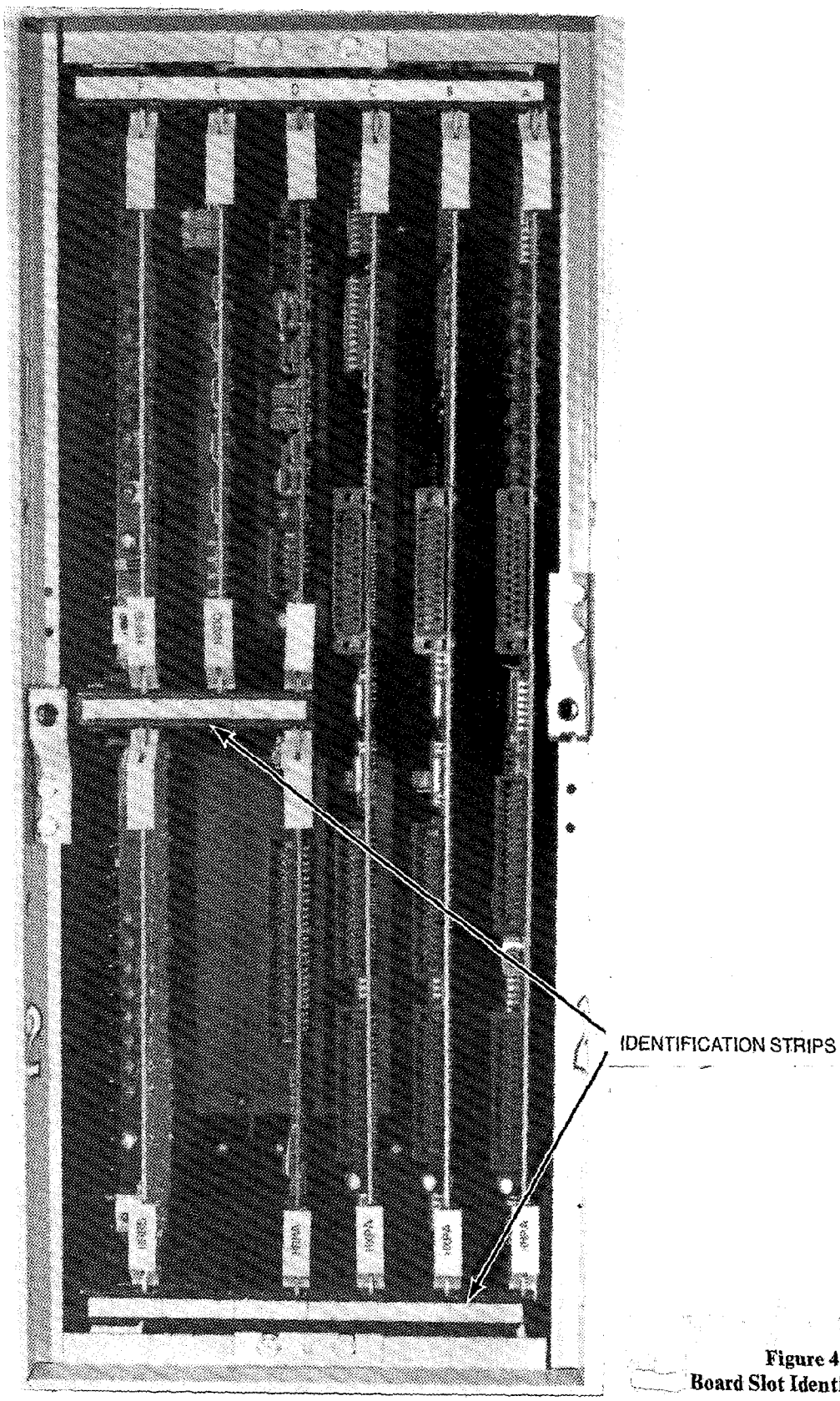
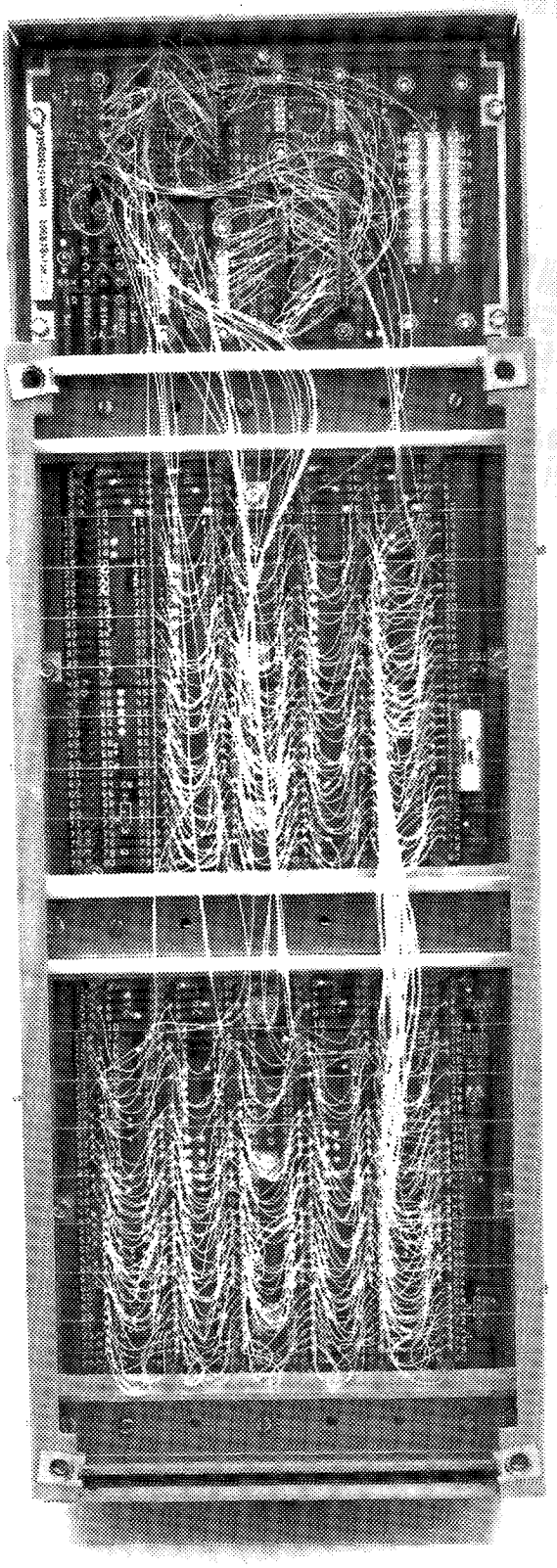


Figure 4
Board Slot Identification



(DSF54056)

Figure 5
Small Electronics Module Backplane Wiring

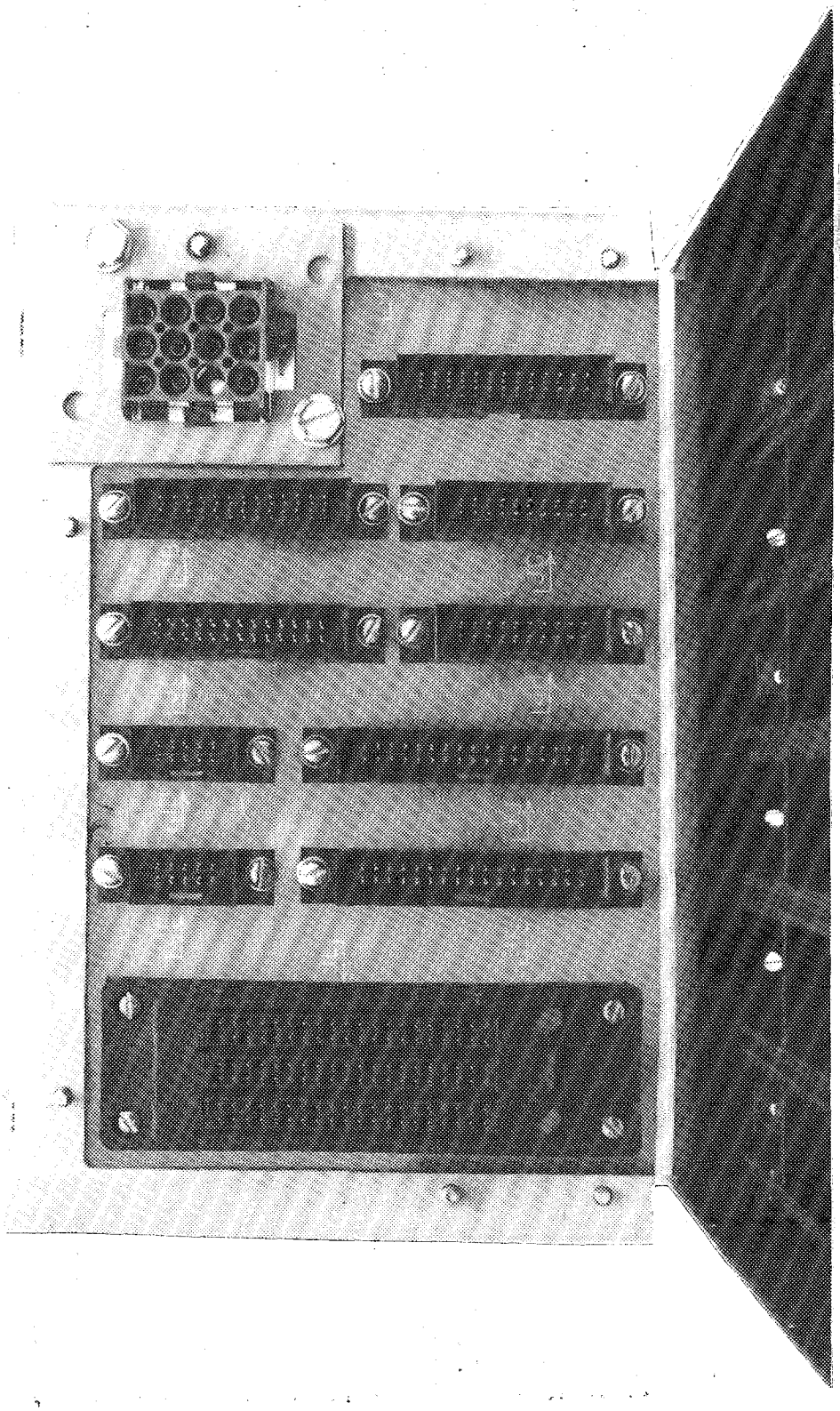
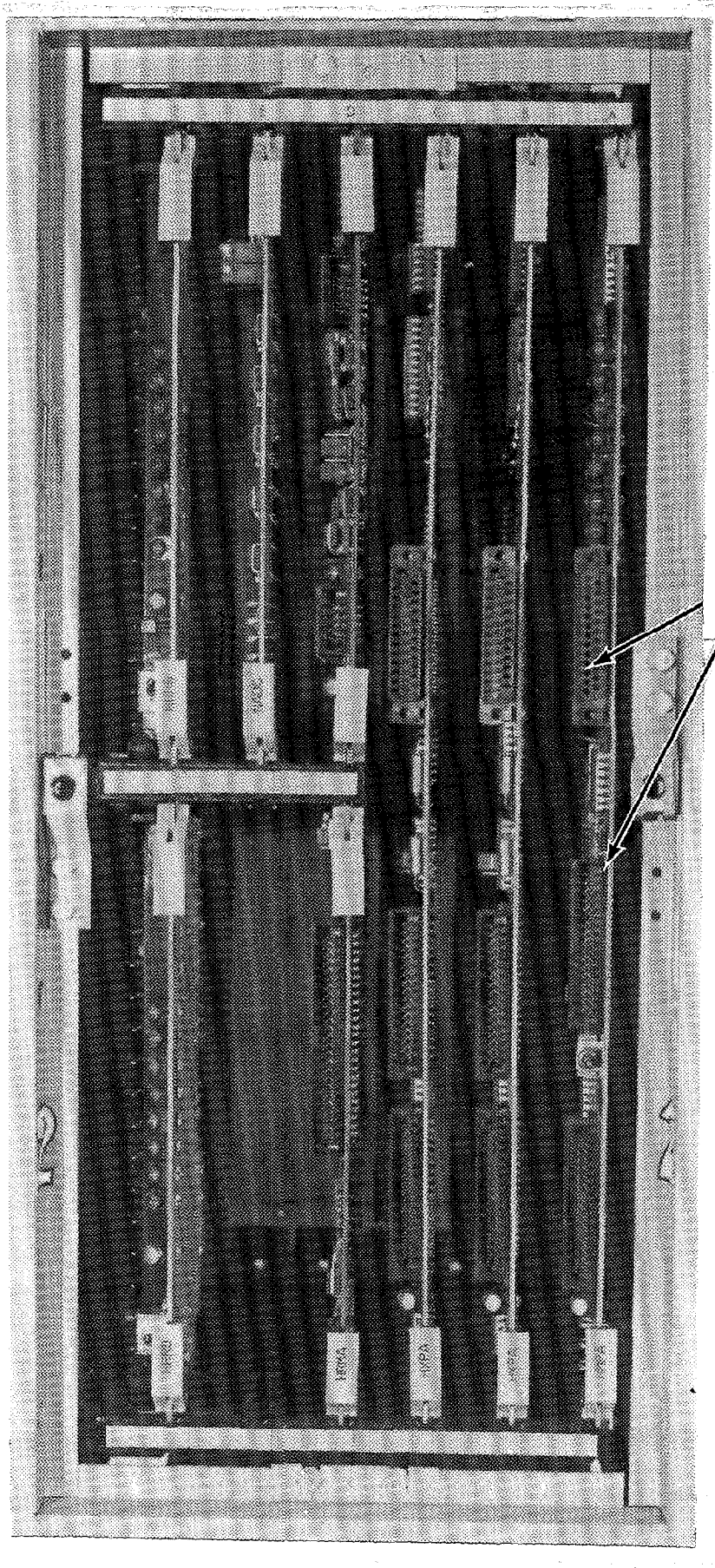


Figure 6
Small Electronics Module Plug Connectors



(DSF54053)

Figure 7
Card Front Connectors

Circuit Boards. Circuit boards that plug into the Data Acquisition System Small Electronics Module come in two sizes referred to as single size and double size boards. A brief description of each type of circuit board is given. Refer to circuit board backsheets for details.

DS3800HMPA Microprocessor. The HMPA is a double size circuit board. It contains the following components:

1. 8080A Microprocessor

2. 4K Bytes EROM

3. 1K Bytes RAM

4. 8 Level priority vectored interrupt structure

5. RS-232C Serial interface

6. 32 Bits digital outputs

7. 16 Bits digital inputs

8. 3 Software controlled timers

DS3800HXPA Microprocessor Expander. The HXPA is a double size circuit board. It contains the following components:

1. 8K Bytes EROM

2. 2K Bytes RAM

3. 8 Priority interrupts

4. RS-232C Serial interface

5. 11 Bits Digital outputs

6. 11 Bits Digital inputs

7. 2 Bits Digital input or output

8. 3 Software controlled timers

DS3800NADC Analog to Digital Converter. The NADC is a single size circuit board. It contains the following components:

1. 8 Channel Analog multiplexer

2. Sample and hold amplifier

3. 12 Bit Analog to digital converter

DS3800HMHA Master Highway Controller. The HMHA is a single size circuit board. It contains the following components:

1. 8085 Microprocessor

2. 2K Bytes dual-ported RAM

3. 2 RS422 Communication channels

DS3800HRMA Dual Ported RAM. The HRMA is a single size circuit board. It contains the following:

1. 16K Bytes dual ported RAM

DS3800HRRB Relay Output. The HRRB is a single size circuit board. It contains the following:

1. 8 Relay contact outputs

Analog Input Rack (AIR)

The Analog Input Rack contains analog input circuit boards (CARDS) and their associated amplifier circuit boards. A ribbon cable connects the Analog Input Rack to the Data Acquisition System Small Electronics Module. The Analog Input Rack is available in both four row and eight row sizes (See Figure 8).

Point Cards. The incoming analog signal connects to a point card. Point cards contain signal conditioning circuitry and a multiplexer. Typically each point card will accommodate four inputs.

Amplifier Cards. Each type of point card requires a differenct type of amplifier card. The amplifier card amplifies the signal to a 10 volt level for the Analog-to-Digital (A/D) card. The A/D card DS3800NADC is mounted in the Data Acquisition System Small Electronics Module.

Circuit Boards. The following is a brief summary of Analog Input Circuit boards:

IC3600AMIN1. This board accepts an input from a 10 ohm copper Resistance Temperature Detector. The required amplifier board is IC3600QEAA3.

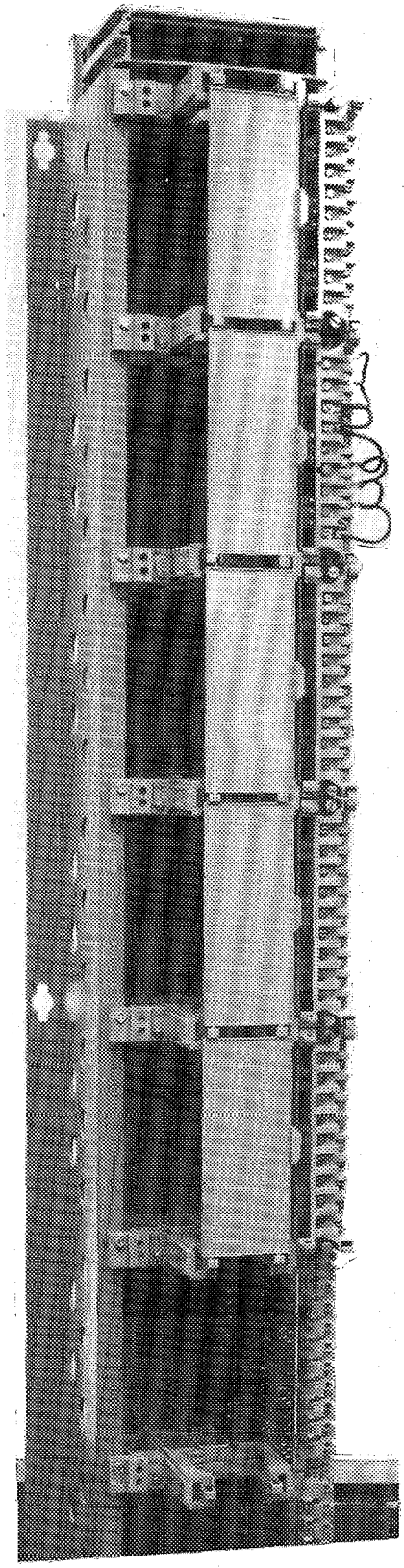
IC3600QMID2. This board accepts an input from a 1-5 milliampere signal. The required amplifier board is IC3600QDAA5.

IC3600QMID3. This board accepts an input from a 4-20 milliampere signal. The required amplifier board is IC3600QDAA2.

IC3600APSJ Isolated Power Supply. This card is used to supply power to the AMIN1 Resistance Temperature Detector card. The IC3600APSJ furnishes eight independent +15 volt, 35 milliampere supplies.

IC3600AEXA Expander Card. Since the amplifier is usually the most expensive card within a channel, it is logical to try to use the same amplifier for more than one channel of like point cards. The point cards must all have the same full scale signal level. Several channels of like signals (identical point cards) can be expanded to one isolated amplifier by using the expander card IS3600AEXA.

IC3600AJBA Jumper Card. If a channel is not to be isolated, a jumper card must be used in the power supply slot. The jumper card connects the normal +/-15 volt bus lines and also the seven control lines directly to their respective channel lines.



(DSF54052)

Figure 8
Analog Input Rack

DS3820PS3 Power Supply

The PS3 power supply accepts a single phase AC voltage (80-132VRMS, 47-63Hz) or a DC input voltage (70-140VDC) at its input and provides four regulated DC output voltages to externally connected loads. The output voltages are +5, +15, +28 and -15 VDC. Refer to backsheets for details (See Figure 9).

SPEEDTRONIC Control Interface

There are several different types of signals that interface between the SPEEDTRONIC control and the Data Acquisition System.

Digital Inputs to the Data Acquisition System from the SPEEDTRONIC Control. Digital signals from the SPEEDTRONIC control are multiplexed using IC3600VMIA cards. Each circuit accepts eight digital signals. Each circuit is read by the Data Acquisition System by enabling that circuit's byte N line where N can be 0 thru F. An IC3600DLIL board is used to interface the Data Acquisition System 20 conductor ribbon cable to the SPEEDTRONIC control.

Digital Outputs to the SPEEDTRONIC Control from the Data Acquisition System. Digital signals from the Data Acquisition System to the SPEEDTRONIC control are interfaced through the IC3600DLIL board. One example of this type of signal is the byte enable lines which select which byte of the multiplexed digital inputs to read.

Contact Outputs from the Data Acquisition System to the SPEEDTRONIC Control. The DS3800HRRB cards in the Data Acquisition System provide contact outputs to the SPEEDTRONIC control and the generator panel. Contacts to the SPEEDTRONIC control are connected to ICS3600SIIE cards mounted in the isolation/filter module.

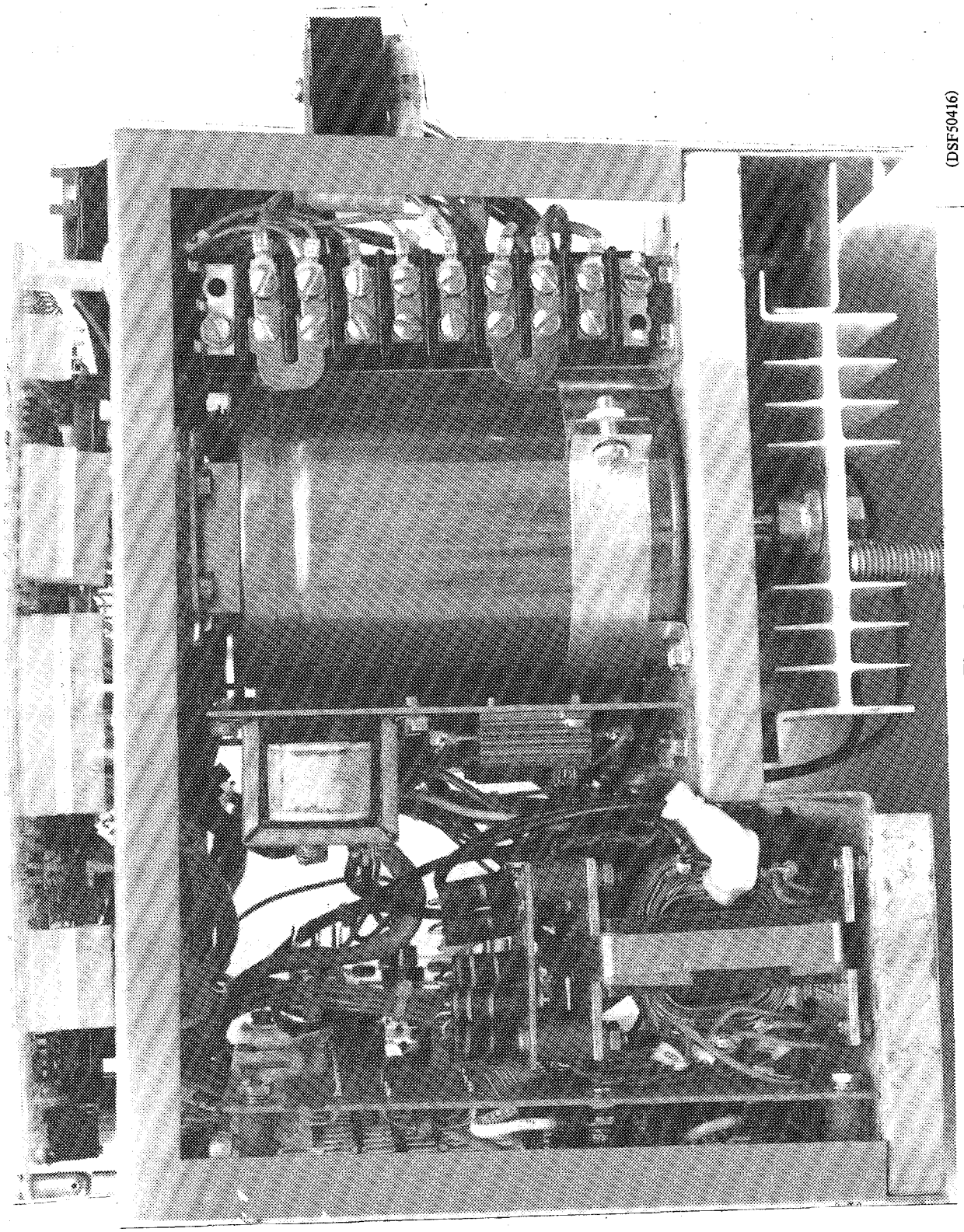
Analog Inputs to the Data Acquisition System from the SPEEDTRONIC Control. Analog inputs to the Data Acquisition System are multiplexed by the ICS3600AMIP card. AN IC3700DLIK board is used to interface the Data Acquisition System 20 conductor ribbon cable to the SPEEDTRONIC control.

DS3820LIMA Isolation Module

The isolation module optically isolates the party line cable. Connection between the isolation module and the DS3800HMHA card in the Data Acquisition System is with a 10 conductor ribbon cable. Connections made to the party line side of the isolation module are terminal board connections. The power required for the DS3820LIMA is +15V, -15V and +5V. Power is supplied from the Data Acquisition System Signal Electronic Module with a separate four conductor cable. The DS3820LIMA is capable of driving a line up to 4000 feet in length. Signal levels are in accordance with RS-422 (See Figure 10).

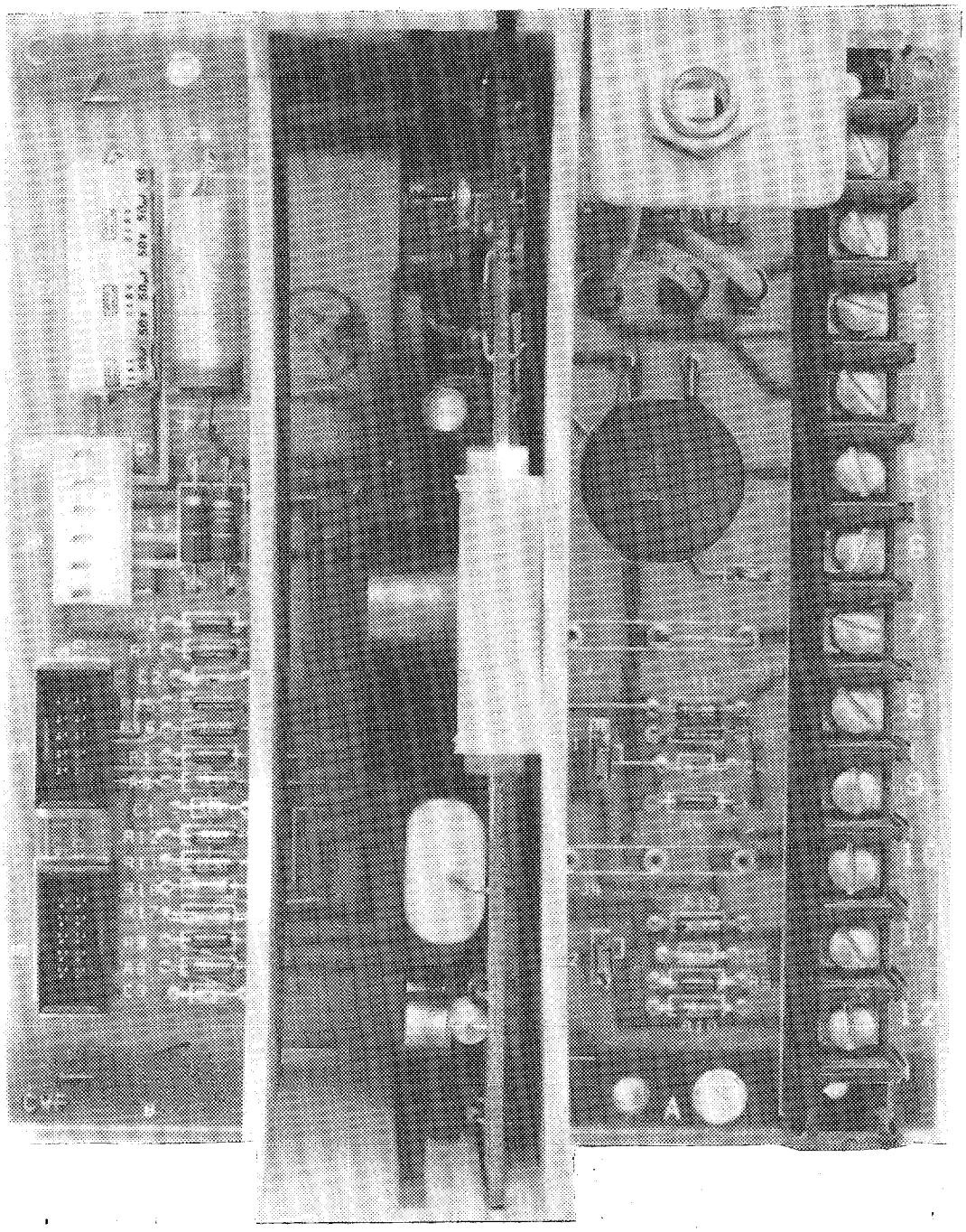
DS3820DDPC Data Acquisition System Display Option

The Data Acquisition System display can be used to read data from the Data Acquisition System memory. Points displayed are analog values and digital values. The display is a five digit LED display. Two ribbon cables connect the Data Acquisition System display with the Data Acquisition System Small Electronics Module (See Figure 11).



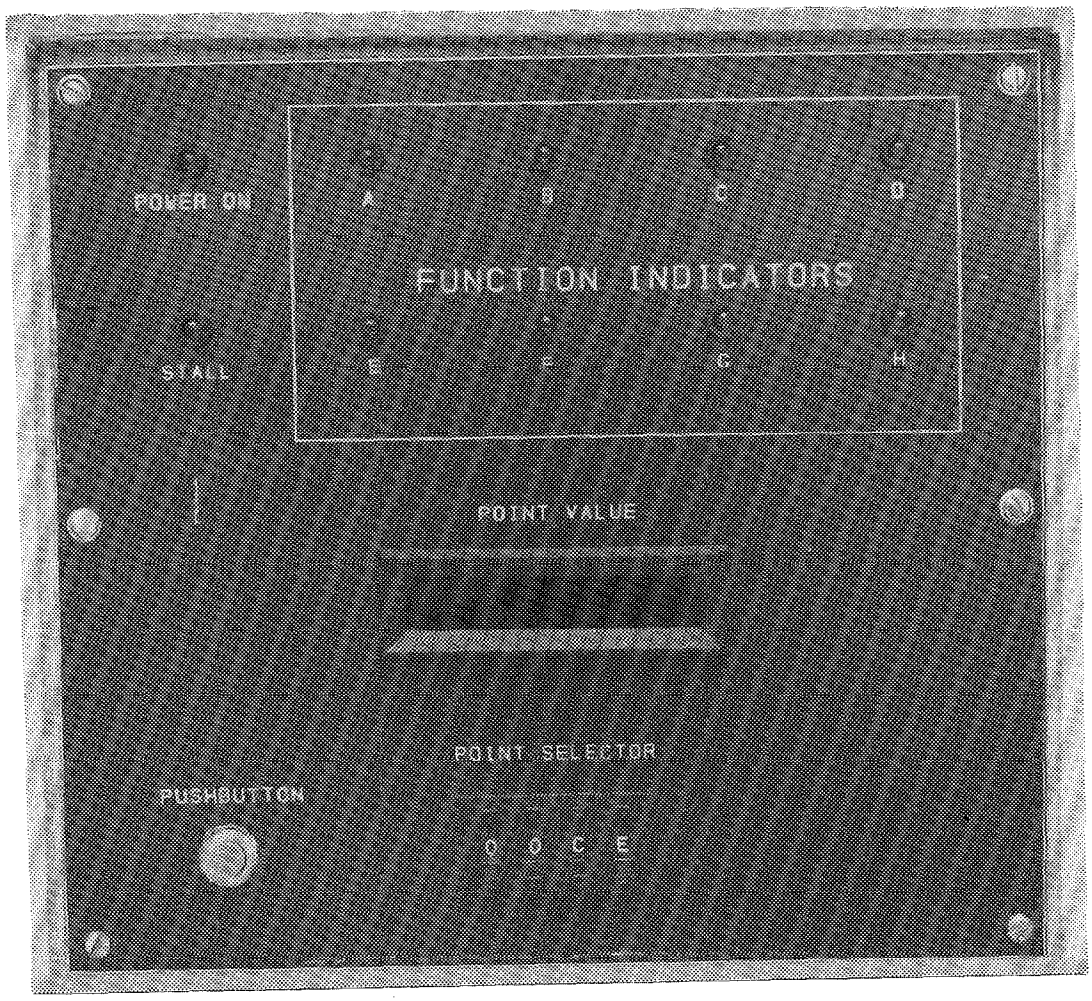
(DSF50416)

Figure 9
PS3 Power Supply



(DSF54060)

Figure 10
DS3820LIMA Isolation Module



(DSF54062)

Figure 11
Optional DAS Display

Table 1
Cable Information

FUNCTION	FROM	TO	CABLE CAT. NUMBER	CONNECTOR TYPE
DAS Power	JA on DAS SEM Power Supply	J1 on DS3820PS3	68A947185G1L__	12 Pin
Digital Inputs and Outputs	JG on DAS SEM	JA on IC3600DLIL Card in SPEEDTRONIC Control	DS3815RCC1BL_	20 Pin
Digital Inputs and Outputs	JB on DAS SEM	JB on IC3600DLIL Card in SPEEDTRONIC Control	DS3815RCE1BL_	26 Pin
Serial Link to ITS	JD on DS3800HXPA in slot 1C/2C of DAS SEM	JC on DS3800HMPA in slot 1A/2A of ITS SEM	277A5736G2L__	25 Pin
Analog Scanner Input/Control	JH on DAS SEM	JA on IC3600DLIK Card in SPEEDTRONIC Control	DS3815RCC1BL_	20 Pin
Analog Scanner Input/Control	JJ on DAS SEM	JA on IC3600DLIJ Card in DAS AIR	DS3815RCG1BL_	34 Pin
Analog Output	JC on DAS SEM	JB on IC3600DLIK Card in SPEEDTRONIC Control	DS3815RCE1BL_	26 Pin
Relay Contact Outputs	Card front connector on DS3800HRRB in DAS SEM	Interface Terminal board	68A947182G1L__ (Down) 68A947182G2L__ (Up)	16 Pin
Serial Communication Link	JA on DS3800HMHA in DAS SEM	JA on DS3820LIMA Module	DS3815RTB1AL_	10 Pin
Power for DS3820LIMA	JE on DAS SEM	JP on DS3820LIMA Module	277A5737G1L__ (Y Conn) 277A5737G6L__ (1-1)	LIMA-6pin SEM-10 Pin
*Power for DS3820DDPC DAS Display	JD on DAS SEM	JC on DS3820DDPC DAS Display	DS3815RTB1AL_	10 Pin
*DAS Display Data Signals	JK on DAS SEM	JB on DS3820DDPC DAS Display	DS3815RTH1AL_	34 Pin

* OPTIONAL

NOTE: The last three digits of the cable catalog number is the length in inches.

Teleprinter Option

The portable teleprinter can be easily carried from unit to unit. The teleprinter can be used to request logs from the Data Acquisition System which will be printed out on the teleprinter. A shelf is provided for the teleprinter to sit on. An AC power plug and an RS-232 cable to the Data Acquisition System are the only connections.

Cables

All the Data Acquisition System components plug together with cables. Table 1 indicates what the cables are used for and where they plug together.

STATION OPERATORS CONSOLE (SOC) DESCRIPTION

Station Communications Controller Medium Electronic Module (SCC MEM)

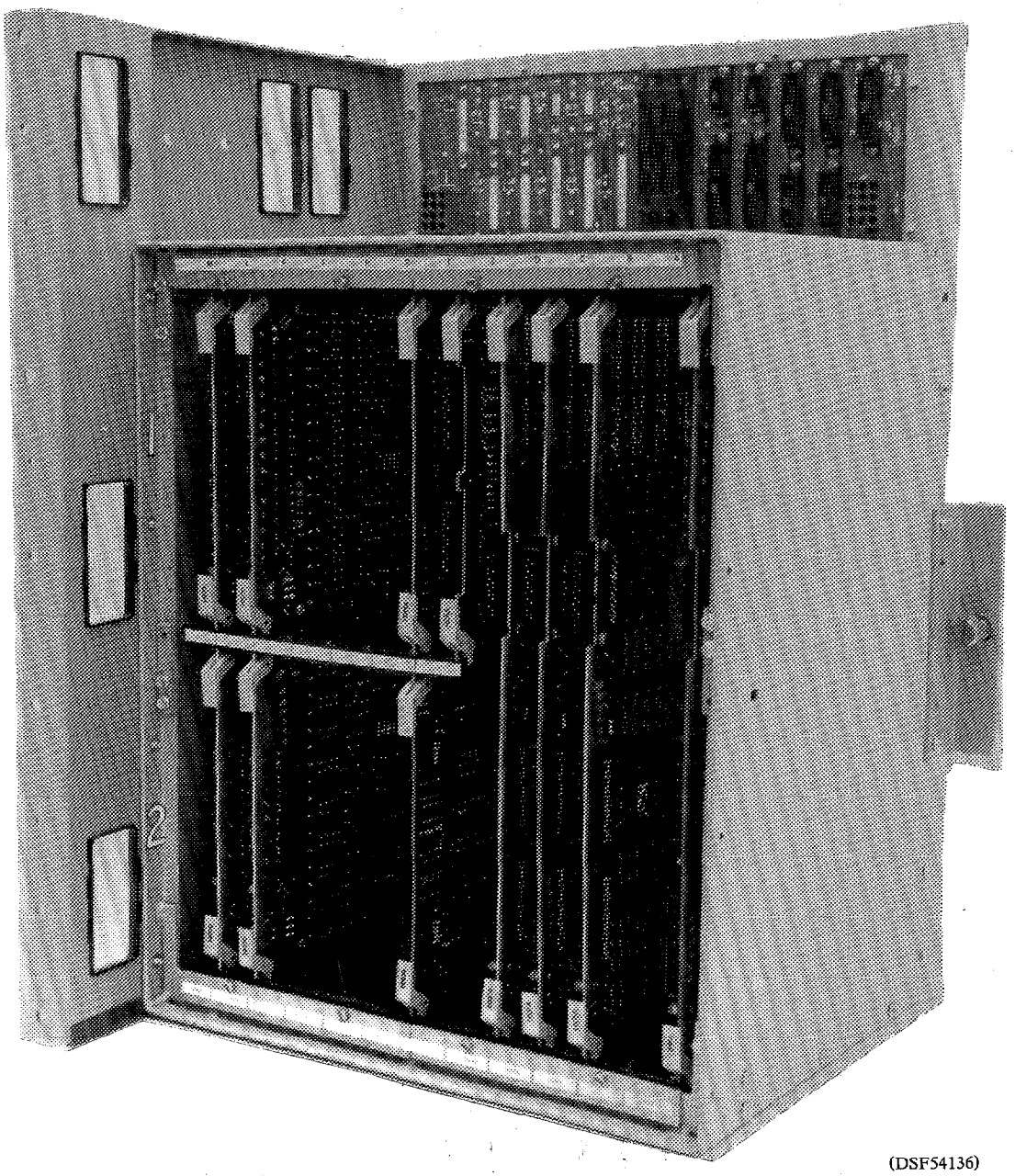
The Station Communications Controller Medium Electronics Module is the module which houses the main processor board and other related boards (See Figure 12). It is physically located in the Station Operator Console. The circuit boards plug into the Medium Electronics Module. An identification strip indicates which boards plug into which slot (See Figure 13).

Internal Connections. All connections between circuit boards and the Medium Electronics Module are made via Plug-in pin connectors located in the rear of the Medium Electronics Module on the backplane. The backplane is a printed wire board which provides all power supply busses (+15, -15, +5 and +28). All other connections to the backplane are wire wrap connections between wire posts (See Figure 14).

External Connections. All external connections to the Station Communications Controller Medium Electronics Module are made either to the top of the Medium Electronics Module or to the printed circuit board fronts via plug connectors.

Connections to Top of Medium Electronics Module. Receptacles are provided at the top of the Medium Electronics Module for plug-in cable connectors. (See Figure 15) These receptacles provide a means of getting power supplies, various analog signals, and display signals into and out of the Medium Electronics Module. Also located at the top of the Medium Electronics Module is a receptacle for use with the Self Verification Module.

Connections to the Front of Circuit Boards. Receptacles are provided on the circuit board fronts for plug-in cable connectors. These are referred to as card front connectors (See Figure 16). Some connectors provide a means of outside communication with the microprocessor. Most communication signals are RS-232 level. The card front connectors make possible communication between the Station Communication Controller Medium Electronics Module and the data logger, alarm CRT and status CRT. Through the use of a limited distance data set it is possible to communicate with a remote location. Other card front connectors are used for contact inputs and outputs.



(DSF54136)

Figure 12
Station Communications Controller MEM

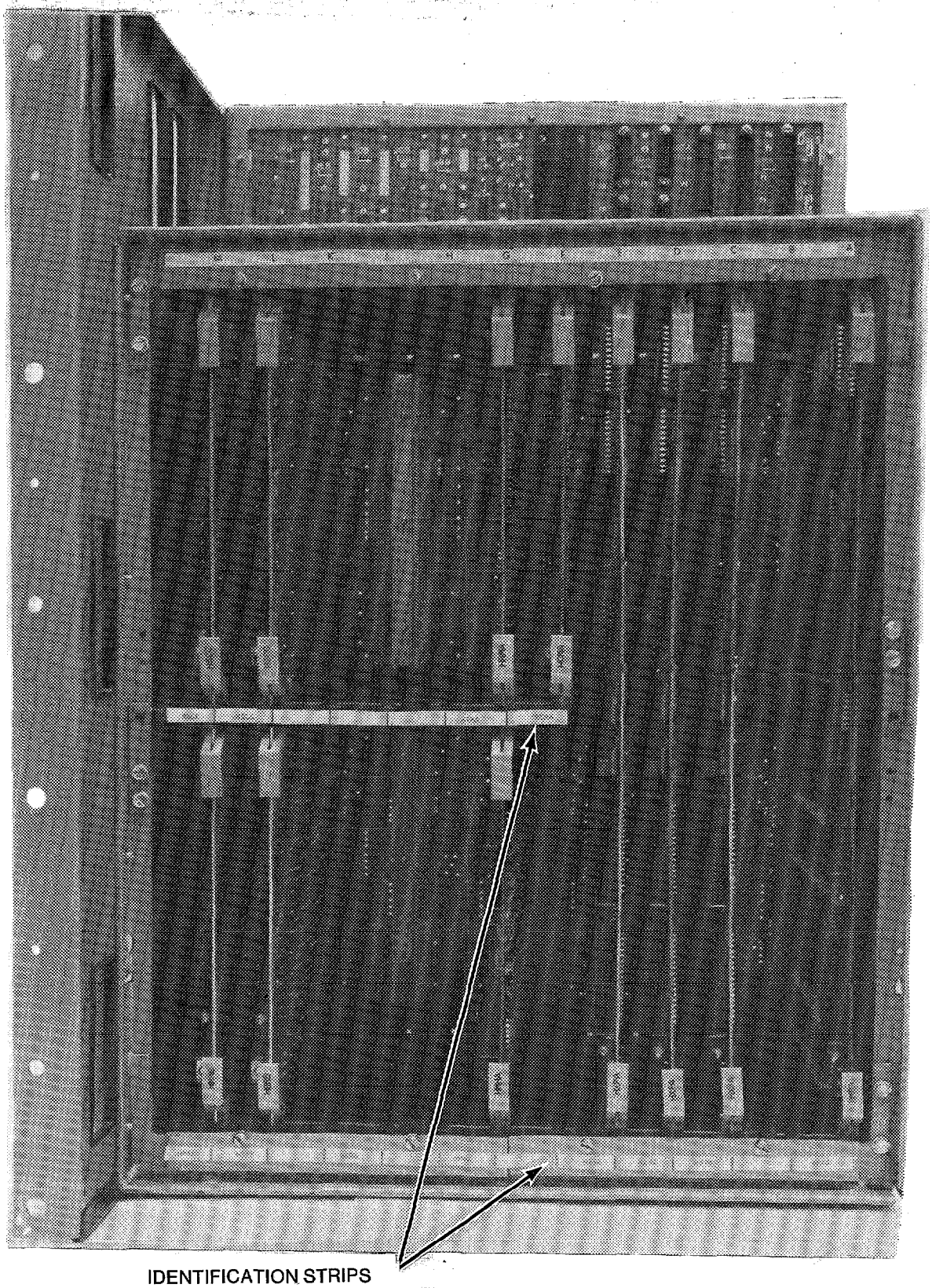
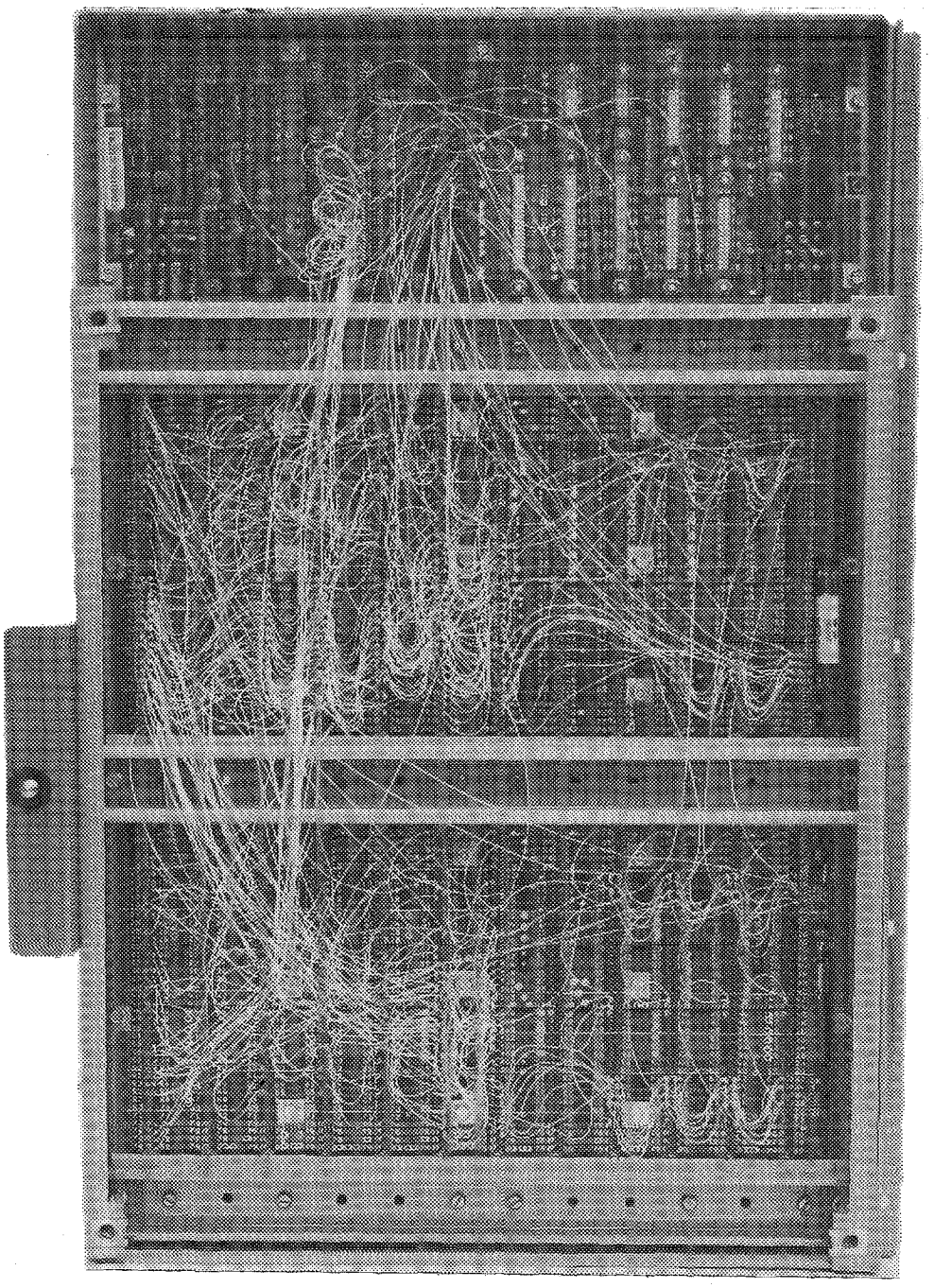
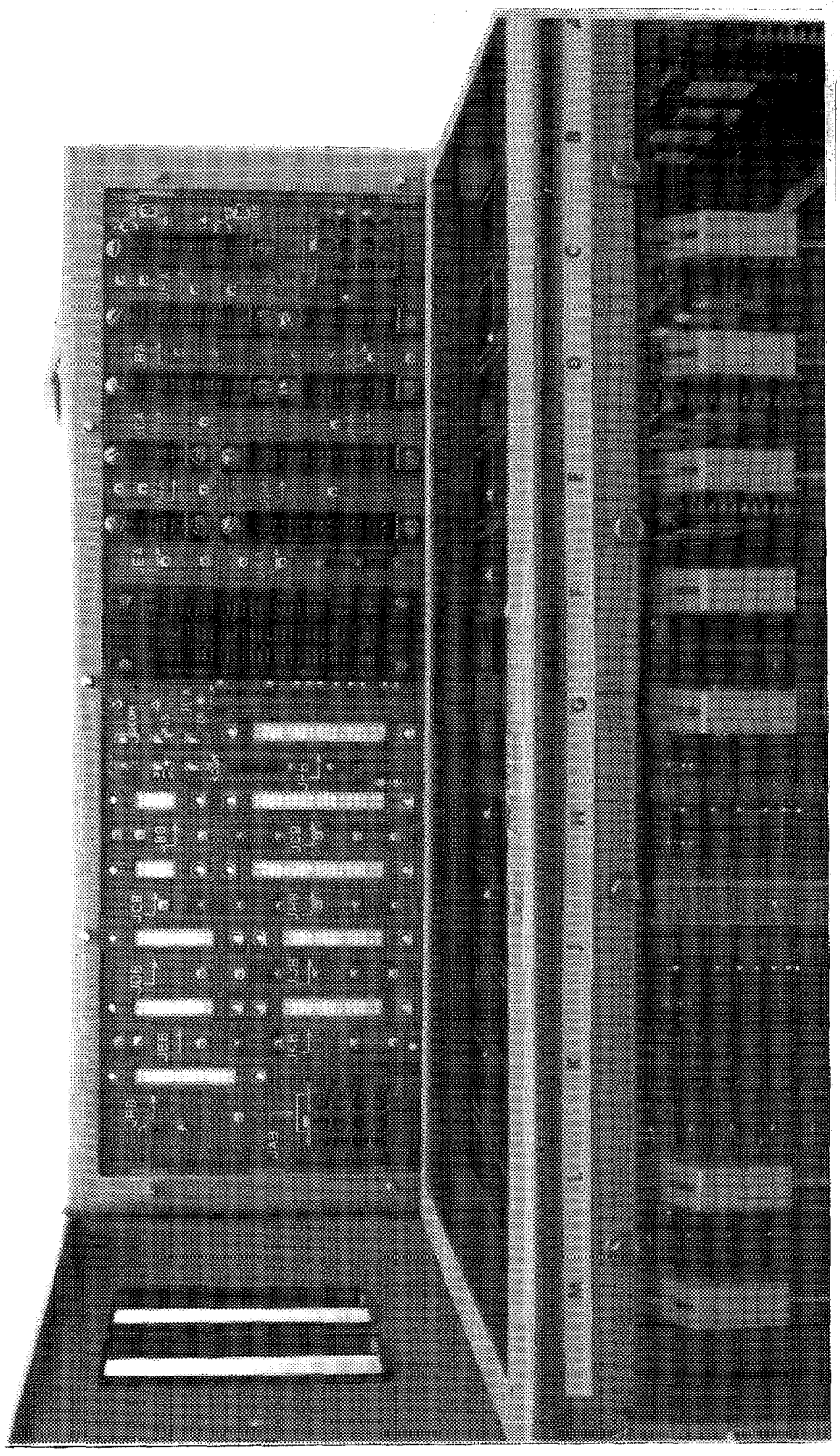


Figure 13
Board Slot Identification



(DSF54141)

Figure 14
Backplane Wiring of MEM



(DSF54142)

Figure 15
Receptacles at Top of MEM

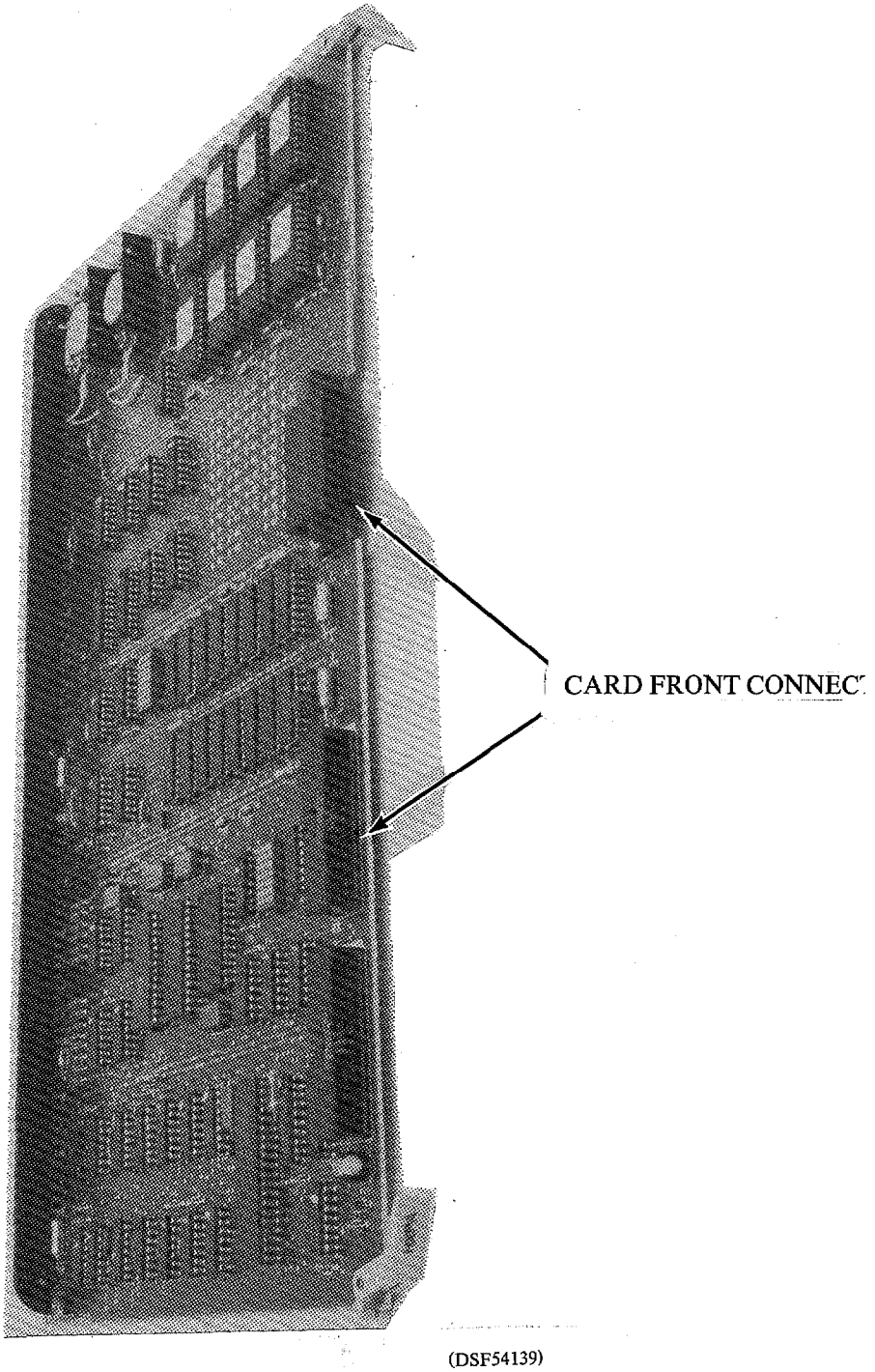


Figure 16
Circuit Board Indicating Card-Front Connectors

Circuit Boards. The circuit boards that plug into the Station Communications Controller Medium Electronics Module come in single size or double size boards. They are held by card slots in the Medium Electronics Module and are connected to the backplane via pin connectors at the back of the card (See Figure 17). The Station Communications Controller Medium Electronics Module contains some description of each type of circuit board. For a more detailed description, refer to the applicable DIRECTO-MATIC ®Plus backsheets.

DS3800HMPA Microprocessor. This is a double size circuit board which houses the main microprocessor chip. It contains the following components:

1. 8080A Microprocessor
2. 4K EROM
3. 1K RAM
4. 8 Level priority vectored interrupt structure
5. RS-232 serial interface
6. 32 Digital output bits
7. 32 Digital output bits
8. 16 Digital input bits
9. 3 Software controlled timers

DS3800HXPA Microprocessor Expander. The HXPA is a double size circuit board which is used as an expander for the HMPA. It contains the following components:

1. 8K Bytes of ROM
2. 2K Bytes of RAM
3. 8 Priority Interrupts
4. RS-232 Serial interface
5. 1 8255 I/O Chip
    a) 11 Bits digital outputs
    b) 11 Bits digital inputs
    c) 2 Bits digital input or output
6. 3 Software Controlled timers

DS3800HMHA Master Highway Controller. The HMHA is a single size circuit board which serves as a communications link between digital systems. It contains the following components:

1. 8085 Microprocessor
2. 2K Bytes dual ported RAM
3. 2 RS-422 Communication channels

DS3800HRMA Dual Ported RAM. This is a single size circuit board which provides memory space for the system. It contains 16K bytes of dual ported RAM memory.

DS3800HRRB Relay Output Card. This is a single size circuit board whose purpose is to provide interface between logic input and relay contact output. It contains 8 relay contact outputs.

DS3800HCMA Dual Communication Controller. This is a single size circuit board. It is a microprocessor based, dual channel, serial communication system. It contains the following components:

1. 8085 Microprocessor
2. 2K Dual ported RAM
3. 2 RS-232 Communication channels

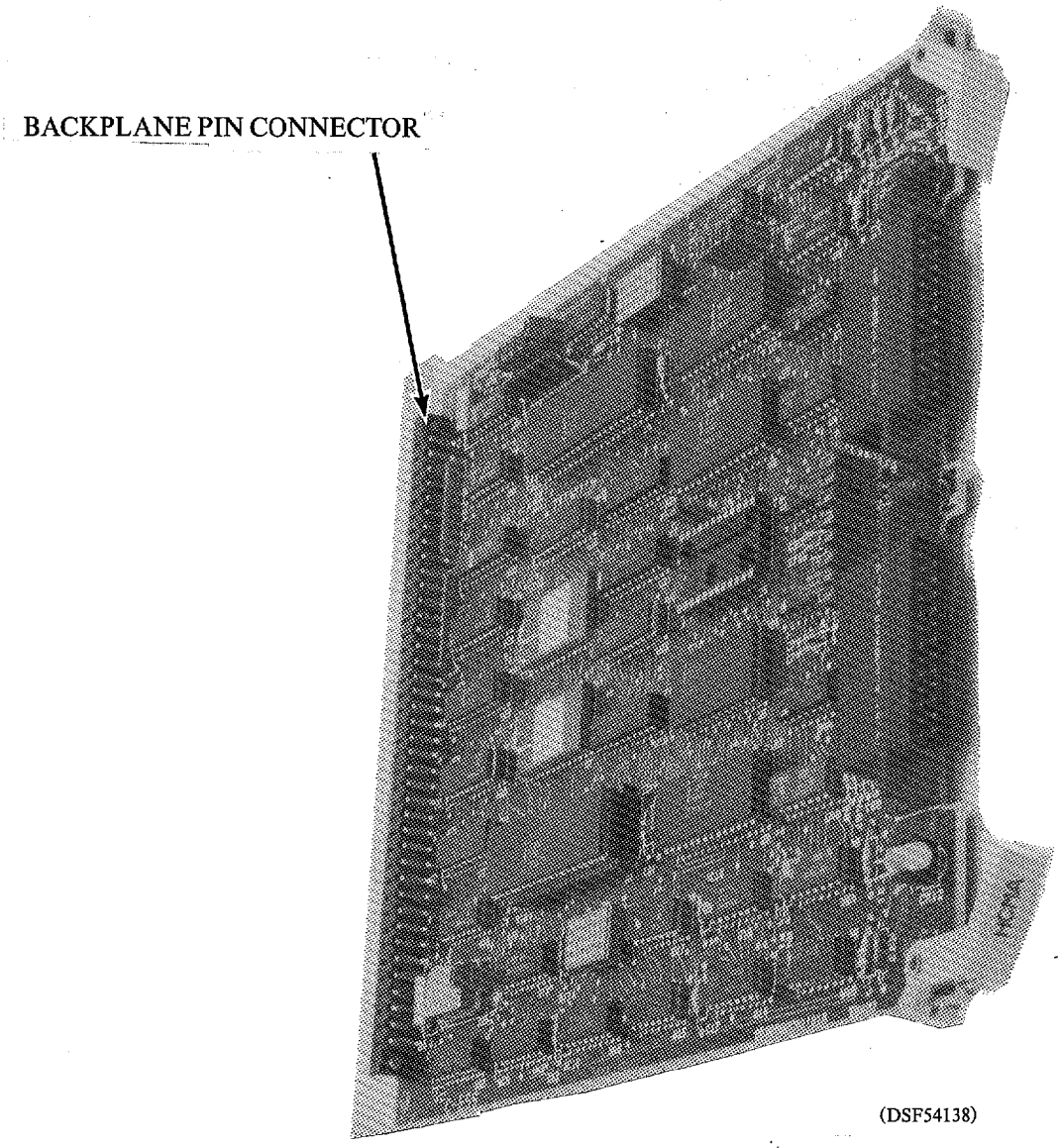


Figure 17
Circuit Board Indicating Backplane Pin Connector

DS3800HSCC Non Isolated Input. This is a single size circuit board which is a 28 VDC, nonisolated, signal conditioning card. It consists of the following components:

1. 16 Filtered 28 VDC inputs
2. 16 Logic output drivers
3. Bus isolation circuit

CRTs

Two CRTs are provided with each Station Operator Console. One is the status CRT. It displays status information for eight units along with time and date. The other CRT is an alarm CRT which displays alarms as they occur for eight units. Each CRT has a 15 inch diagonal screen. The alarm CRT is connected to the HCMA card front connector located in the Station Communication Controller Medium Electronics Module via an RS-232 cable. The status CRT is connected to a second channel on the same card, also via an RS-232 cable. Each CRT requires a single phase 120 VAC source. The following is a brief description of the CRT display.

Status CRT. The Status CRT (See Figure 18) includes a maximum of 24 lines of display at a maximum of 80 characters per line. Each line consists of a name field and a data field. If communication with a unit is lost, that unit's data fields are blanked. The header line has the unit number headers. The unit that is selected by the unit selector has its unit number displayed in inverse video format. The time and date is also displayed on the header line.

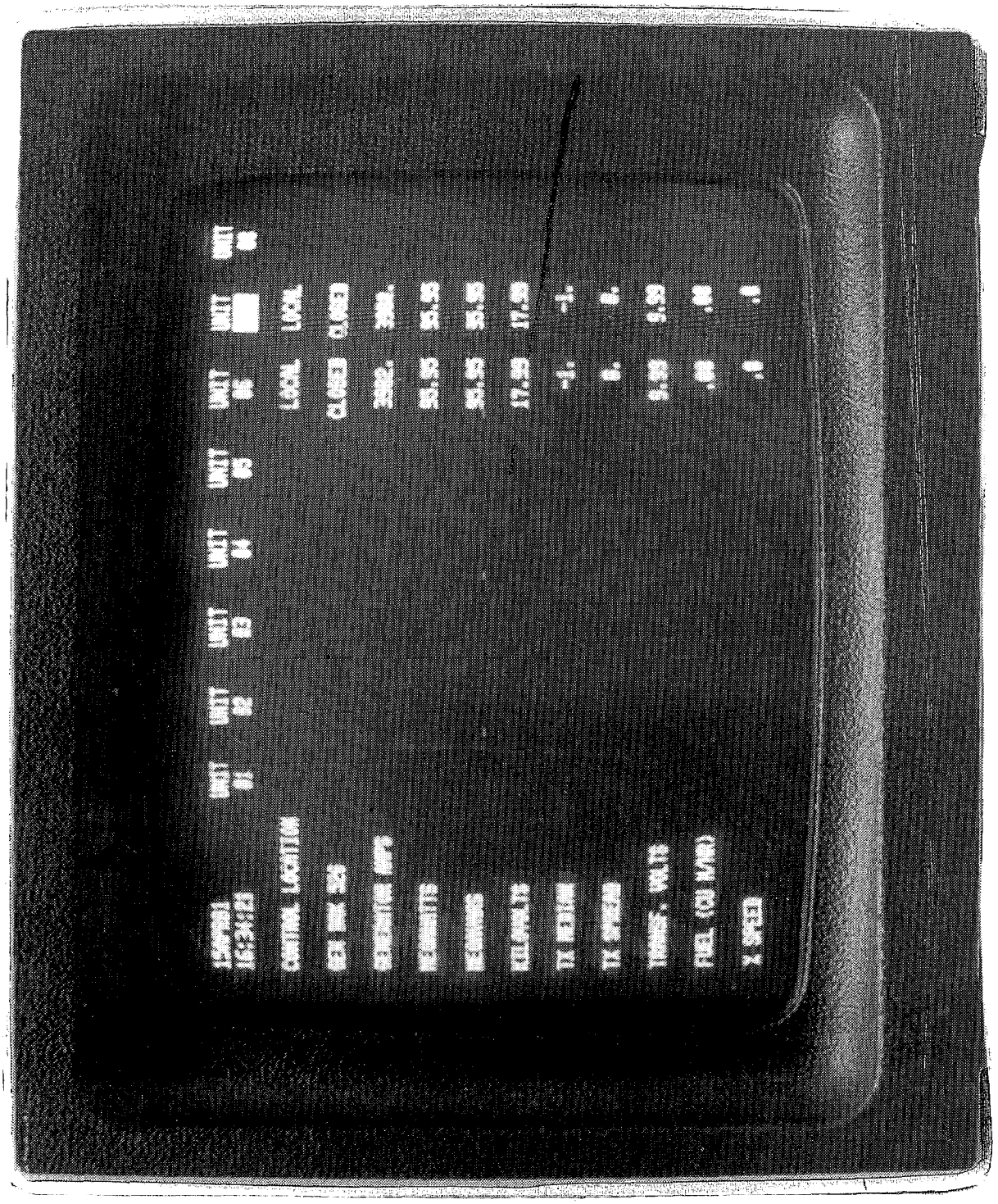
Alarm CRT. The Alarm CRT (See Figure 19) notifies the operator of alarm conditions at a particular unit. When an alarm occurs, the alarm message is written and the unit number field of the message begins to flash. The beeper also begins to sound. The date and time of the alarm is also written on the screen. After an alarm condition has cleared, pushing the clear pushbutton will clear the alarm message from the screen. The scan rate of alarms is 50 times a second.

Power Supplies

The station operator's console houses two power supplies: the PS3 and PS-28A. The PS3 provides bus power to the Station Communications Controller Medium Electronics Module. The PS-28A supplies 28 VDC for light and relay driver operation. The following is a brief description of the power supplies. For a more detailed description, refer to the DS3820PS3 and IC3600EPSM backsheets.

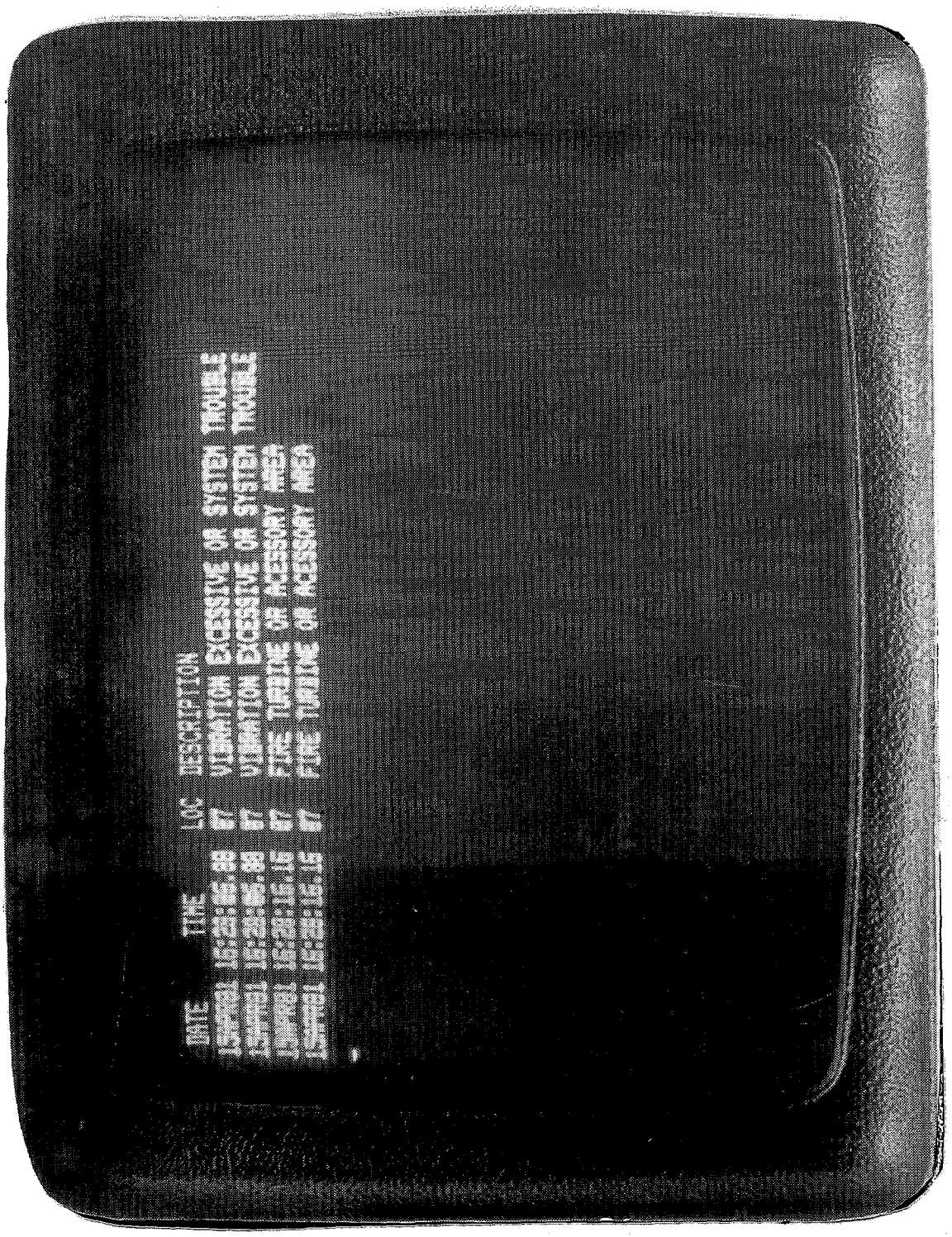
DS3820PS3 Power Supply. The PS-3 accepts a single phase AC voltage, 80 to 132 VRMS at 47 to 63 Hz. It may be converted to a DC input, 70 to 140 VDC by changing jumpers on the power supply terminal board. Refer to back sheets. The PS-3 provides four regulated DC output voltages to externally connected loads. The output voltages are +5 VDC, +15 VDC, -15 VDC and 28 VDC. Output voltage adjustments are available on the DS3800NPSC for P5, P15 and N15 busses. The status light located on DS3800NPSC will be on when all output voltages are normal (See Figure 20). PS-3 is equipped with fault detection circuits that will shut down the power supply if there is an overvoltage or overcurrent on any output. The power supply cannot be reset until the fault is removed.

PS-28A. The PS-28A power supply accepts a 105 to 135 VDC source at 1 ampere (FU1). It provides an unregulated 28 VDC, 2 ampere supply for driving lights and relays. The output 28 volt bus varies directly with the input voltage. Test jacks are provided for measuring the 28 VDC bus. The power supply is overload protected by a 3 ampere fuse (FU-2), and short circuit protected by automatic shut off of inverter (See Figure 21).



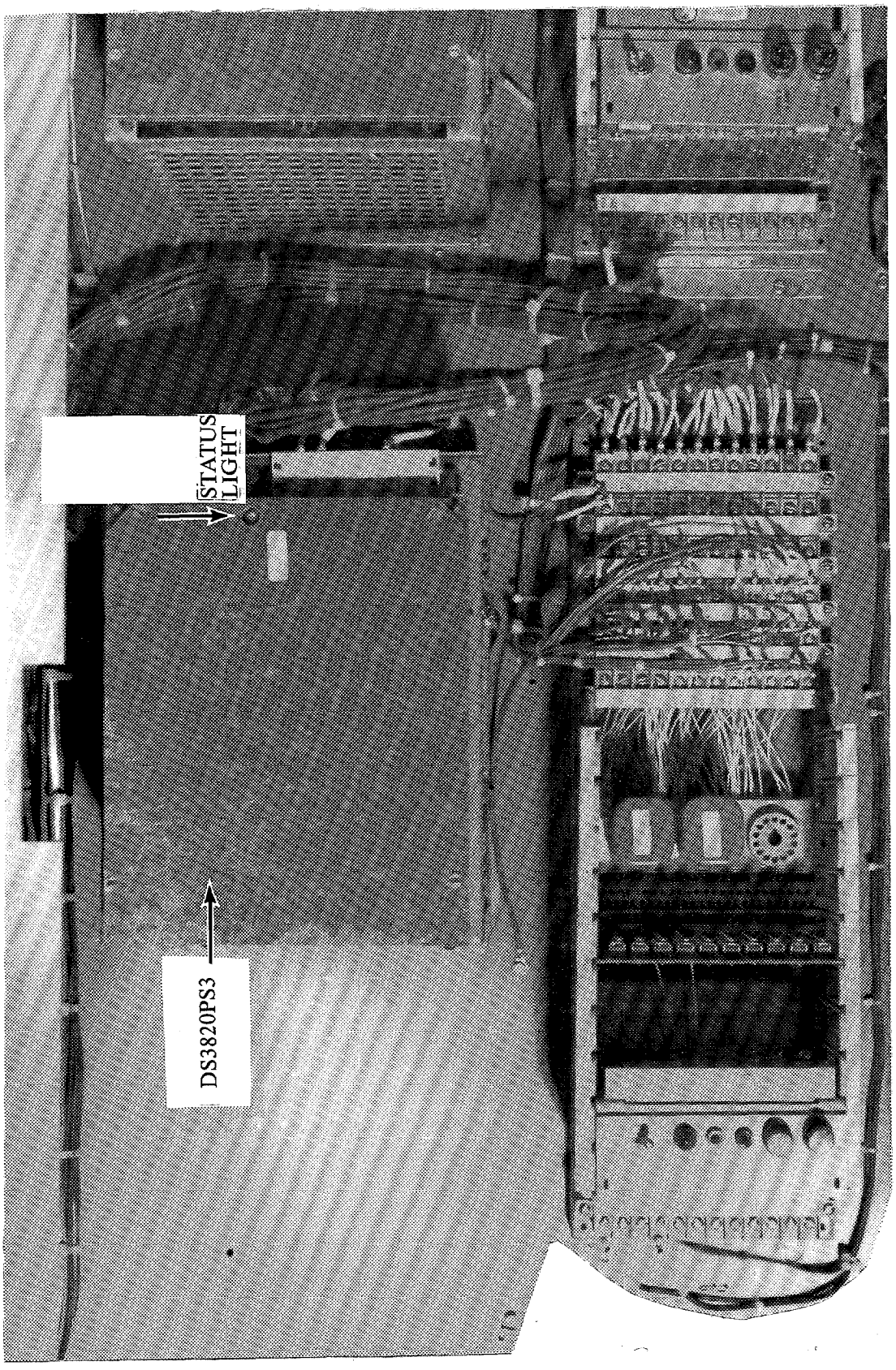
(DSF54227)

Figure 18
Typical Status CRT Display



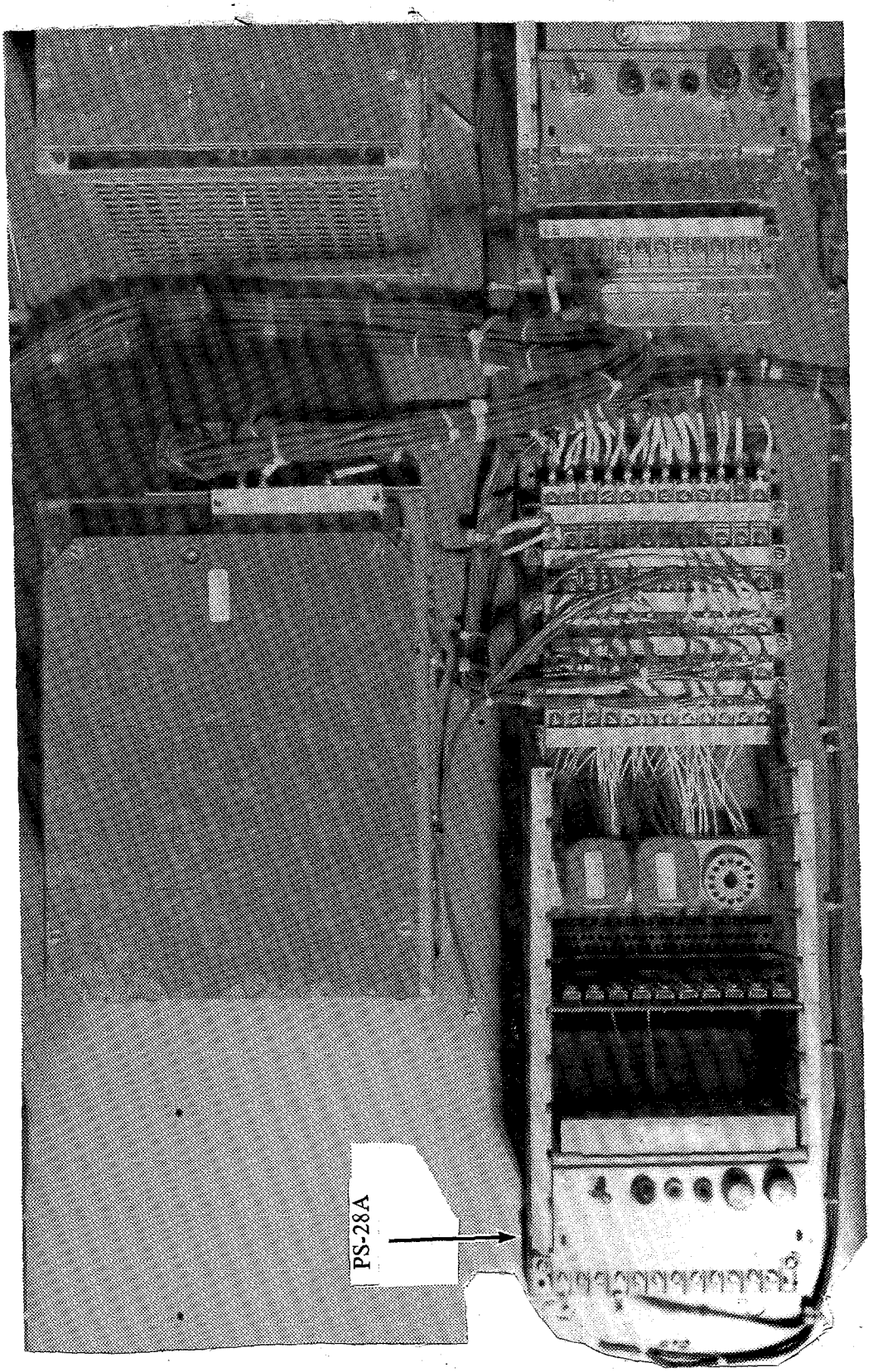
(DSF54228)

Figure 19
Typical Alarm CRT Display



(DSF54225)

Figure 20
DSD3820PS3 Power Supply

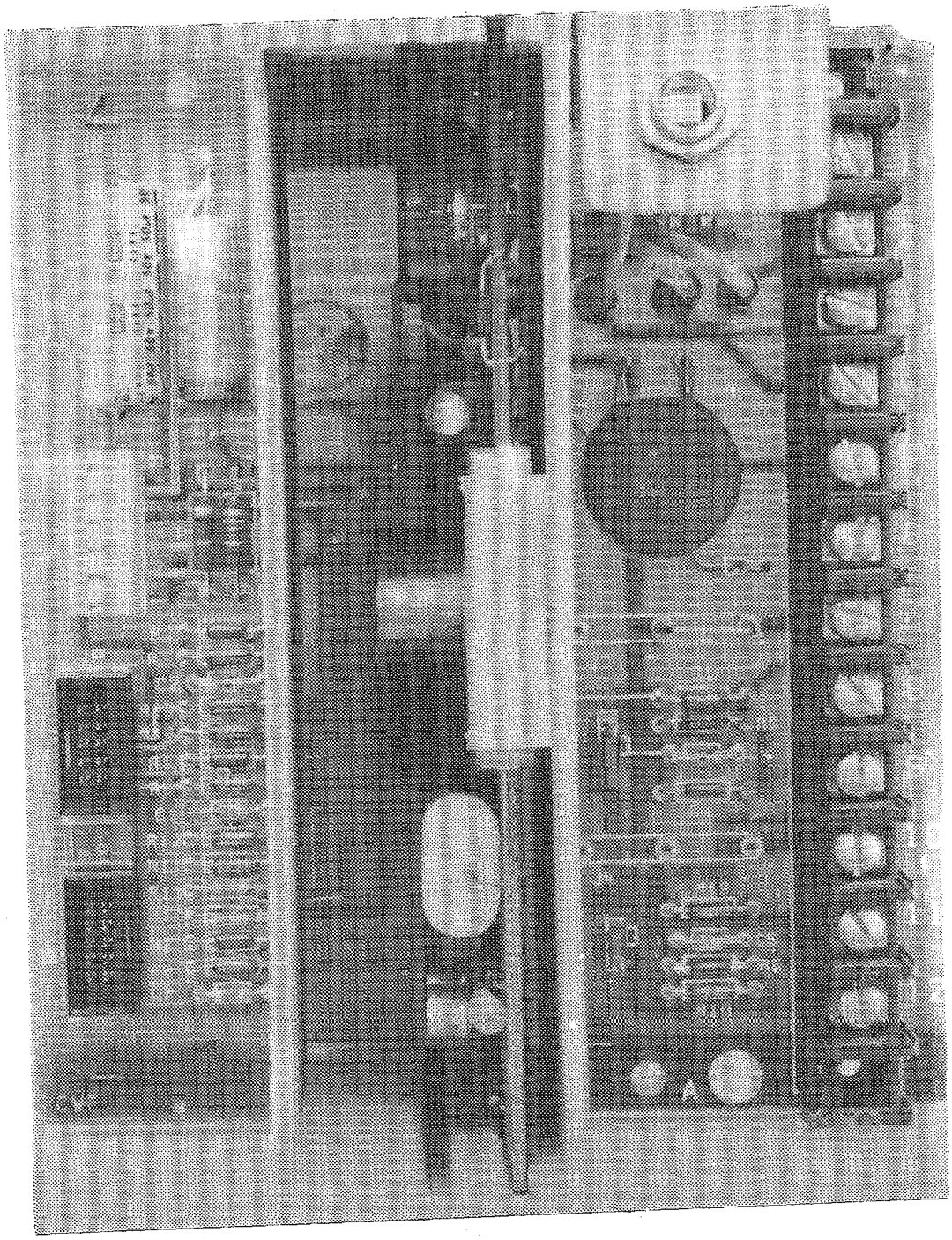


(DSF54225)

Figure 21
PS-28A Power Supply

Teleprinter Option

The teleprinter is used for setting time and requesting logs. Logs available for printout are turbine logs, generator logs and vibration logs.
The teleprinter has two plug-in connectors. The communication line is RS-232 level and plugs into the HMPA card front connector located in the Station Communication Controller Medium Electronics Module. The second connector is the power cord. The printer may be plugged into a standard 120 VAC outlet. These are the only connections required by the teleprinter. The teleprinter may easily be moved and be used with the Data Acquisition System. The teleprinter should be set up to print 30 characters per second (300 baud). It should be set to operate in the full duplex mode which will permit it to transmit and receive data simultaneously. The printer should also be set for upper case letters. The teleprinter uses the standard ASCII code system.



(DSF54060)

Figure 22
DS3820LIMA Isolation Module

DS3820LIMA Isolation Module

The isolation module optically isolates the party line (See Figure 22). The connection between the isolation module and the DS3800HMHA card in the Station Communication Controller Medium Electronics Module is with a 10 conductor ribbon connector. This is known as the low (non-isolated) side. The connection to the party line side of the isolation module consists of terminal board connections, located on the isolation module. This is known as the high (isolated) side. The DS3820LIMA isolation module is capable of driving a line up to 4000 feet at a maximum speed of up to 50K baud. Signal levels are in accordance with RS422. Power is supplied to the isolation module via a six pin connector from the Station Communication Controller Medium Electronics Module. Supply voltages are P15V, N15V and P5V. A phone jack is also provided on the isolation module.

Limited Distance Data Set (LDDS)

The LDDS permits communication between digital equipment at local and remote locations. Signals are transmitted over twisted pair cable (i.e., telephone lines). The LDDS is needed when the user wishes to transmit digital signals (i.e., RS-232) over distances greater than 50 feet. This is an industry standard. The exact distance at which the LDDS becomes necessary is dependent on local environment and application. A LDDS is also required at the receiving end to interface between the transmission line and the digital equipment.

The maximum distance signals that may be driven is dependent on signal speed. The LDDS used with the DATA-TRONIC™ system is capable of speeds up to 9600 baud and distances up to 20 miles on standard 19 to 26 gauge twisted pair cable. The LDDS requires an external 110 VAC power source. When the LDDS is connected to the remote data set, the loop back switch may be used to test the local data set, transmission line, and the data recovery and transmission sections of the remote data set. Refer to the manufacturer's manual for detailed instruction in the use of the limited distance data set.

Supervisory Control and Data Acquisition (SCADA) Link Option

The SCADA link option consists of a serial RS-232 data link to customer's load dispatch computer. This enables the load dispatch computer to raise or lower megawatts and megavars. The link also enables the load dispatch computer to request both analog and digital data for all turbines. Normally limited distance data sets are used to provide the drive for the line. The SCADA serial link uses the RS-232 port on the HXPA card slot 1C/2C in the Medium Electronics Module. An RS-232 cable connects this port to the limited distance data set. The other limited distance data set is mounted in the customer's equipment.

Data Analyzer and Processor (DAP) Link Option

The DAP link option consists of a serial RS-232 data link to a maintenance computer. This enables the maintenance computer to request analog, digital, alarm, pre-trip and post-trip data for all turbines. Normally limited distance data sets are used to provide the drive for the line. The DAP serial link uses the RS-232 port on the HXPA card in slot 1D/2D in the Medium Electronics Module. An RS-232 cable connects this port to the limited distance data set. The other limited distance data set is mounted with the maintenance computer.

Logger Interface Option

The logger interface option consists of a serial RS-232 data link to a data logger. This enables the logger to request analog, digital and alarm data for all the turbines. The logger serial link uses an RS-232 port on the DS3800HCMA in slot 2F of the Medium Electronics Module.

Cables

The Station Communication Controller Medium Electronics Module and its related components are connected by plug-in cables. The following table indicates the type of cable used, its function, its origin and its destination (See Tables 2 and 3).

Redundant Option

The redundant option provides the user with two independently operating systems. Each Station Operator's Console with the redundant option includes two Station Communication Controller-Medium Electronics Modules, each one with its own power supplies (PS-3 and PS-28A). The two Station Communication Controller Medium Electronic Modules allow two independent communication links with the SCADA and Data Analyzer and Processor lines. Two independent Data Acquisition System party lines are also provided, one going to each Station Communication Controller. The teleprinter, alarm CRT and status CRT are shared between the 2 Medium Electronic Modules. A selector switch located on the Station Operator's Console is used to select which Medium Electronics Module the teleprinter, alarm CRT and status CRT are in communication with. This same selector switch also selects which Medium Electronics Module's contact outputs (HRRB) will be used for controlling functions such as start, stop, raise, lower, etc. There is one communication link between the two Medium Electronics Modules. JA on the HXPA card Medium Electronics Module A is connected to JB on the HXPA card Medium Electronics Module B, and JB on the HXPA card Medium Electronics Module A is connected to JA on the HXPA card Medium Electronics Module B. It is used for time setting and synchronization, and data exchange.

Under normal operation, both Station Communication Controller Medium Electronics Modules are energized, Station Communication Controller Medium Electronics Module A or Station Communication Controller Medium Electronics Module B would be selected via the selector switch. Both Medium Electronics Modules operate as if they were the one selected. They each scan their respective Data Acquisition System party lines and maintain data in memory. Each one receives contact inputs from the Station Operators Console, and each ones contact outputs (HRRB) operate, however, only the Medium Electronics Module selected has its output contact circuits energized. The Medium Electronics Module that is selected will have controlling functions through its contact outputs (HRRB). It will light the console lights and send control signals to the turbines when instructed to do so.

The teleprinter, alarm CRT and status CRT would be in communication with the selected Medium Electronics Module. All alarms, data and system status will be displayed and/or printed through the selected Medium Electronics Module.

The following is a brief description of the operation of the reduntant option in various situations. This is not an attempt to depict all possible operating situations; only a few examples are chosen.

Example 1: Station Communications Controller Medium Electronics Module A is selected. A failure occurs in Medium Electronics Module B. Medium Electronics Module A has all control and display functions. A communication trouble alarm will appear on the CRT. Station Communications Controller Medium electronics Module B may be de-energized and removed from service. When Medium Electronics Module B is returned to service, it will begin to scan all inputs and request a data dump from Medium Electronics Module A. All historical type data will be communicated to Medium Electronics Module B and stored. Both Medium Electronics Modules will be time-synchronized and continue normal operation.

Table 2
Station Communication Controller Medium Electronics Module and Related Cabling for Redundant Configuration

FUNCTION	FROM	TO	CABLE CAT. NUMBER	CONN. TYPE
SCC MEM Power Supply	PS-3 J-1 PS-3 J-2	SCC-MEM JAA SCC-MEM JAB	68A947185G1L__ 68A947185G1L__	12 Pin 12 Pin
Serial Data to Alarm CRT (RS-232)	SCC MEM HCMA(1F)JA Terminal Board	Terminal Board Alarm CRT	277A5738G2L__ 277A5761G1L__	25 Pin 25 Pin
Send Data to Status CRT (RS-232)	HCMA (1F) JB Terminal Board	Terminal Bd. Status CRT	277A5738G9L__ 277A5761G1L__	25 Pin 25 Pin
* Serial Data to Data Logger (RS-232)	SCC MEM HCMA (2F) JA	Data Logger HCMA (1E)	206B6818G3A__	25 Pin
Party Line Comm Link (RS-422)	SCC MEM HMHA (2G) JA	LIMA Module JA	DS3815RTB1AL__	10 Pin
Power For LIMA	SCC MEM JEA	LIMA Module JP	277A5727G6L__	6 Pin on LIMA 10 Pin on MEM
* Serial Data to SCADA (RS-232)	SCC MEM HXPA (1D/2D) JD	LDDS DS512	277A5725G2L__	25 Pin
* DAP Serial Comm (RS-232)	SCC MEM HXPA(1D/2D) JD	LDDS DS512	277A5735G2L__	25 Pin
Contact Inputs	Terminal Bd.	SCC MEM HSCC Card Front	68A947182G2L__	17 Pin
Contact Outputs	SCC MEM HRRB Card Front	Interface term board	68A947182G12L__	17 Pin
Serial Data to Teleprinter	HMPA(1A/2A) Terminal Bd.	Terminal Bd Teleprinter	277A5738G1L__ 277A5272G1L__	25 Pin 25 Pin
Parallel Data Link	HXPA(1C/2C) JA HXPA(1C/2C) JB	Other MEM HXPA (1C/2C) JB Other MEM HXPA (1C/2C) JA	277A5272G1L__ 277A5272G1L__	34 Pin 34 Pin

* Optional

NOTE: The last three digits of the cable catalog number is the length in inches.

Table 3
SCC Cables For Non-Redundant Configuration

FUNCTION	FROM	TO	CABLE CAT. NUMBER	CONNECTOR
SCC MEM Power Supply	PS3 J1 PS3 J2	SCC MEM JAA SCC MEM JAB	68A947185G1L__ 68A947185G1L__	12 Pin 12 Pin
Serial Data to CRT (RS-232)	SCC MEM HCMA (IF) JA	Alarm CRT	206947185G1L__	25 Pin
Serial Data to Status CRT (RS-232)	SCC MEM HCMA (IF) JB	Status CRT	206B6818G1A__	25 Pin
* Serial Data to Data Logger	HCMA (2F) JA	Data Logger HCMA (IE)	206B6818G3A__	25 Pin
Party Line Communication Link (RS422)	SCC MEM HMHA (2G) JA	LIMA Module JA	DS3815RTB1AL__	10 Pin
Power for LIMA	SCC MEM JEA	LIMA Module JP	277A5737G6L__	6 Pin on LIMA 10 Pin on MEM
*Serial Data to SCADA (RS 232)	HXPA (1D/2D) JDS	LDDS DS512	277A5737G2L__	25 Pin
Contact Inputs	Terminal Board	SCC MEM MSCC Card Front	687A947182G12L__	17 Pin
Contact Outputs	SCC MEM HRRB Card Front	Terminal Board	68Al947182G12L__	17 Pin

* Optional

NOTE: The last three digits of the cable catalog number is the length in inches.

Example 2: Station Communications Controller Medium Electronics Module A is selected. Failure occurs in Medium Electronics Module A. Medium Electronics Module A has all control and display functions. When the failure occurs, the Station Operator's Console indicating lights will go off. The time display on the CRTs will freeze. No alarm will be displayed because the alarm CRT is in communication with Medium Electronics Module A and it is unable to deliver the message. The operator must recognize this and select Medium Electronics Module B. Once Medium Electronics Module B has been selected, it now has all control and display functions. The indicating lights on the Station Operator's Console will re-energize, a communication trouble alarm will appear on the alarm CRT and time display will be functioning. Medium Electronics Module A may be removed from service for repair. When returned to service, it will begin scanning all inputs, and request a data dump from Medium Electronics Module B. All historical type data will be communicated and stored in Medium Electronics Module A. Both medium Electronics Modules will be in time synchronization and resume normal operation. These examples demonstrate how the redundant option allows the Station Operator's Console to continue its normal operation if a problem occurs with one of the Medium Electronics Modules.

INSTALLATION NOTES

Data Acquisition System Installation

To prepare for the Data Acquisition System installation, you should be familiar with the various parts of the Data Acquisition System. Also verify that all parts of the Data Acquisition System are connected in accordance with the system elementary drawing.

Party Line Connectors. Refer to system elementary for instructions regarding the party line connectors.

Party Line Terminators. Each DS3820LIMA module has two Berg jumpers which can jumper an impedance across both the send and receive circuits of the party line. This termination impedance needs to be jumpered in at the ends of the party line. The Berg jumpers are labled J1 and J2 (See Figure 23). In Figure 24, the party line terminators would be jumpered in only at the Station Operators Console and Data Acquisition System #8. The terminators should be jumpered out at all other locations.

In Figure 25, the party line terminators would be jumpered in at the Station Operator's Console and Data Acquisition Systems #1 and #8. The terminators should be jumpered out at all other locations.

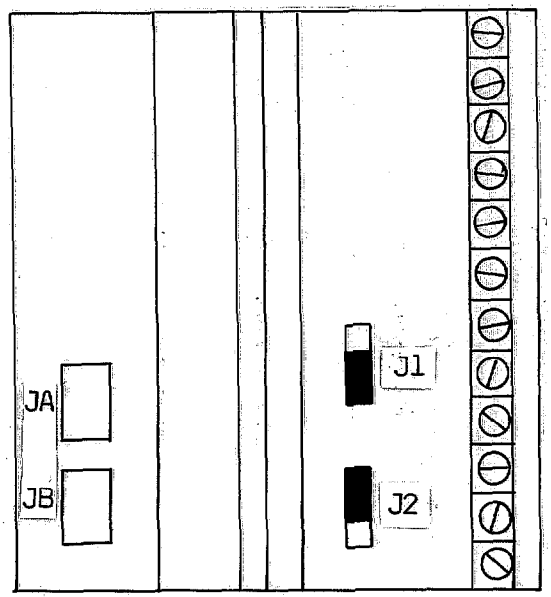
Verify that the Berg jumper positions on all cards are correct. Refer to front sheets of the elementary diagram.

Verify that the Data Acquisition System is operating normally. Refer to the Troubleshooting Section for a description of normal operation. If the Data Acquisition System is not operating normally, refer to the Troubleshooting Guide paragraph.

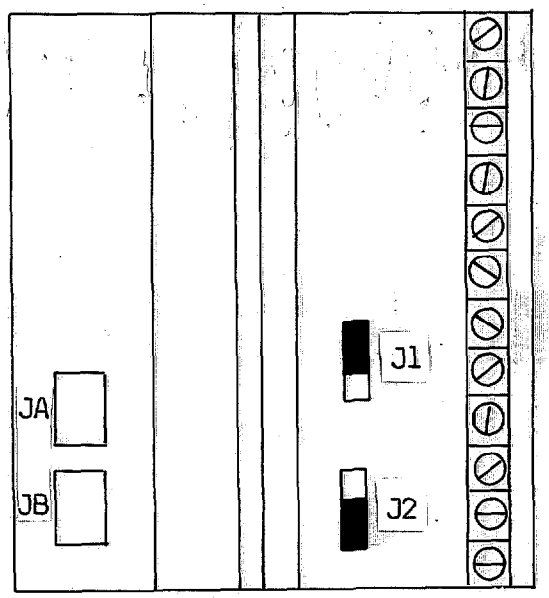
Verify that the analog input rack is operating by either of two methods:

1. Use the optional Data Acquisition System display to view individual analog points. Analog values are displayed in engineering units.

2. Run the Self Verification Module test described in the paragraph entitled Data Acquisition System Analog Input.



BERG JUMPER POSITIONS
TO JUMPER IN TERMINATION
IMPEDANCE



BERG JUMPER POSITIONS
TO JUMPER OUT TERMINATION IMPEDANCE

Figure 23
DS3820LIMA Isolation Module Berg Jumpers

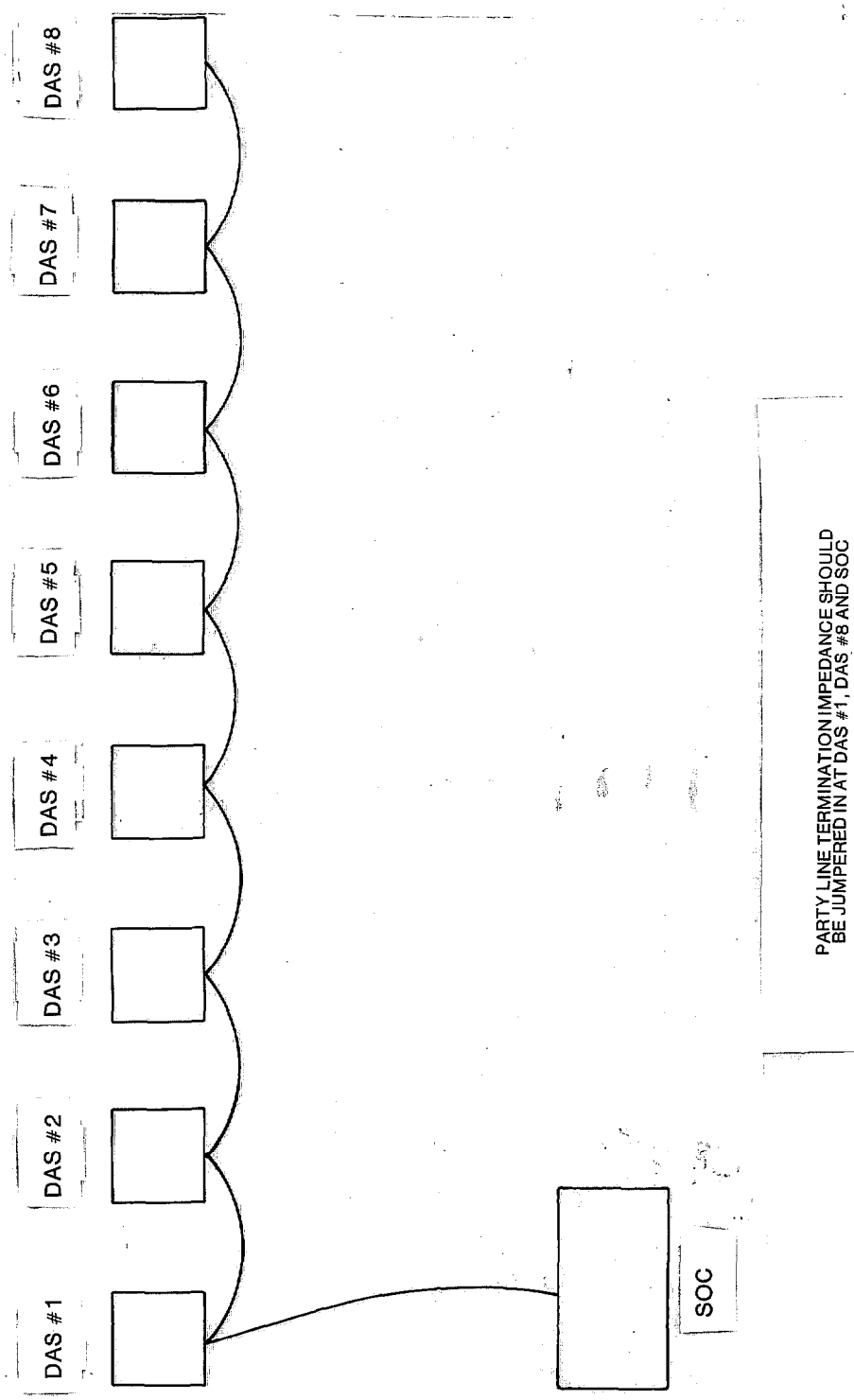


Figure 24
Party Line Terminators

Station Operator's Console Installation

Familiarize yourself with the paragraph explaining the Station Operator's Console in this manual.

Verify that all parts of the Station Operator's Console are connected in accordance with the elementary drawing.

Party Line Connector. Refer to the Station Operator's Console elementary diagram for instructions regarding the party line connector.

Party Line Terminator. Refer to Installation Notes (Party Line Terminators) for the set up of the party line termination impedance.

Unplug all cards in the Station Communications Controller Medium Electronics Module. Perform the power supply checkout described in the Power Supply instructions herein.

Verify that the Berg jumper positions on all cards are correct. Refer to frontsheets of the Station Operator's Console elementary diagram.

Verify that the Station Operator's Console is operating normally. Refer to the Troubleshooting Section, Normal Operating Procedures.

OPERATOR INSTRUCTIONS

The purpose of these instructions is to describe the Station Operator's Console from an operator's point of view. A description of each device and how to operate that device is included. This is not intended as a tutorial on operation of the turbine.

Device Summary

Unit Select Switch. The Unit Select Switch (See Figure 26) is used to select which unit the switches, pushbuttons and indicating lights control or monitor. See the paragraph entitled "Unit Selection" for details.

Master Control Switch. Start position initiates a start sequence for the selected unit provided all permissives are satisfied. Stop position initiates a stop sequence for the selected unit provided the unit is running.

Load Control Switch. Base selects base temperature control for the selected unit. Peak selects peak temperature control for the selected unit.

Base Indicating Light. Base temperature control has been selected for the selected unit when this lamp is illuminated.

Peak Indicating Light. Peak temperature control has been selected for the selected unit when this lamp is illuminated.

Watts Switch. Raise position will cause the selected unit to pick up load. Lower position will cause the selected unit to shed load. Unit will only raise or lower watts while the switch is held in the raise or lower position.

Gas Monitor Switch. Monitor position selects gas temperature monitoring function. Override position deselects the gas temperature monitor log function.

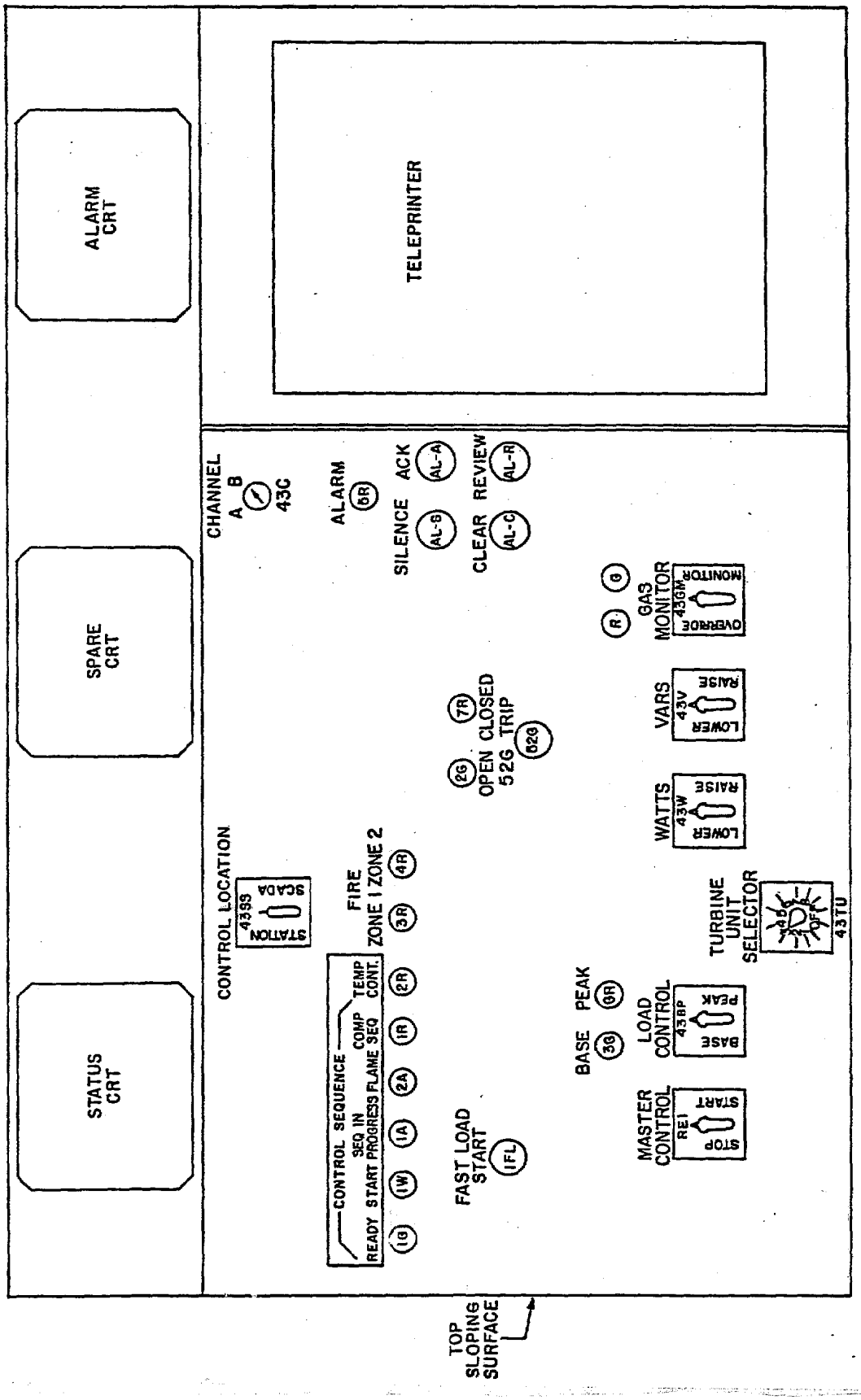


Figure 26
Station Operator's Console Devices

Override Indicating Light. Gas temperature monitoring system for the selected unit is deselected when this lamp is illuminated.

Monitor Indicating Light. Gas temperature monitoring function for the selected unit has been selected when this lamp is illuminated.

Fast Load Start Pushbutton. Pushing this pushbutton will cause the generator to load at the fast rate for the selected unit.

52G Trip Pushbutton. Pushing this pushbutton will trip the 52G circuit breaker for the selected unit.

52G Open Indicating Light. 52G Breaker is open for the selected unit when illuminated.

52G Closed Indicating Light. 52G Breaker is closed for the selected unit when this lamp is illuminated.

Alarm Indicating Light. An alarm has occurred on any one of the units when this lamp is illuminated.

Silence Pushbutton. Pushing the silence pushbutton will silence the beeper.

Acknowledge Pushbutton. Pushing the acknowledge pushbutton will stop the flashing of the unit field on the alarm CRT.

Clear Pushbutton. Pushing the clear pushbutton will clear alarms from the alarm CRT that have been acknowledged and are no longer in alarm condition.

Review Pushbutton. Pushing the review pushbutton will cause the alarm CRT to scroll such that all current alarms may be viewed.

Channel Select Switch. Position "A" will select channel "A". See the paragraph entitled "Channel Selection" for details.

Control Location Switch. Station position selects the Station Operator's Console for control of the selected unit. SCADA position selects the customer's load dispatch for control of the selected unit. See the paragraph entitled "Control Location" for details.

Ready Indicating Light. The selected unit is ready for a start command when this lamp is illuminated.

Sequence in Progress Indicating Light. The selected unit is in a starting or shutdown procedure when this lamp is illuminated. Ratchet or turning gear operation will also be indicated as a sequence in progress.

Flame Indicating Light. This lamp will be illuminated when flame has been detected in the combustion chambers of the selected unit.

Complete Sequence Indicating Light. The selected unit has completed the starting sequence when this lamp is illuminated.

Temperature Control Indicating Light. This lamp is illuminated when the selected unit is under temperature control.

Fire Zone 1 Indicating Light. This lamp is illuminated when fire is detected in the load tunnel area of the selected unit.

Fire Zone 2 Indicating Light. This lamp is illuminated when fire is detected in the turbine compartment or the generator auxiliary compartment of the selected unit.

Status CRT. The status CRT displays status information for all eight units and also time and date. See paragraph entitled "Status CRT" for details.

Teleprinter. The teleprinter is used for setting time and date and requesting logs. See "DAS Maintenance Instructions" for details.

Beeper "A". When an alarm occurs, the beeper will sound until the silence pushbutton is pressed.

Beeper "B" (Part of Redundant Option). When an alarm occurs, the beeper will sound until the silence pushbutton is pressed.

Channel Selection (Part of Redundant Option)

The Station Operator's Console is equipped with two sets of electronics and communication links. They are designated as channels "A" and "B". Although only one channel is in control at a given time, both channels are continually updated with data from the unit. The operator selects the desired chanel with the channel select switch. When the other channel is selected, the operator will see the status CRT become rewritten. No other change should be noticed. If the selected channel should malfunction, it is the operator's responsibility to select the other channel. No automatic switchover is provided.

Unit Selection

The Unit select switch is a twelve position switch. One position is provided for each of eight units and four off positions are provided. When the operator selects a particular unit, the switches and lights on the console control or monitor only that unit. For example, if unit 2 is selected, the status indicating lights reflect the status of unit 2 and the switches and pushbuttons control only unit 2. If the unit select switch is in one of the four off positions, the status lights will all be off and the control switches and pushbuttons are disabled.

Control Location

There are three separate control locations for each unit. Control location for each unit is indicated on the status CRT.

CAB or Local. The 43/REMOTE switch on the SPEEDTRONIC control panel allows either local or remote control. If LOCAL is selected in the CAB, neither the station operator nor SCADA can have control. If REMOTE is selected in the CAB, control is given to the Station Operator Console.

Station or Central. The station operator at the Station Operator's Console has control of a unit only if REMOTE has been selected in the CAB. Once control for a unit has been given to the station operator, he may either control the unit from the Station Operator's Console or give control to SCADA.

Supervisory Control and Data Acquisition (SCADA). The control location switch on the Station Operator's Console allows either station or SCADA control. When SCADA control is selected, SCADA will have control of that unit. At any time the station operator may elect to have control of that unit by selecting station control. The operator in the CAB may at any time take control away from either SCADA or station.

Status CRT

The status CRT includes a number of lines of information. Each line consists of a name field and then a data field for each of eight units. If communication is lost with a particular unit, all of that unit's data fields are blanked. Each line of a typical status CRT is described in detail.

Header Line. The header line has the unit number headers. The data fields for that unit line up directly below that unit header. The unit that is selected by the unit selector switch has its unit number displayed in inverse video format. Time and date are also displayed on the header line. The format is as follows:

_ _ _ _ : _ : _
YEAR MONTH DAY HOUR MINUTE SECOND

Control Location. Describes control location for each unit.

CAB or local = Control location is at the SPEEDTRONIC turbine control panel in the CAB.

STA or Central = Control location is at the Station Operator's Console.

SCADA = Customer's load dispatch has control.

Gen Brk 52G. This shows status of 52G breaker.

OPEN = Breaker is open.

CLOSED = Breaker is closed.

Megawatts. This shows the megawatts being produced by the unit. One megawatt equals one million watts.

Megavars. This shows the megavars being produced by the unit. Positive values reflect lagging vars. Negative values reflect leading vars. One megavar equals one million vars.

Kilovolts. This shows line kilovolts for each unit. One kilovolt equals one thousand volts.

EXH Median. This indicates the median exhaust temperature sensed by the thermocouples at the turbine exhaust.

Temp Spread. This indicates the spread between the highest and lowest reading exhaust thermocouple.

Gas P2. This indicates gas pressure in pounds per square inch (PSI).

Gas Fuel Flow. This shows gas fuel flow in units such as cubic meters per hour or standard cubic feet per day.

Speed-RPM. This indicates turbine speed in revolutions per minute (RPM).

Alarm CRT

The alarm CRT notifies the operator of alarm conditions at a particular unit. An alarm message consists of:

_ _ _ _ :_ :_ _ _ _ _ _ _ _ _
YR MTH DAY HR MIN SEC FRACTION UNIT DESCRIPTION
OF SEC OF ALARM

When an alarm occurs, the alarm message written on the screen and the unit number field of the message begins to flash. Also, the beeper begins to sound. Pressing the silence pushbutton stops the beeper from sounding. Pushing the acknowledge pushbutton will stop the flashing of the unit number. Pushing the clear pushbutton will clear all alarm messages for those alarms which are no longer in an alarm condition. In case more alarms occur than will fit on the screen, the previous alarms may be viewed by pushing the review push-button. As the review pushbutton is held down, the alarm CRT is scrolled. A maximum of 96 alarms are stored. Subsequent alarms after 96 are lost.

Header Line. The header line is as follows:

DATE TIME LOC DESCRIPTION

The date the alarm occurred lines up under the date field.

The time the alarm occurred lines up under the time field. Time is given in hours, minutes, seconds and fractions of seconds. Scan rate of alarms is 50 times per second. Therefore, the actual resolution of the time lag is 1/50th of a second or .02 seconds.

The unit number indicates the unit that has detected the alarm. Most alarm types are sensed at the units. However, there are several alarms sensed at the Station Operator's Console. These alarms have "STA" in the unit number field.

Alarm Messages. The following is a typical set of alarm messages. Most are self explanatory. The first 60 messages correspond to the 60 drops of the SPEEDTRONIC annunciator. Unused drops are also indicated.

Motor control center or BATT charger AC undervoltage.

Unused drop #1

Cooling water system trouble or cooling water pressure low.

Lube level high or low.

Hydraulic supply pressure low or auxiliary hydraulic pump running.

Turning gear trouble.

Battery undervoltage or DC motor undervoltage.

Turbine underspeed.

Auto sync trouble or bus undervolt. Don't use auto sync.

Generator stator temperature high.

Auxiliary motor overload.

Power supply trouble. (This refers to SPEEDTRONIC power supply.)

Speed circuit abnormal or VCE gagged.

Aux lube pump running or emergency lube pump running.

Master protective or hydraul protective trouble.

Position servo trouble.

Battery ground.

Fuel skid trouble.

Exciter trouble.

Generator breaker tripped or generator field ground.

Starting means trouble.

Unused drop #21.

Lube Pressure low.

Lube header temperature high or low.

Sequences incomplete.

Temperature control abnormal.

Comp bleed valve pos trouble or IGV Pos trouble.

Combustion trouble.

Turbine air inlet trouble.

Fuel pressure high or low.

Fire.

Unused drop #31.

CO2 system inoperative or #3 bearing temperature.

Failure to fire or loss of flame or system trouble.

Hydraulic trip circuit pressure low.

Exhaust temperature trip or trouble.

Vibration excessive or system trouble.

Overspeed trip or system trouble.

Turbine or control compt temp high.

Generator differential.

Fire load tunnel area.

Unused drop #41.

Unused drop #42.

Generator bearing temperature high.

Generator overvoltage.

Generator purge CO2 low pressure.

Hydrogen temperature high.

Cooling water level low or cooling fan vibration excessive.

PT trouble.

Exhaust frame cooling air pressure low.

Computer failed.
(This means the Integrated Temperature System computer in the SPEEDTRONIC control has stalled.)

Wheelspace trouble.

Unused drop #52

Seal fluid pressure trouble.

Seal drain liq level high or casing liq level high.

Generator purge system on manual control.

Hydrogen control panel undervoltage (AC supply).

Hydrogen pressure high or low (Generator).

Hydrogen pressure low (skid).

Hydrogen purity trouble.

Unused drop #60.

Unused drop #61.

Unused drop #62.

Unused drop #63.

Communication channel trouble — see status display.
(This indicates that one or more Data Acquisition System units do not respond. The Data Acquisition System that does not respond will have blanks in its data field on the status CRT.)

Comumunication card failure — Slot 2G.
(This indicates that the communication card for the channel in use has failed.)

Other communication channel trouble.
(This indicates that the channel not in use has experienced a communication card failure or a stalled processor.)

Turbine Control

The following functions enable the operator to control a turbine:

Start. The operator may start a turbine in the following manner:

1. Select the desired unit with the unit selector switch.
2. Check to see that the status display column for that unit is not blanked. If that column is blank, communication has not been established with that unit.
3. Check to see that the control location is station or central on the status CRT.
4. Check to see that the ready light is illuminated.
5. Check all status lights and status CRT to be sure all desired conditions are present.
6. Turn the master switch to START position. The START indicating light should light.

Fast Load Start. The operator may fast-load start a turbine in the following manner:

1. Select the desired unit with the unit selector unit.
2. Check to see that the status display column for that unit is not blanked. If that column is blank, communication has not been established with that unit.
3. Check to see that the control location is station or central on the status CRT.
4. Check to see that the ready light is illuminated.
5. Check all status lights and status CRT to be sure all desired conditions are present.
6. Push the fast load start pushbutton. The start indicating light should light.

Stop. The operator may stop a turbine in the following manner:

1. Select the desired unit with the unit selector switch.
2. Check to see that th status display column for that unit is not blanked. If that column is blank, communication has not been established with that unit.
3. Check to see that the control location is station or central on the status CRT.
4. Turn the master switch to the stop position.

Base Load Select. The operator may select base load in the following manner:

1. Select the desired unit with the unit selector switch.
2. Check to see that the status display column for that unit is not blanked. If that column is blanked, communication has not been established with that unit.
3. Check to see that the control location is station or central on the status CRT.
4. Turn the load control switch to the base position. Base indicating light should light.

Peak Load Select. The operator may select peak load in th following manner:

1. Select the desired unit with the unit selector switch.
2. Check to see that the status display column for that unit is not blanked. If that column is blank, communication has not been established with that unit.
3. Check to see that the control location is station or central on the status CRT.
4. Turn the load control switch to the peak position. Peak indicating light should light.

Raise Watts. The operator may raise megawatts in the following manner:

1. Select the desired unit with the unit selector switch.
2. Check to see that the status display column for that unit is not blanked. If that column is blank, communication has not been established with that unit.
3. Check to see that the control location is station or central on the status CRT.
4. Hold the watts switch in the raise position until the megawatts on the status CRT reaches the desired level, then release the switch.

Lower Watts. The operator may lower megawatts in the following manner:

1. Select the desired unit with the unit selector switch.
2. Check to see that the status display column for that unit is not blanked. If that column is blank, communication has not been established with that unit.
3. Check to see that the control location is station or central on the status CRT.
4. Hold the watts switch in the lower position until the megawatts on the status CRT reaches the desired level, then release the switch.

Gas Monitor. The operator may select gas monitor in the following manner:

1. Select the desired unit with the unit selector switch.
2. Check to see that the status display column for that unit is not blanked. If that column is blank, communication has not been established with that unit.
3. Check to see that the control location is station or central on the status CRT.
4. Turn the gas monitor switch to the monitor position. The green monitor indicating light should light. The red override indicating light should be off.

Gas Override. The operator may select gas override in the following manner:

1. Select the desired unit with the unit selector switch.
2. Check to see that the status display column for that unit is not blanked. If that column is blank, communication has not been established with that unit.
3. Check to see that the control location is station or central on the station CRT.
4. Turn the gas monitor switch to the override position. The red override indicating light should light. The green monitor light should be off.
5. The override function is inhibited if the turbine is at zero speed.

Generator Control

The following functions enable the operator to control the generator:

Raise VARS. The operator may raise megavars in the following manner:

1. Select the desired unit with the unit selected switch.
2. Check to see that the status display column for that unit is not blanked. If that column is blank, communication has not been established with that unit.
3. Check to see that the control location is station or central on the status CRT.
4. Hold the VARS switch in the raise position until the megavars on the status CRT reach the desired level, then release the switch.

Lower VARS. The operator may lower megavars in the following manner:

1. Select the desired unit with the unit selector switch.
2. Check to see that the status display column for that unit is not blanked. If that column is blank, communication has not been established with that unit.
3. Check to see that the control location is station or central on the status CRT.
4. Hold the vars switch in the lower position until the megavars on the status CRT reaches the desired level then release the switch.

52G Breaker Control

52G Trip. The operator may trip the 52G breaker in the following manner:

1. Select the desired unit with the unit selector switch.
2. Check to see that the status display column for that unit has not been established with the unit.
3. Check to see that the control location is station or central on the status CRT.
4. Push the 52G trip pushbutton. The green indicating light labeled "OPEN" should light up. The red indicating light labeled "CLOSED" should extinguish.

Teleprinter Functions

Teleprinter Set Up. The teleprinter should be set up as follows:

1. 300 Baud
2. Full Duplex
3. Upper case letters

Time Set Procedure. To perform the Time Set procedure, proceed as follows:

1. Hold down "CONTROL" key and type a "Z". This alerts the processor that an operator input is coming.

2. Type in "C" and then the "RETURN" key. The current time will print out. If the operator elects not to change the time, a "RETURN" will exit the time change routine. If the operator desires to change the time, follow step 3. This tells the processor that the time change procedure is desired.

3. Type in the time in exactly the following format. Then type a "RETURN".

```
80    03 05      15    03 22

Y     M D     SS H     M S
E     O A     PP O     I E
A     N Y     AAU      N C
R     T       CC R     U O
      H       EE       T N
                       E D
```

The above example is equivalent to:

March 5, 1980 3:03 P.M. 22 Seconds

If an illegal entry is attempted, the current time is typed out and the operator is given a chance to re-enter the new time. If no entry is made within forty five seconds, the time set procedure is terminated.

When a legal time entry is made, the Station Operator's Console updates the time of each unit once every minute. Thus, within one minute, all eight units will be updated with the new time.

Turbine Log. The operator may request a turbine log with the following sequence of keystrokes.

1. Hold down "CONTROL" key and type a "Z". This alerts the processor that an operator input is coming.

2. Type in "T" "space", two digit turbine number, and then depress the "RETURN" key. EXAMPLE: "T", "SPACE", "0", "1", "RETURN" will yield the turbine log for unit 1.

Generator Log. The operator may request a generator log with the following sequence of keystrokes.

1. Hold down "CONTROL" key and type a "Z". This alerts the processor that an operator input is coming.

2. Type in "G" , "space", two digit number, and then the "RETURN" key.

Vibration Log. The operator may request a vibration log with the following sequence of keystrokes.

1. Hold down "CONTROL" key and type a "Z". This alerts the processor that n operator input is coming.
2. Type in "V", "space", two digit turbine number, and then the "RETURN" key.

DATA ACQUISITION SYSTEM MAINTENANCE INSTRUCTIONS

DS3820PS3 Power Supply

The DS3820PS3 power supply provides all power for the Data Acquisition System Small Electronics Module, DS3820LIMA isolation module, and Analog Input Rack. It accepts a single phase AC voltage (80-132VRMS, 47-63Hz) or a DC input voltage (70-140VDC) at its input. Output voltages are +5V, +28V, +15V and -15V.

CB1 Circuit Breaker. A circuit breaker is supplied for disconnect purposes as well as short circuit protection. The circuit breaker is rated at 15 amperes.

Status Light. A status light mounted on the DS3800NPSC board indicates that all output voltages are normal. This light is on when power is applied and all voltages are correct. Trouble is indicated when the light is off.

Test Points and Adjustments. Test points and adjustments are available on the DS3800NPSC board. In order to get at the test points and adjustments, the DS3800NPSC board must be swung out. First undo the two plastic retainers which hold the board in place. The board will then swing out to make the test points and adjustments accessible (See Figures 27 and 28).

CAUTION

ADJUST ONLY THE POTENTIOMETERS INDICATED. OTHER POTENTIOMETERS MUST NOT BE SET IN THE FIELD. IF THE SETTING OF R2, R3, OR R7 IS INADVERTENTLY DISTURBED, THE POWER SUPPLY SHOULD BE TURNED OFF IMMEDIATELY AND RETURNED TO THE FACTORY FOR READJUSTMENT.

Preliminary Checks. To perform the preliminary checks, proceed as follows:

1. Unplug cards in Data Acquisition System Small Electronics Module and Analog Input Rack and all cables except the power supply cable to the Data Acquisition System Small Electronics Module.

2. Check for proper input voltage per the following Power Supply Setup instructions.

3. Apply power and check output voltages for "ballpark" values to prevent card damage. See Power Supply Setup instructions for test points.

Power Supply Check. To check out the power supply, proceed as follows:

1. De-energize the power supply and plug cards into the Data Acquisition System Small Electronics Module and Analog Input Rack. Re-energize the power supply.

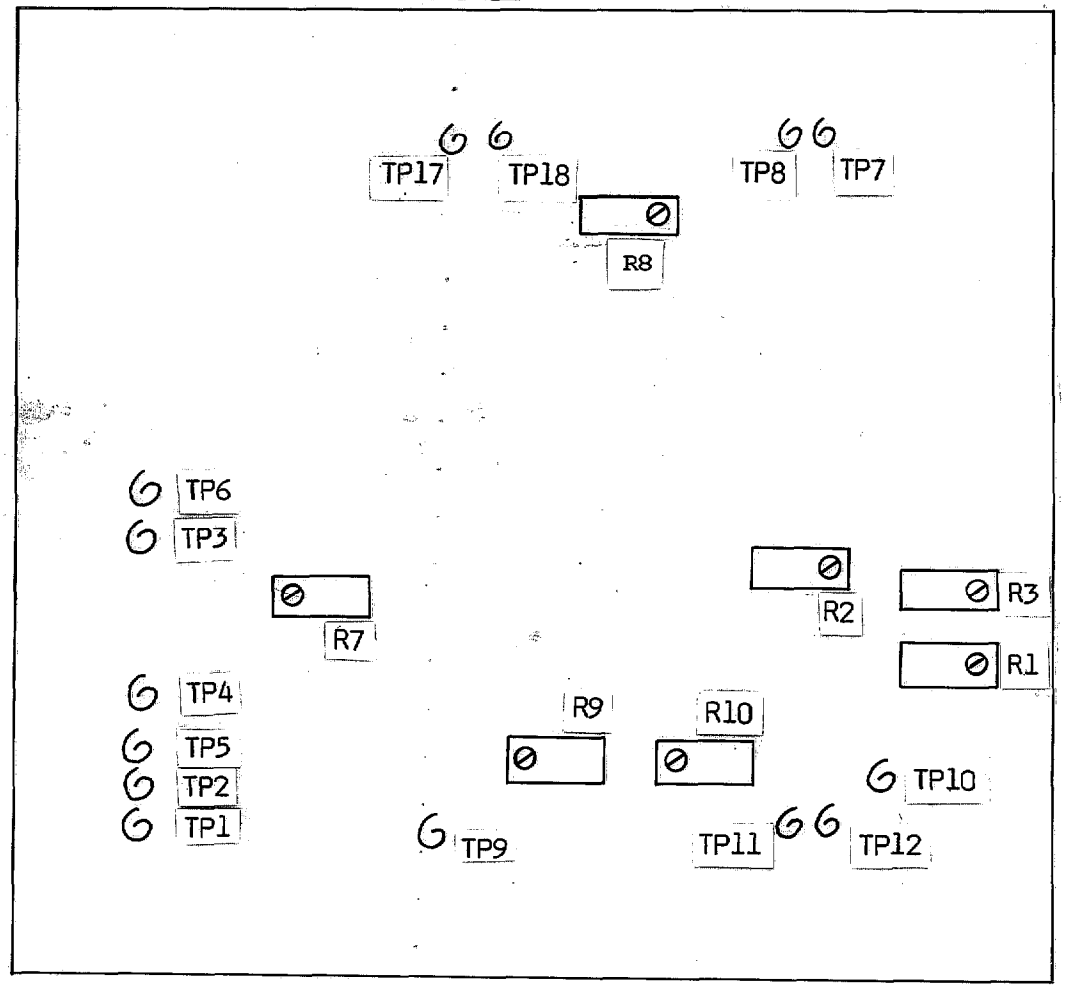
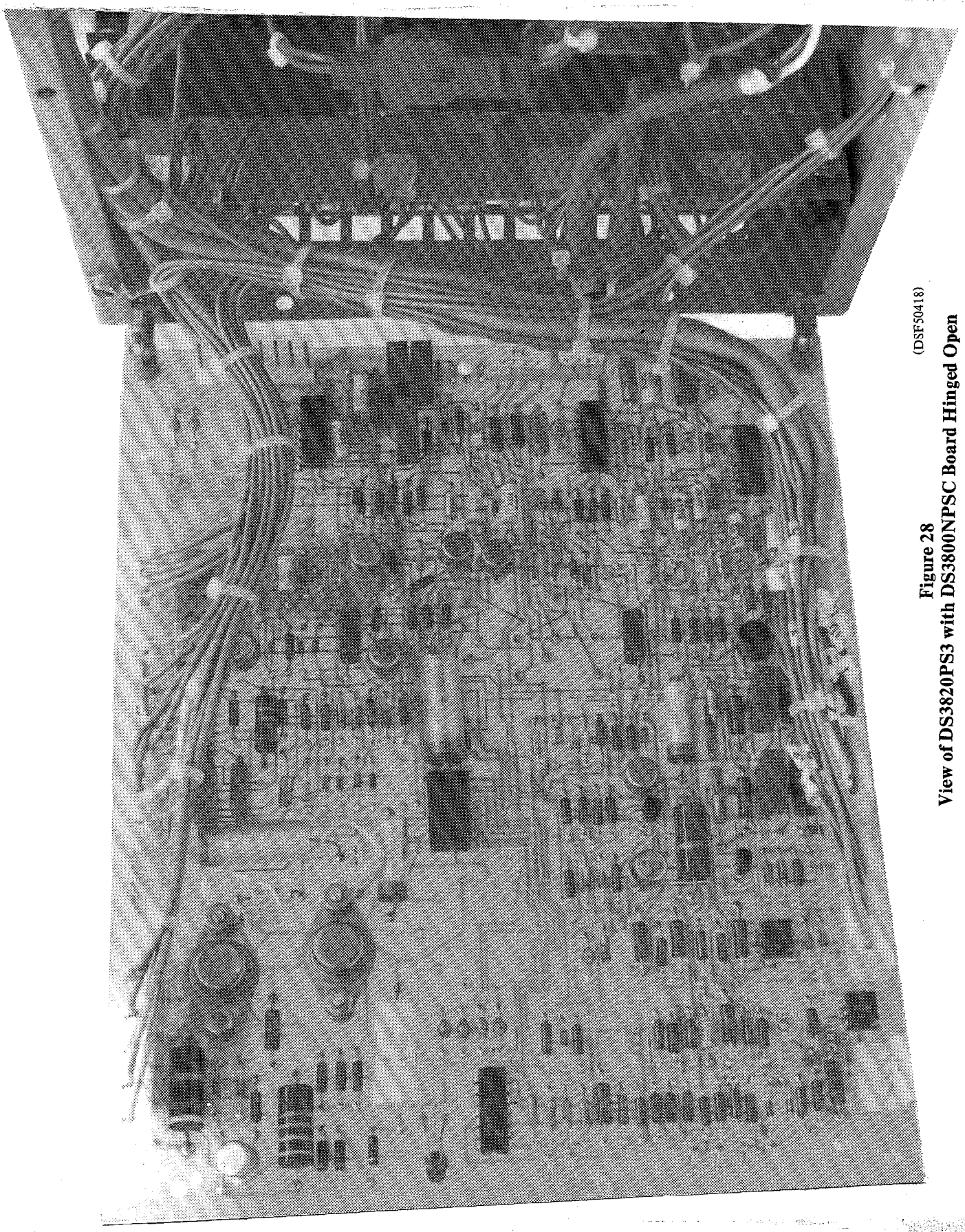


Figure 27
Potentiometer and Test Point Locations on DS3820NPSC



(DSF50418)

Figure 28
View of DS3820PS3 with DS3800NPSC Board Hinged Open

NOTE

It is important that the following adjustments be made with all the cards plugged in. This is necessary because voltage feedback is taken from the Data Acquisition System Small Electronics Module and will change when the cards are unplugged.

2. Check for the following power supply voltages under load:

SUPPLY	TEST POINT	VALUE	ADJUST
P5	TP2 to TP3	5.00+.05VDC	R1
P15	TP4 to TP3	15.00+.05VDC	R9
N15	TP5 to TP3	-15.00+.05VDC	R10
P28	TP1 to TP3	28.00+.70VDC	R8

3. Check for the following power supply currents:

SUPPLY	TEST POINT	CALIBRATION	MAXIMUM
P5	TP15 to TP16	.005 Volts/Amp	.150V
P15	TP9 to TP10	.100 Volts/Amp	.300V
N15	P11 to TP12	.100 Volts/Amp	.300V
P28	TP7 to TP8	.470 Volts/Amp	.500V

Fault Detection. There are multiple fault detection circuits (on the DS3800NPSC) to prevent internal/external load faults from damaging the PS3 during normal operation. The ower supply will shutdown if there is an overvoltage on any one output greater than 20% of the nominal value for all voltages except the P5. Excess is less than 40% of nominal value. It will also shut down if there is a short on any output. The power supply cannot be reset until the load short(s) is removed.

Visual Diagnostic Aids

DS3800HMPA. A description of the card-mounted LEDs is as follows:

"I'm OK" LED. The HMPA has a yellow LED which is lit when parity between the HMPA and other boards is good. This is known as the "I'm OK" indicator. This LED will go out if the processor stalls.

Teleprinter Receive LED. There are eight red LEDs on the front of the HMPA. The bottom one will light when it receives an input from the teleprinter.

Teletype Transmit LED. The second LED up from the bottom will light when the HMPA transmits to the teleprinter.

Integrated Temperature System-to-Data Acquisition System Receive LED. The third LED up from the bottom will light when the HMPA receives characters from the Integrated Temperature System on the serial Integrated Temperature System-to-Data Acquisition System Link.

DS3800HXPA. There is one card-mounted LED on this card:

Parity LED. The HXPA has a yellow LED which is lit when parity between the HXPA and the HMPA boards is good.

DS3800HMHA. The following three LEDs are mounted on this card:

"I'm OK". This is a yellow LED which, when lit, indicates that the HMHA processor is not stalled. This light will go out if the processor stalls.

Channel A Traffic Indicator. The top red LED, when lit indicates activity on communication channel A.

Channel B Traffic Indicator. The bottom red LED, when lit, indicates activity on communication channel B.

DS3800HRRB. There are two LEDs mounted on this card:

A yellow LED lights up when parity between the HRRB and the HMPA is good.

A red output LED is provided for each of eight circuits on the HRRB. The red light is lit when the corresponding relay is energized.

Data Acquisition System Display (Option)

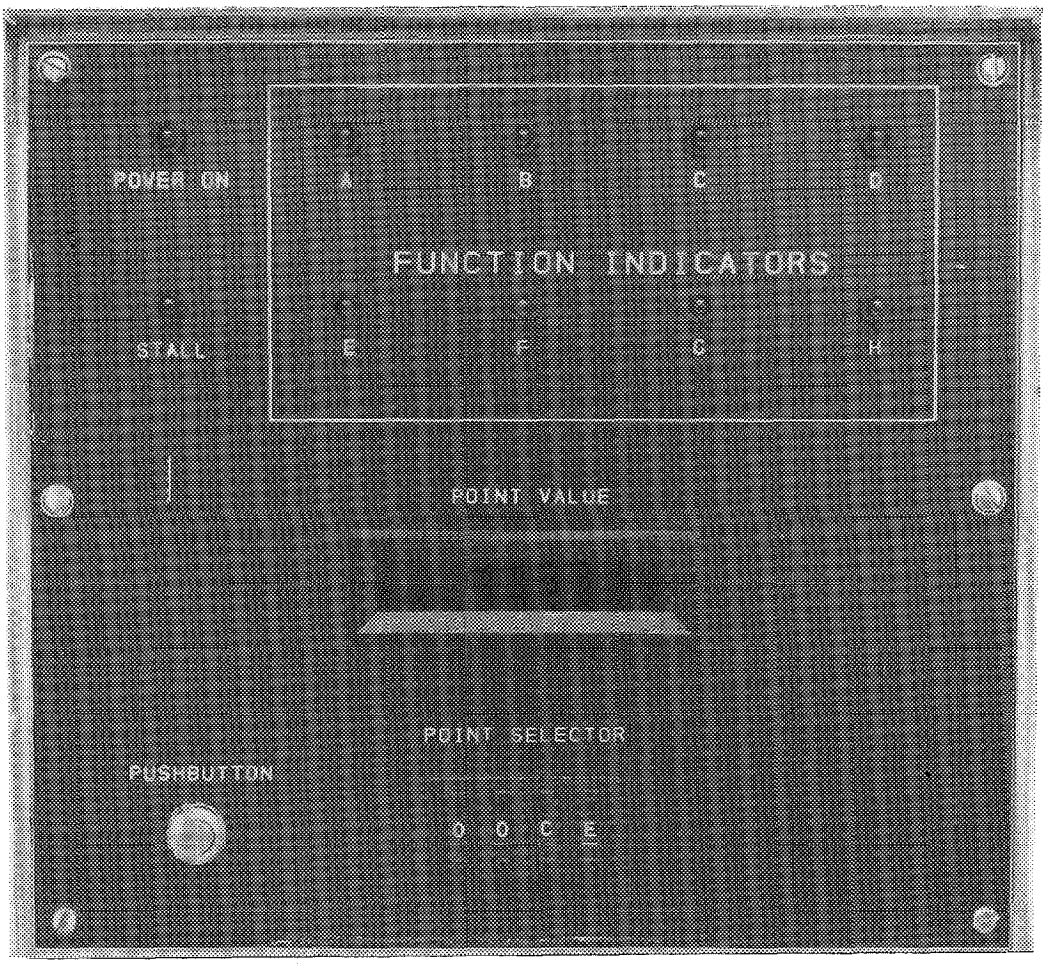
Data within the Data Acquisition System may be viewed with the Data Acquisition System display. Each value that may be displayed has an associated point number. To display a value, simply dial in the point number assigned to that value with the thumbwheels. Point numbers are listed in frontsheets of the elementary diagram (See Figure 29).

Analog Values. Analog values are displayed already scaled in engineering units.

Digital Values. Digital values are displayed one byte at a time. Digital signals are stored in the Data Acquisition System with eight digital signals (bits) making one byte. What is displayed is the hexadecimal value of the byte. For example, let us say the following byte of digital data is stored in the Data Acquisition System: 11110011. The value F3 would be displayed by the data acquisition system display.

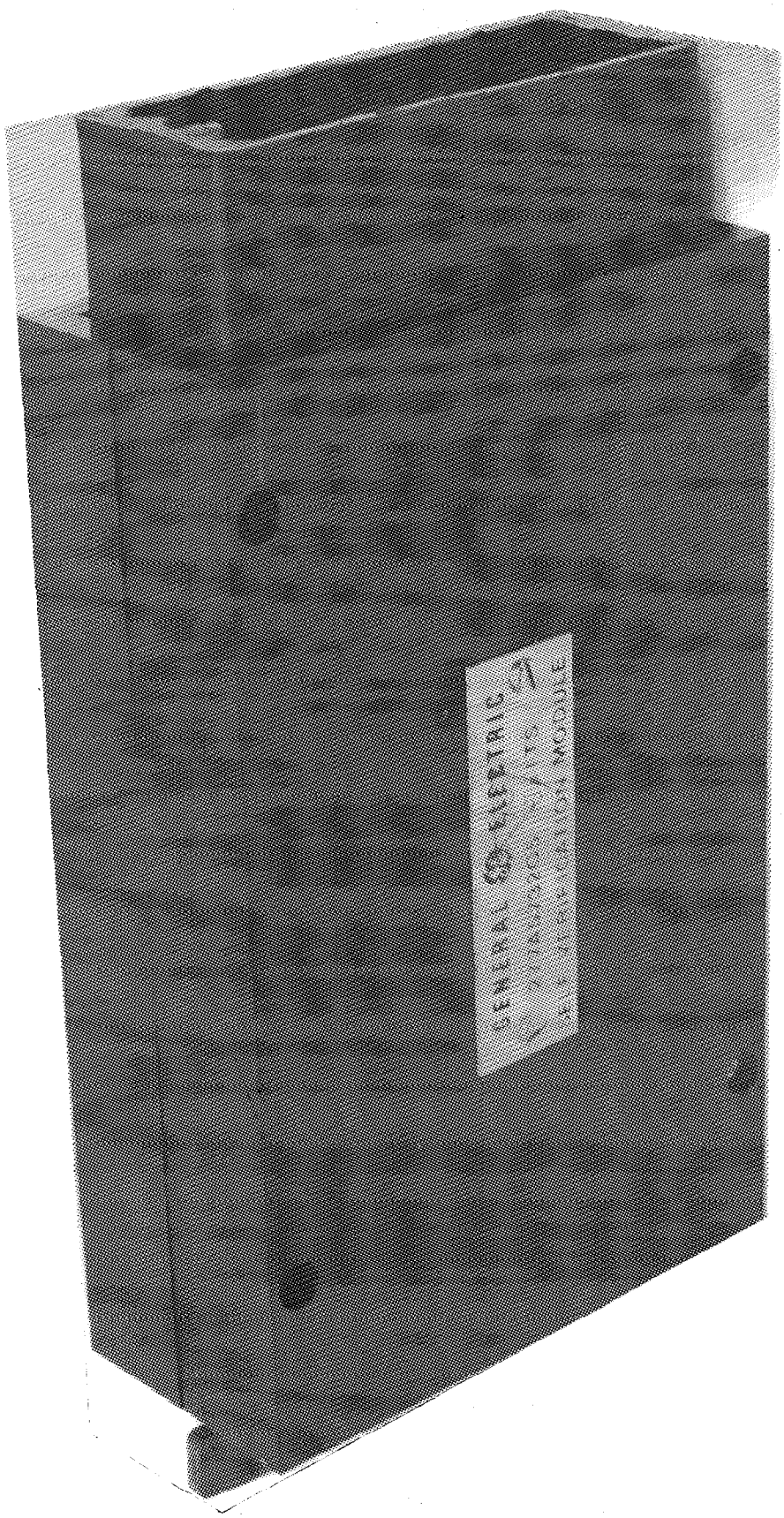
Self Verification Modules

The Self Verification Module (See Figures 30 and 31) is a tool that can be used to aid in locating defective hardware within the Data Acquisition System. There are three Self Verification Modules available.



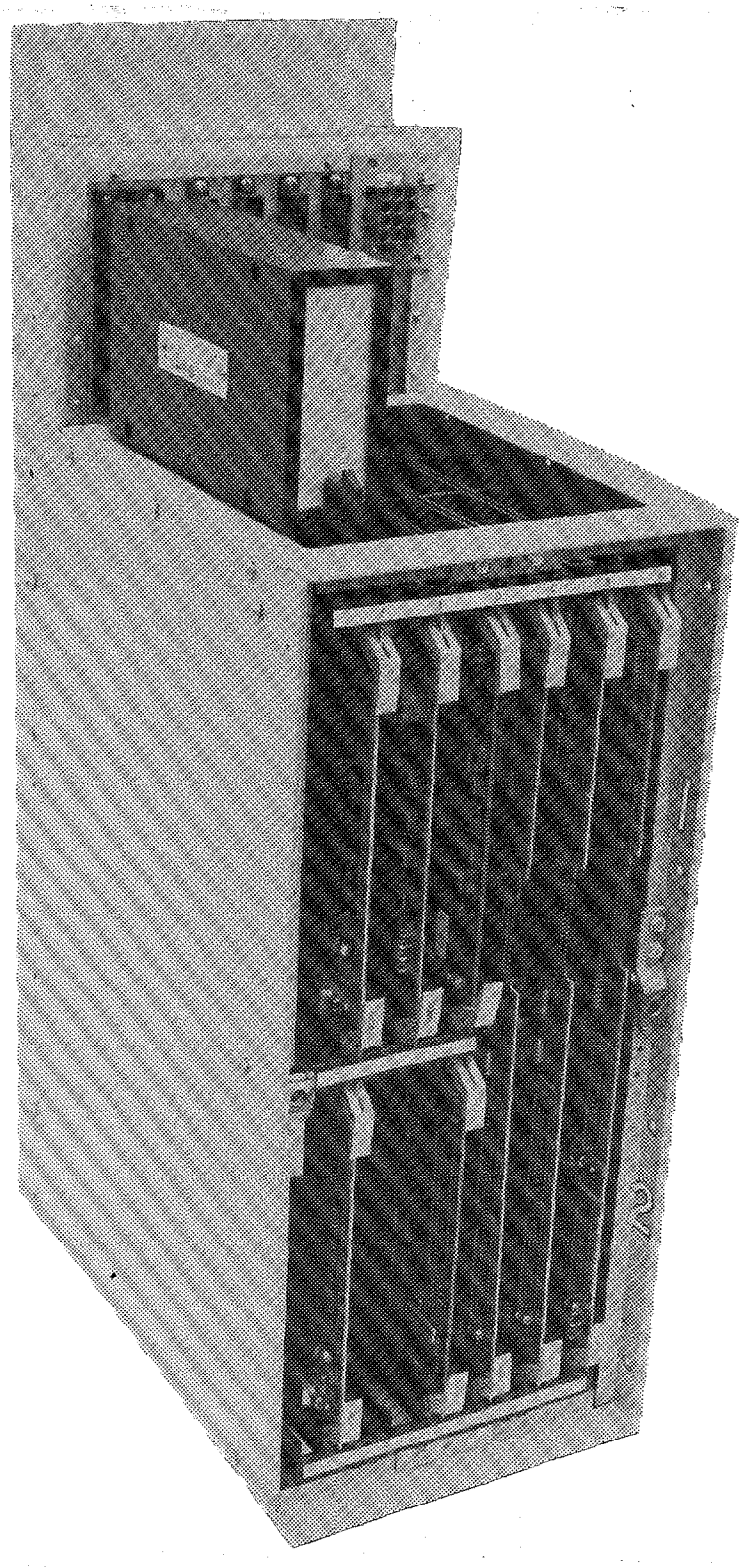
(DSF54062)

Figure 29
DAS Display



(DSF54055)

Figure 30
Self Verification Module



(DSF54057)

Figure 31
Self Verification Module Plugged into SEM

Self Verification Module 277A5732G6. This Self Verification Module performs the following tests:

1. CPU check of the HMPA.
2. PROM checksum on HMPA and both HXPA's. (Refer to elementary diagram frontsheets to determine which PROM addresses are assigned to which PROM.)
3. RAM test on HMPA and both HXPA's.
4. Timer test on HMPA and both HXPA's.
5. Stall test.
6. Parallel I/O loopback test on HMPA and both HXPA's.
7. Interrupt test.
8. Analog loopback - This test requires an NDAC board to execute. If trouble is suspectred in the A/D board (NADC), an NDAC board could be borrowed from an Integrated Temperature System module not currently in use.
9. HALT test.
10. Integrated Temperature System Display
11. Analog Input Rack.

You can set up and use the Self Verification Module by following these steps:

1. Turn all Data Acquisition System power off. Verify that all cables are connected correctly per the assembly prints.

2. Load the hardware verification program SVM005A by plugging the diagnostic module containing 68A9172 PROMS programmed with SVM005A into the Small Electronics Module maintenance test socket JL (68A9172P0085, 68A9172P0086). When the Self Verification Module is fully inserted, rotate the lock lever 1/4 turn clockwise to the vertical position to lock the module in place.

CAUTION

THE SELF VERIFICATION MODULE/OR THE SIGNAL ELECTRONICS MODULE CAN BE DAMAGED IF THE PINS IN THE SELF VERIFICATION MODULE PLUG ARE BENT. BEFORE INSERTING THE SELF VERIFICATION MODULE, CAREFULLY INSPECT THE PINS TO ASSURE THAT THEY ARE STRAIGHT. ALL PINS IN A HORIZONTAL ROW SHOULD ALIGN ALONG THE HORIZONTAL ROW WITHIN ± 1/64 INCH. PINS WHICH ARE OUT OF ALIGNMENT SHOULD BE REPAIRED BEFORE THE SELF VERIFICATION MODULE IS USED. WHEN INSERTING THE SELF VERIFICATION MODULE, USE MINIMUM FORCE! IF THE SELF VERIFICATION MODULE SEEMS TO "HANG" ABOUT 1/8 INCH FROM FULL INSERTION, DO NOT FORCE SINCE A PIN IS PROBABLY NOT ENTERING A HOLE PROPERLY.

3. Disconnect the cables connected to the HMPA (Data Acquisition System Small Electronics Module 1A/2A) connectors JA and JB. Connect JA to JB using test cable DS3815RG1CL020X00.

4. Disconnect the cables connected to the HXPA1 (Data Acquisition System Small Electronics Module 1C/2C) connectors JA and JB. Connect JA to JB using test cable DS3815RCG1CL020X00.

5. Disconnect the cables connected to the HXPA2 (Data Acquisition System Small Electronics Module 1B/2B) connectors JA and JB. Connect JA to JB using test cable DS3815RCG1CL020X00.

6. Disconnect the cables to the Data Acquisition System display from Small Electronics Module connector JD and JK. Temporarily connect the ITS display jack JB to the DAS small electronics module jack JK with a 34 conductor ribbon cable 3S3815RCG1BL—-.

7. Temporarily install a DS3800NDAC Digital-to-analog converter card in Data Acquisition System Small Electronics Module slot 2E. Temporarily disconnect the cable connected to Small Electronics Module jack JJ and install test cable 277A6581G1 from Small Electronics Module jack JJ to Small Electronics Module jack JC. (NOTE: This is required only to test the DS3800NADC analog converter card. If this test is needed, an NDAC board could be borrowed from an Integrated Temperature System not currently in use.)

8. Verify the position of all Berg jumpers on the DS3800HRMA (Data Acquisition System Small Electronics Module slot 2D) and DS3800HMHA (Data Acquisition System Small Electronics Module slot 1D) cards per the systems elementary.

9. Turn power back on.

10. Make the following settings on the Integrated Temperature System display (see Figure 32):

    Set LIST thumbwheel switch = 0
    Set HUNDREDS thumbwheel switch = 1
    Set TENS thumbwheel switch = 4
    Set ONES thumbwheel switch = 5
    Set KEYSWITCH = NORMAL

These settings will run the following tests in this order:

TEST	PARAMETERS
1 (CPU)	None
4 (PROM)	20K
5 (RAM)	5K (HMPA and 2-HXPA)
6 (Timer)	HMPA and 2-HXPA (Due to wiring on the Data Acquisition System Small Electronics Module backplane, the timer should fail twice in 6-2 (1st HXPA) and once in 6-1 (HMPA timer 2)).
7 (Stall)	I'm OK (amber) light on HMPA should go out one second after stall test start
8 (Parallel I/O Loopback)	HMPA and 2-HXPA
9 (Analog Loopback)	Does not run "HALT" test
C (Analog loopback)	None

11. Initiate "ALLTEST" by cycling the HMPA RESET switch to initialize the processor and start execution of the diagnostic program. The "INCR" will have to be pushed in test 6 (Timer test) for each of three failures. Test C will fail if there is no DS3800NDAC card in Data Acquisition System Small Electronics Module slot 2E. If desired, each test (1, 4, 5, 6, 7, 8, 9 and C) can be run individually by selecting that test with the "LIST" thumbwheel switch instead of running "ALLTEST". Any failures that occur other than those noted above in test 6 and test C should be investigated and repaired before continuing. Refer to 278A1549 for details on operating the diagnostic program and interpreting failure information.

12. Turn power off. Disconnect the test cables installed above in steps 3, 4, 5 and 7. Reconnect the cables removed in those steps. Remove the DS3800NDAC card from Data Acquisition System Small Electronics Module slot 2E. Leave the Integrated Temperature System display connected. Reapply power.

13. You can now run the remaining diagnostic routines (Not part of "ALLTEST") one at a time by setting the LIST thumbwheel switch as specified and cycle the ENTER pushbutton to start the test.

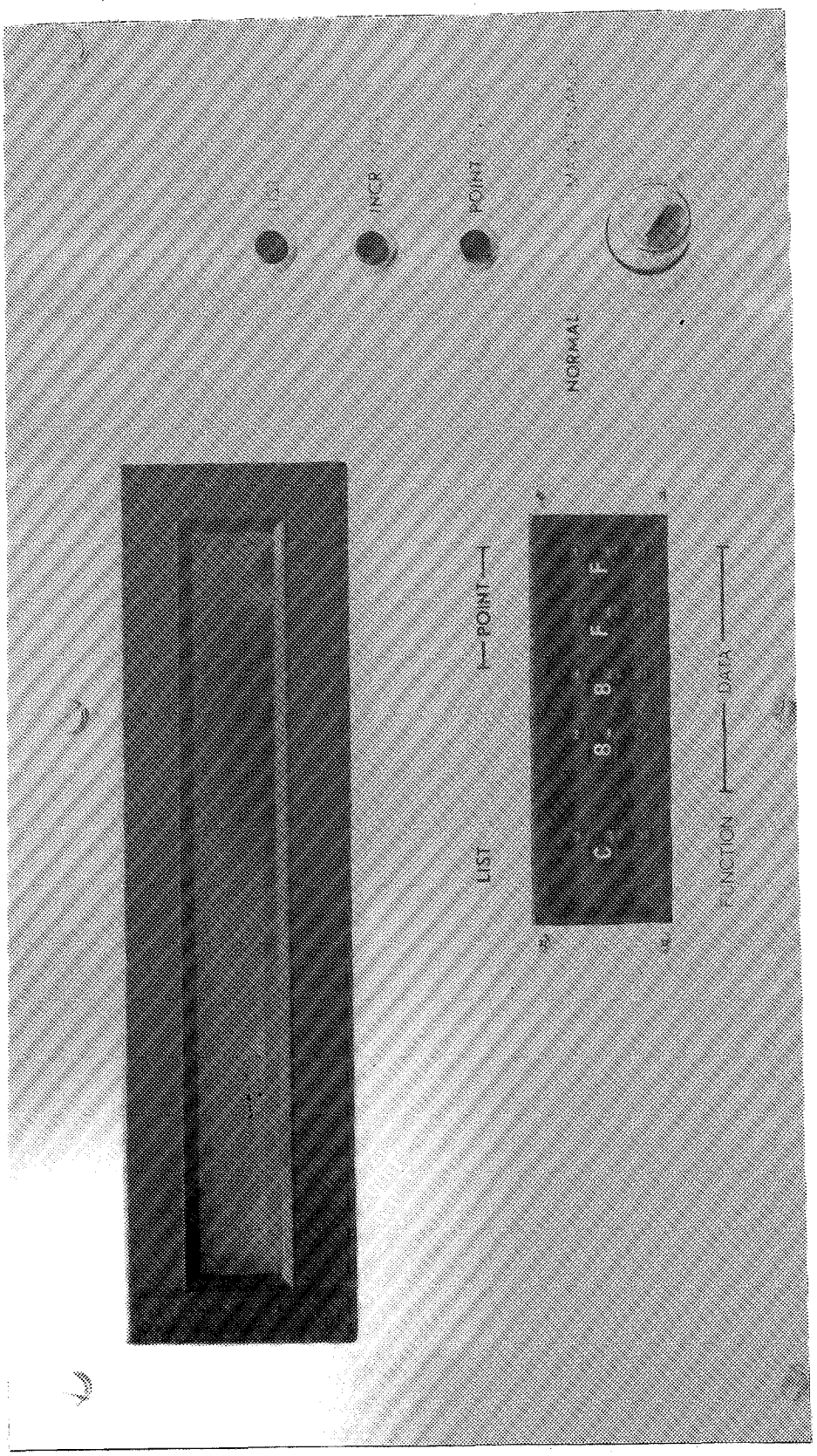


Figure 32
Integrated Temperature System (ITS) Display

14. To run the **HALT test**, set the LIST thumbwheel switch = 9 (Interrupt) and Keyswitch = maintenance. Start the test and wait 5 seconds after the message HALT TEST-RESTART appears on the display. Verify the HMPA lights = 93H (1001 0011) and not EEH (1110 1110). Cycle the HMPA RESET switch to resume testing.

15. To run the **DISPLAY test**, set the LIST thumbwheel switch = 2 and Keyswitch = NORMAL. Cycle ENTER pushbutton to start the display test. (Cycle the INCR pushbutton (Hold for 1 second) to cause program to switch to the SINGLE STEP mode where one new character will be displayed each time the INCR pushbutton is cycled, if desired. Cycle the ENTER pushbutton to return to AUTO-INCREMENT mode.) Observe the display long enough to assure that each character occurs properly in the following sequence:

CHARACTER	DESCRIPTION
A thru Z	Alphabet
[	Left square bracket
	Tilde ("squiggle")
]	Right square bracket
{	Left braces
}	Right braces
!	Exclamation point
"	Quote (double)
#	Number sign
$	Dollar sign
%	Percent sign
&	Ampersand
'	Apostrophe
(	Left parenthesis
)	Right parenthesis
*	Asterisk
+	Plus sign
,	Comma
-	Hyphen (Minus sign)
.	Period
/	Slash
0 thru 9	Numbers
:	Colon
;	Semicolon
<	Left angle bracket ("Less than")
>	Right angle bracket ("Greater than")
?	Question mark
@	"At" sign

After the display character set has been verified, set Keyswitch = maintenance and cycle FUNCTION pushbutton (hold for at least 1 second) to switch the display test to the echo mode. In this mode the display will show (flashing):

L XHTO Maintenance - where:
L = LIST thumbwheel switch position
X = THOUSANDS thumbwheel switch position
H = HUNDREDS THUMBWHEEL switch position
T = TENS thumbwheel switch position
0 = ONES thumbwheel switch position

The lights adjacent to the pushbuttons will light when the corresponding pushbutton is pushed. Check all positions of all thumbwheel switches and verify that the display and the thumbwheel switch setting match. Verify that each pushbutton, singly and in combination with the others, lights the appropriate light. Set Keyswitch = NORMAL to return to the DISPLAY CHARACTER mode.

16. To run the **DAS analog input test**, set LIST thumbwheel switch = A and Keyswitch = NORMAL. Cycle the ENTER pushbutton to start the analog input test. The display will read SET POINT ADDR after the address line tests are finished. Set ONES thumbwheel switch and TENS thumbwheel switch for the channel and point to be displayed. Cycle the ENTER pushbutton to read in the address and display the point data. (If the Keyswitch is placed in MAINTENANCE and the ENTER pushbutton is cycled once, approximately once per second a new point address will be read from the thumbwheel switch and another set of readings will be taken from that point. If the thumbwheel switch is changed, the next set of readings will be taken using the new thumbwheel switch setting. If this feature is used, it is not necessary to cycle the ENTER pushbutton after setting the thumbwheel switches to the point address to take a set of readings.) The display will show the first hexadecimal count, the fifth hexadecimal count and the fifth decimal count. The first count must be within 3 counts of the fifth count. The decimal count must be ±5 counts of the value given in the table below. Repeat the procedure for all points per the table. Refer to the systems elementary to determine whether or not a point is supplied on the job.
Using the systems elementary, determine which combination of input cards are used for each channel and point. The table lists the calibration input and required output (on the display) for the various combinations of input cards applied on a Data Acquisition System.

POINT CARD/ AMP CARD	TEST INPUT	DISPLAY OUTPUT
AMIN*/QEAA3	10.00/14.80 ohms	+0000/+2040
AMIP/-None-	+/-9.00 VDC	+/-1843
	0.00 VDC	+0000
QMID2/QDAA3	+/-4.50 MADC	+/-1843
	Open Circuit	+0000
QMID2/QDAA5	+/-0.900 MADC	+/-1843
	+/-Open Circuit	+0000
QMID3/QDAA2	+/-18.00 MADC	+/-1843
	Open Circuit	+0000

* Do not short the input of the AMIN card. Be sure both return paths are tied together.

Fill out and refer to the table below when running this test.

CH. PT	POINT CARD	AMP CARD	INPUT	DISPLAY OUTPUT
0, 0	____	____	____	_____
0, 1	____	____	____	_____
0, 2	____	____	____	_____
0, 3	____	____	____	_____
0, 4	____	____	____	_____
0, 5	____	____	____	_____
0, 6	____	____	____	_____
0, 7	____	____	____	_____
0, 8	____	____	____	_____
0, 9	____	____	____	_____
0, A	____	____	____	_____
0, B	____	____	____	_____
0, C	____	____	____	_____
0, D	____	____	____	_____

0, E				
0, F				
1, 0				
1, 1				
1, 2				
1, 3				
1, 4				
1, 5				
1, 6				
1, 7				
1, 8				
1, 9				
1, A				
1, B				
1, C				
1, D				
1, E				
1, F				
2, 0				
2, 1				
2, 2				
2, 3				
2, 4				
2, 5				
2, 6				
2, 7				
2, 8				
2, 9				
2, A				
2, B				
2, C				
2, D				
2, E				
2, F				
3, 0				
3, 1				
3, 2				
3, 3				
3, 4				
3, 5				
3, 6				
3, 7				
3, 8				
3, 9				
3, A				
3, B				
3, C				
3, D				
3, E				
3, F				

Self Verification Module 277A5732G3. This Self Verification Module performs the following tests:

1. Power-up verification
   A) CPU test
   B) HMPA RAM test
   C) PROM checksum test
2. HRMA RAM test
3. HCMA dual-ported RAM test
4. HMHA dual-ported RAM test
5. Digital input scan test
6. 200-Level output test (HRRB)
7. 200-Level input test (HSCC)
8. Integrated Temperature System display test
9. Data Acquisition System display decimal and sign test
10. Data Acquisition System display memory after blanking
11. Data Acquisition System hex digit display test
12. Data Acquisition System LED rotation test
13. Data Acquisition System Thumbwheel switch test
14. Memory display test

Refer to drawing 278A4901 for complete operating instructions for SVM 277A5732G3.

Self Verification Module 277A5732G7. This Self Verification Module performs the following tests:

1. USART chip test on HMPA and HXPA #1 (Data Acquisition System Small Electronics Module slot 1C/2C)
2. Serial printer/keyboard test
3. Module to module loopback

Follow these steps to set up the system to run the above tests:

1. Turn power off. Disconnect the cables to the Data Acquisition System display from the Small Electronic Module connectors JD and JK. Temporarily connect the Integrated Temperature System display jack JB to the Data Acquisition System Small Electronic Module jack JK with a 34-conductor ribbon cable DS3815RCG1BL___.

2. Load the hardware verification program XSVM006A by plugging in the diagnostic module 277A5732G3 into the Small Electronics Module maintenance test socket JL.

3. Turn power back on. Verify that the power-up sequence occurs. The display should read G7-SEL TEST.

4. Refer to the operating instructions 278A4908 for XSVM006A and the instructions in the following paragraphs:

TEST	PARAMETERS
USART Chip Test	HMPA and HXPA #1
Serial Printer/Keyboard Test	HMPA and Teleprinter
Module to Module Test	HXPA and Cable

To run the **USART chip test**:

1. Disconnect the cable connected to the HMPA (Data Acquisition System Small Electronics Module 1A/2A) serial jack JC and install test plug 277A6580G1. Disconnect the cable connected to HXPA #1 (Data Acquisition System Small Electronics Module 1C/2C) serial jack JD and install test plug 277A6580G2.

2. Set LIST thumbwheel switch = 1 and push the ENTER pushbutton to enter the test. The display should read "1 - SEL card". Set the ONES thumbwheel switch = 0 and push the ENTER pushbutton again. The display should read HMPA. Depress the ENTER pushbutton again to start the test. If any part of the test fails, replace the HMPA card and repeat the test.

3. Set the LIST thumbwheel switch = 1 - and depress the ENTER pushbutton to enter the test. The display should read "1 - SEL card". Set the ONES thumbwheel switch = 1 and depress the ENTER pushbutton again. The display should read HXPA #1. Depress the ENTER pushbutton again to start the test. If any part of the test fails, replace the HXPA card and repeat the test.

4. Remove the test plugs and re-connect the cables.

To run the **serial printer/keyboard test**:

1. Verify that the teleprinter is connected to the HMPA card (Data Acquisition System Small Electronics Module 1A/2A). Set the teleprinter for 300 baud (High Speed on Texas Instruments Silent-700) and on-line. Turn on teleprinter power.

2. Set LIST thumbwheel switch = 3 and depress the ENTER pushbutton. The display should read "#3 SEL baud". Set the TENS thumbwheel switch = 3 and depress the ENTER pushbutton again. The display should read "Baud = 300". Depress ENTER pushbutton again and the display should read "#3SEL card". Set the ONES thumbwheel switch = 0 and depress ENTER pushbutton. The display should read "HMPA". Set the keyswitch = NORMAL and depress ENTER pushbutton to start the test. The teleprinter should begin printing a test pattern of 80-character lines. Continue this mode long enough to verify that the typed characters are correct and the line feed and carriage return functions are working.

3. Set the Keyswitch = MAINT. The test pattern should stop. Switch the teleprinter to LOCAL mode, advance the paper with the RETURN key, and return the teleprinter to LOCAL mode. This sequence may need to be repeated during this mode of the test as the printhead reaches the right margin, since the echo mode will not produce a carriage return or line feed output. Test each key on the keyboard by striking it and verifying that the printed character matches the key. The function keys (tab, del, return, backspace, etc) should echo the two character sequence 'X' ("up-arrow, letter"), where X = letters indicated in the following list:

KEY	LETTER
Backspace	H
Tab	I
Linefeed	J
Return	M
Rubout	Space (rubout = "Delete" Key)

4. Hold the CONTROL key down and strike the following keys. Verify for each that the printed echo is 'KEY' ("Up-Arrow, Key), where key is the printing key struck:

@, Keys a thru Z (Alphabet, [, ,], _ ('at' sign, alphabet, left bracket, backslash, right bracket, up-arrow, and underscore).

5. When the printer and the keyboard have been verified as fully functional, set the keyswitch to NORMAL. Returning to the test pattern output, and set LIST thumbwheel switch to 4 to exit the test.

To run the **module to module loopback**:

1. Verify the cable from the Data Acquisition System Small Electronics Module to the Integrated Temperature System is connected to the Data Acquisition System HXPA #1 (Data Acquisition System Small Electronics Module 1C/2C) jack JD. Disconnect that cable from the Integrated Temperature System HMPA jack JC connection, and install test plug 277A6580G2 in the end of the cable which was connected to the Integrated Temperature System HMPA card to produce a loopback connection for the module to module test.

2. Set LIST thumbwheel switch = 4 (module to module test), TENS thumbwheel switch = 7 (9600 baud), and ONES thumbwheel switch = 1 (HXPA #1). Push the ENTER pushbutton five times (verifying the settings as they are displayed) to start the test. Allow the test to run for about 5 minutes. If the TOTAL ERROR message still shows zero errors, the test has been successfully completed. If errors are counted, or if a USART timeout occurs, the test has failed. If a USART timeout occurs, and the HXPA card passes the USART chip test ff1, the cable is either defective or not properly terminated. If the error counters accumulate errors of any kind, the cable is probably picking up electrical noise, or has a loose connection. This test can be run on the same cable using the Integrated Temperature System Small Electronics Module (plug cable into HMPA card and terminate Data Acquisition System end with 277A6580G1. Set ONES thumbwheel switch = 0) if desired to verify cable problems versus Signal Electronic Module problems.

3. Turn power off. Remove the diagnostic module. Disconnect the Integrated Temperature System display from the Data Acquisition System Small Electronics Module, and reconnect the Data Acquisition System display to Data Acquisition System Small Electronics Module, and reconnect the Data Acquisition System Small Electronics Module cables that were disconnected, including the USART cable to the Integrated Temperature System Small Electronics Module.

4. Turn power back on. Verify that the yellow status LEDs illuminate on all the cards in the Data Acquisition System Small Electronics Module which have them (HMPA, HXPA's, HMHA, and HRRB's).

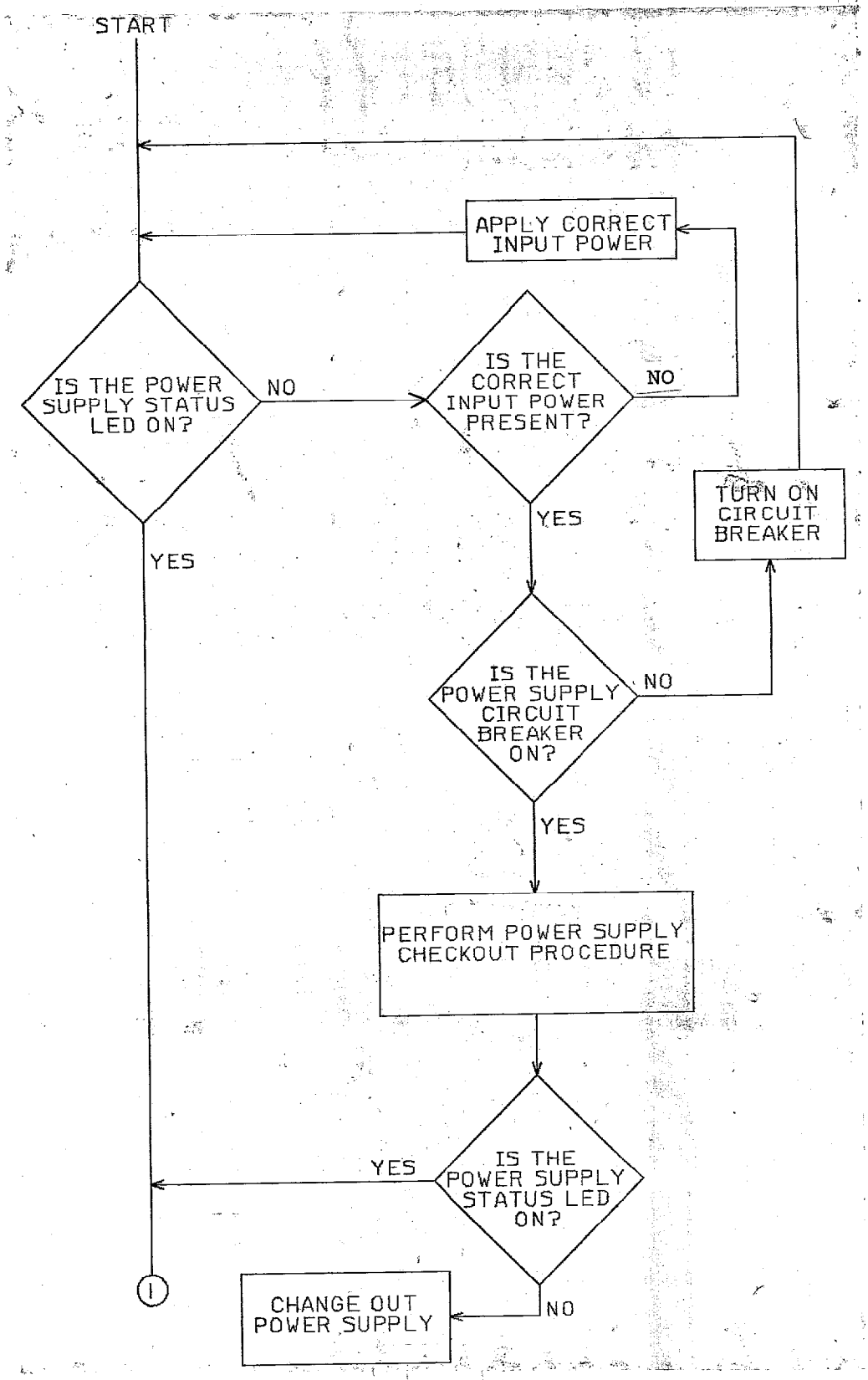
Trouble Shooting

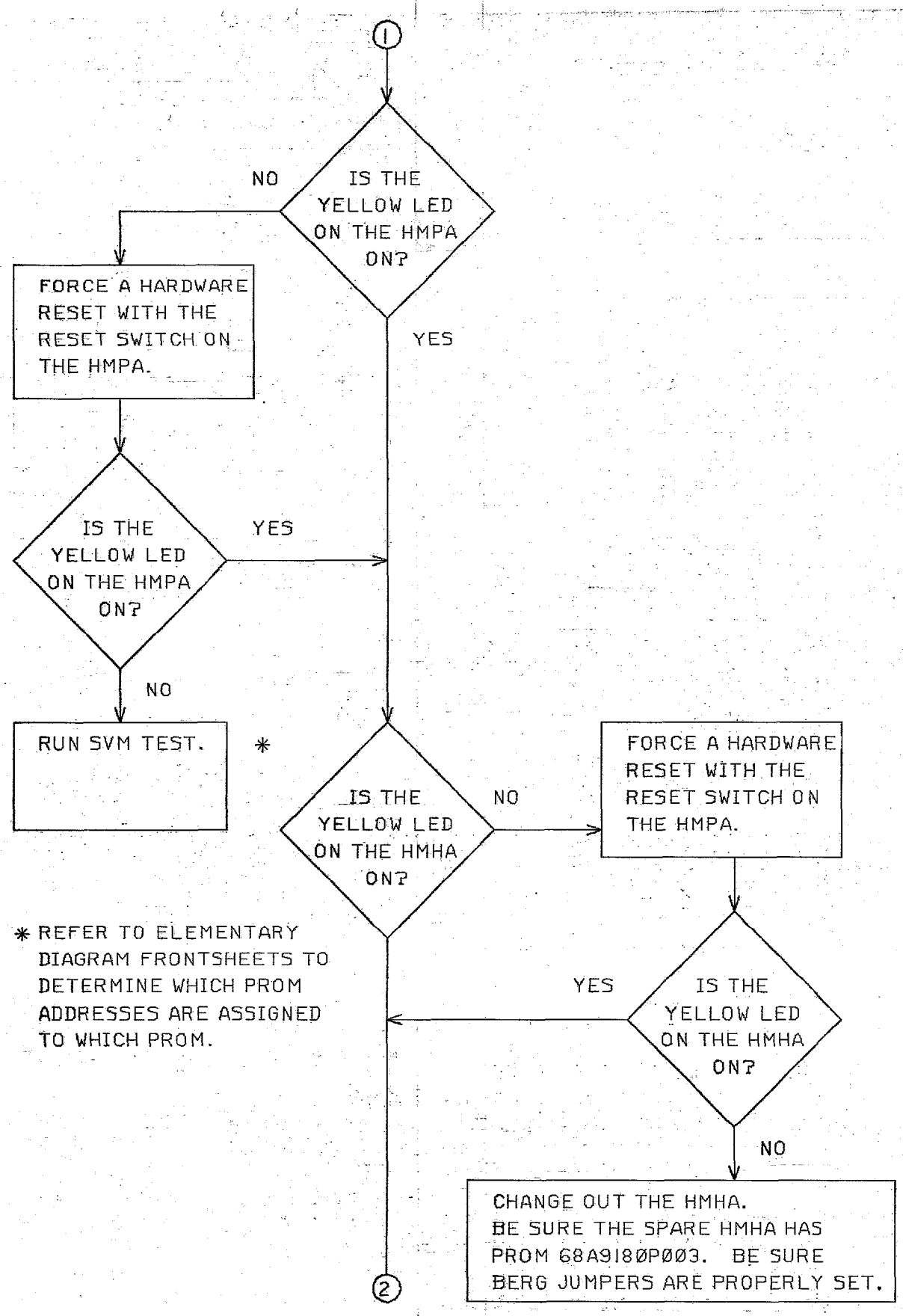
Normal Operation. During normal operation, the following conditions are present:

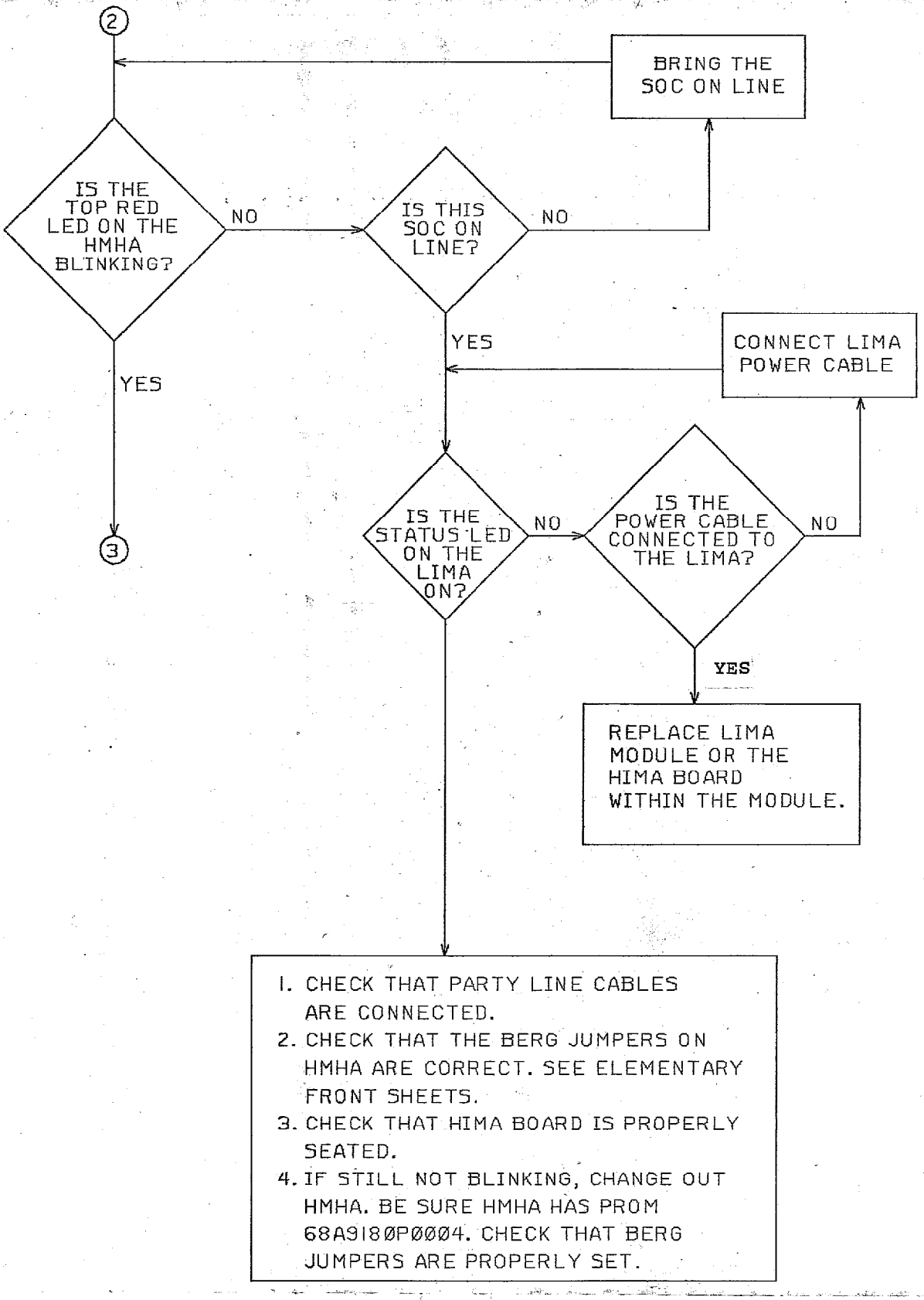
- Yellow status LED on - Power supply is on.
- Yellow status LEDs on HMPA, HXPA, HRRB, and HMHA are all on.
- The top red LED on the HMHA board should be blinking. This indicates party line traffic. The bottom red LED on the HMHA board should be blinking only if this is a redundant system with two party lines.
- The third red LED from the bottom on the HMPA board should be lit for part of each second. This indicates the Integrated Temperature System-to-Data Acquisition System link is functioning normally.
- Yellow STATUS LED on DS3820LIMA module is on.
- When a hardware reset is generated with the reset switch on the HMPA, the following sequence occurs:

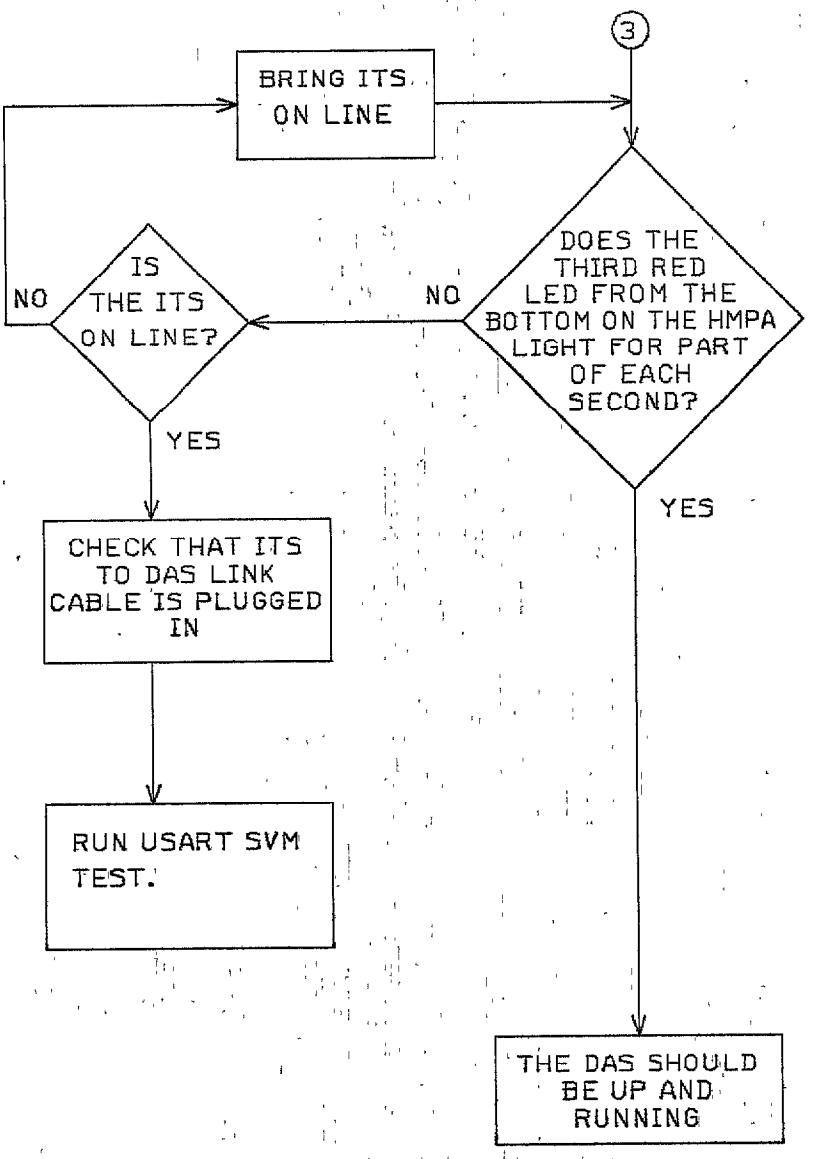
    1. The yellow LEDs on HMPA, HXPA, HRRB and HMHA boards go out.
    2. All eight red LEDs on the HMPA and the two red LEDs on the HMHA come on.
    3. The Yellow LED on the HMHA comes on and the two red LEDs on the HMHA go out.
    4. The eight red LEDs on the HMPA go out and the yellow LEDs on the HMPA, HXPA and HRRB come on.
    5. The entire above sequence takes approximately four seconds.

Troubleshooting Guide. The following guide may be used if the Data Acquisition System is not operating normally. It is intended as a guide and does not cover every possible contingency.




1
IS THE YELLOW LED ON THE HMPA ON?
NO
YES
FORCE A HARDWARE RESET WITH THE RESET SWITCH ON THE HMPA.
IS THE YELLOW LED ON THE HMPA ON?
YES
NO
RUN SVM TEST.
*
IS THE YELLOW LED ON THE HMHA ON?
NO
FORCE A HARDWARE RESET WITH THE RESET SWITCH ON THE HMPA.
IS THE YELLOW LED ON THE HMHA ON?
YES
NO
CHANGE OUT THE HMHA. BE SURE THE SPARE HMHA HAS PROM 68A9180P003. BE SURE BERG JUMPERS ARE PROPERLY SET.
2
* REFER TO ELEMENTARY DIAGRAM FRONTSHEETS TO DETERMINE WHICH PROM ADDRESSES ARE ASSIGNED TO WHICH PROM.


2
BRING THE SOC ON LINE
IS THE TOP RED LED ON THE HMHA BLINKING?
NO
IS THIS SOC ON LINE?
NO
YES
CONNECT LIMA POWER CABLE
YES
3
IS THE STATUS LED ON THE LIMA ON?
NO
IS THE POWER CABLE CONNECTED TO THE LIMA?
NO
YES
REPLACE LIMA MODULE OR THE HIMA BOARD WITHIN THE MODULE.
1. CHECK THAT PARTY LINE CABLES ARE CONNECTED.
2. CHECK THAT THE BERG JUMPERS ON HMHA ARE CORRECT. SEE ELEMENTARY FRONT SHEETS.
3. CHECK THAT HIMA BOARD IS PROPERLY SEATED.
4. IF STILL NOT BLINKING, CHANGE OUT HMHA. BE SURE HMHA HAS PROM 68A9180P0004. CHECK THAT BERG JUMPERS ARE PROPERLY SET.


3
BRING ITS ON LINE
DOES THE THIRD RED LED FROM THE BOTTOM ON THE HMPA LIGHT FOR PART OF EACH SECOND?
NO
IS THE ITS ON LINE?
NO
YES
YES
CHECK THAT ITS TO DAS LINK CABLE IS PLUGGED IN
RUN USART SVM TEST.
THE DAS SHOULD BE UP AND RUNNING

STATION OPERATORS CONSOLE MAINTENANCE INSTRUCTIONS

Power Supplies

The Station Operator's Console has two power supplies:

PS3 is identical to the Data Acquisition System power supply and supplies all bus voltages to the Station Operator's Console Medium Electronics Module.

PS-28A provides a 28 VDC source for contact inputs to the HSCC board and contact outputs for the HRRB board.

DS3820PS3 Power Supply. The DS3820PS3 power supply provides all power for the Station Communication Controller Medium Electronics Module, DS3820LIMA isolaton module, and Analog Input Rack. It accepts a single phase AC voltage (80-132 VRMS, 47-63 Hz) or a DC input voltage (70-140 VDC) at its input. Output voltages are +5V, +28V, +15V and -15V.

CB1 Circuit Breaker. A circuit breaker is supplied for disconnect purposes as well as short circuit protection. The circuit breaker is rated at 15 amperes.

Status Light. A status light mounted on the DS3800NPSC board indicates that all output ovoltages are normal. This light is on when power is applied and all voltages are correct. Trouble is indicated when the light is off.

Test Points and Adjustments. Test points and adjustments are available on the DS3800NPSC board. In order to get at the test points and adjustments, the DS3800NPSC board must be swung out. First undo the two plastic retainers which hold the board in place. The board will then swing out to make the test points and adjustments accessible.

CAUTION

ADJUST ONLY THE POTENTIOMETERS INDICATED.
OTHER POTENTIOMETERS MUST NOT BE SET IN THE FIELD.
IF THE SETTING OF R2, R3 or R7 IS INADVERTENTLY DISTURBED,
THE POWER SUPPLY SHOULD BE TURNED OFF IMMEDIATELY AND
RETURNED TO THE FACTORY FOR READJUSTMENT.

Preliminary Checks. The preliminary checks are performed as follows:

1. Unplug cards in Station Communication Controller Medium Electronics Module and Analog Input Rack and all cables except the power supply cable to the Station Communication Controller Medium Electronics Module.

2. Check for proper input voltage per instructions in the following paragraph (Power Supply Setup).

3. Apply power and check output voltages for "ballpark" values to prevent card damage.

Power Supply Setup. Proceed as follows:

1. De-energize the power supply and plug cards into the Station Communication Controller Medium Electronics Module and Analog Input Rack. Re-energize the power supply.

   NOTE: It is important that the following adjustments be made with all the cards plugged in. This is necessary because voltage feedback is taken from the Station Communication Controller Medium Electronics Module and will change when the cards are unplugged.

2. Check power supply voltages under load:

SUPPLY	TEST POINT	VALUE	ADJUSTMENT
P5	TP2 to TP3	5.00±.05VDC	R1
P15	TP4 to TP3	15.00±.05VDC	R9
N15	TP5 to TP3	-15.00±.05VDC	R10
P28	TP1 to TP3	28.00±.70VDC	R8

3. Power supply currents may be measured:

SUPPLY	TEST POINT	CALIBRATION	MAXIMUM
P5	TP15 to TP16	.005 Volts/Amp	.150V
P15	TP9 to TP10	.100 Volts/Amp	.300V
N15	TP11 to TP12	.100 Volts/Amp	.300V
P28	TP7 to TP8	.470 Volts/Amp	.500V

Fault Detection. There are multiple fault detection circuits (on the DS3800NPSC) to prevent internal/external load faults from damaging the PS3 power supply during normal operation. The power supply will shutdown if there is an overvoltage on any one output greater than 20% of nominal for all voltages except the P5 - excess is less than 40% of nominal. It will also shut down if there is a short on any output. The power supply cannot be reset until the load short(s) is removed.

PS-28A Power Supply. The PS-28A power supply houses the IC3600EPSM DIRECTO-MATIC II circuit board. Its function is to convert a 105 volts to 135 VDC into an unregulated 28 VDC source (See Figure 33).

SW-1 Switch. A manually operated contact switch is provided for disconnect purposes. This switch does not provide short circuit protection.

FU-1 Fuse. FU-1 provides short circuit protection for the EPSM board. It is rated at 1 ampere. Should a short circuit occur, its cause must be located and rectified. FU-1 must be checked for continuity and replaced if ncessary.

FU-2 Fuse. FU-2 provides overload protection to the output 28 VDC bus. It is rated at 3 amperes. Should the output current exceed this level, FU-2 will interrupt the current flow. The cause of the overload must be determined and corrected. FU-2 must be replaced.

Fault Protection. The converter (EPSM) is short circuit proof. If a short circuit occurs on the output bus, the base drive to the inverter will be at zero volts, causing the inverter to shut off. When the short circuit is removed, the inverter will re-start.

Initial Start-Up. Before any power is applied to PS-28A the following should be completed:

1. Check input voltage. Verify that it is between 105 and 135 VDC.

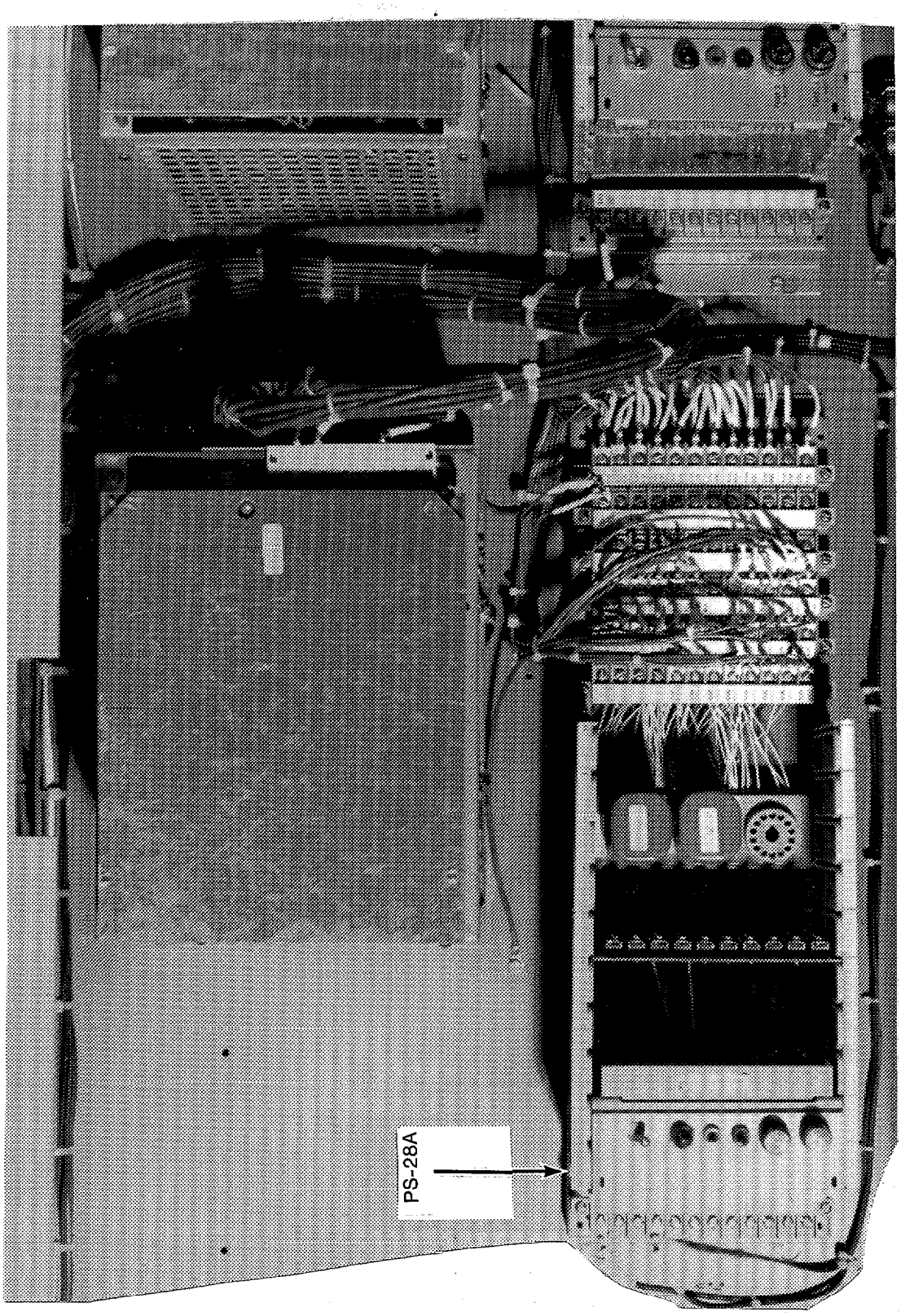
2. Check FU-1 and FU-2 for continuity and proper rating. (FU-1: 1 ampere; FU-2: 3 amperes)

3. Check outputs for any short circuits or grounds. If any are found, they must be cleared before energizing PS-28A.

Once the above steps have been completed, the supply voltage may be applied by closing switch 1. The output bus should be checked for proper voltage. Test jacks are provided on the 28 VDC bus for this purpose. No voltage adjustments are provided. The output 28 VDC bus varies directly with the input voltage. After the output voltage has been verified, deenergize PS-28A. Reconnect all loads to the output bus. PS-28A may now be put into service. When the power supply is reenergized, the output voltage should be checked again under load.

Visual Diagnostic Aids

Some of the cards in the Station Operator's Console Medium Electronics Module have various LED's mounted on the card fronts. These LED's can be used to aid in locating and diagnosing problems with the Station Operator's Console and its associated equipment, should they occur.



(DSF54225)

Figure 33
PS-28A Power Supply

The following is a description of the LED's and their function:

DS3800HMPA (See Figure 34) The HMPA has a yellow status LED which is lit when parity between the HMPA and other boards is good. This is known as the "IMOK" (I'm OK) light. This light will go out if the processor stalls.

There are eight red LEDs on the front of the HMPA. Their function is as follows:

1. Teleprinter receive LED. This is the bottom red LED. It will light when it receives an input from the teleprinter.

2. Teleprinter transmit LED. This is the second red LED up from the bottom. It will light when the HMPA transmits to the teleprinter.

3. SCADA receive LED. This is the third red LED up from the bottom. It will light when it receives an input from the SCADA link.

4. SCADA transmit LED. This is the fourth red LED up from the bottom. It will light when the HMPA transmits to the SCADA link.

5. Data Analyzer and Processor Receive LED. This is the sixth red LED up from the bottom. It will light when it receives an input from the Data Analyzer and Processor link.

6. Data Analyzer and Processor Transmit LED. This is the sixth red LED up from the bottom. It will light when the HMPA transmits to the Data Analyzer and Processor link.

7. Parallel Link Receive LED. This is the seventh red LED up from the bottom. It will light when it receives an input from the parallel link (optional, used with redundant option).

8. Parallel link transmit LED. This is the top red LED. It will light when the HMPA transmits on the parallel link (optional, used with redundant option).

DS3800HXPA. The HXPA card has a PARITY LED (yellow) which is lit when parity between the HXPA and the HMPA boards is good.

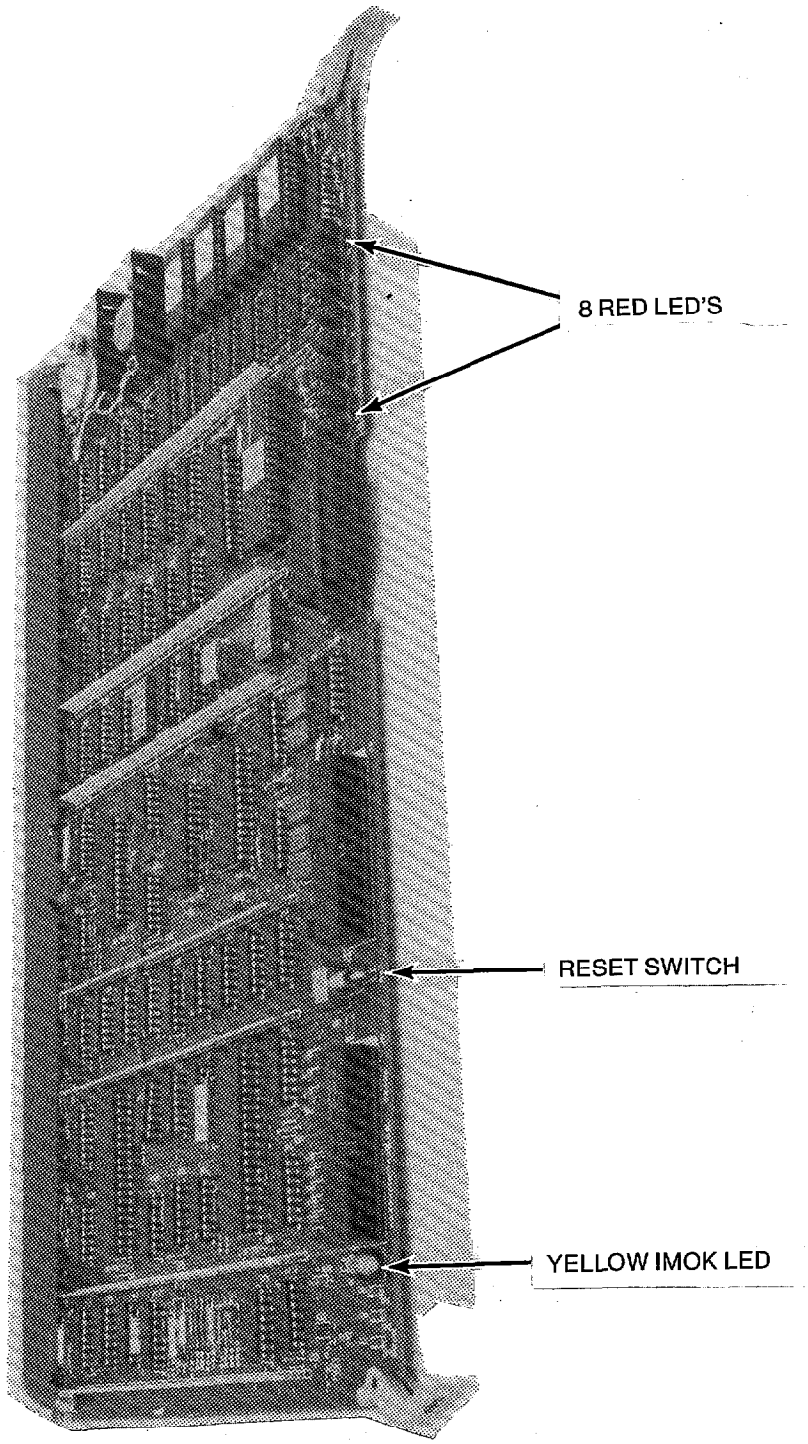
DS3800HCMA (See Figure 35) The HCMA card has an IMOK LED (yellow) which is lit when the CPU is not stalled and is executing its program. This light will go out if the processor stalls.

The top red LED on this card is a channel A traffic indicator and is lit to indicate the existence of traffic on channel A. The bottom red LED on this card is a channel B traffic indicator and is lit to indicate the existence of traffic on channel B.

DS3800HMHA (See Figure 36) The HMHA card has an IMOK LED (yellow) which, when lit, indicates that the HMHA processor is not stalled. This light will go out if the processor stalls.

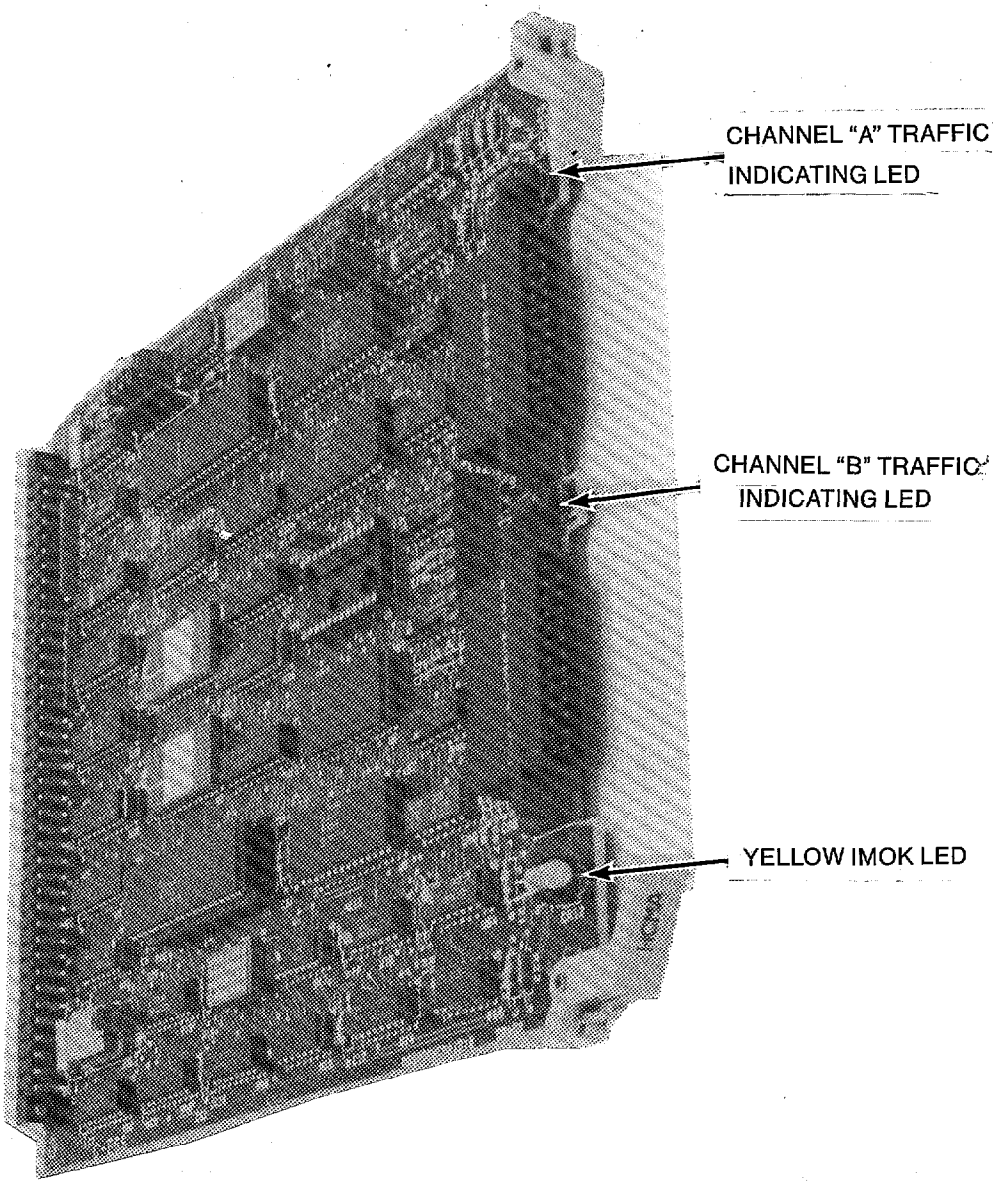
The top red LED on this card is a channel A traffic indicator and is lit to indicate the existence of traffic on channel A. The bottom red LED on this card is a channel B traffic indicator and is lit to indicate the existence of traffic on channel B.

DS3800HSCC (See Figure 37) This card has a PARITY LED (yellow) which, when lit, indicates that the parity between the HSCC and HMPA is good.



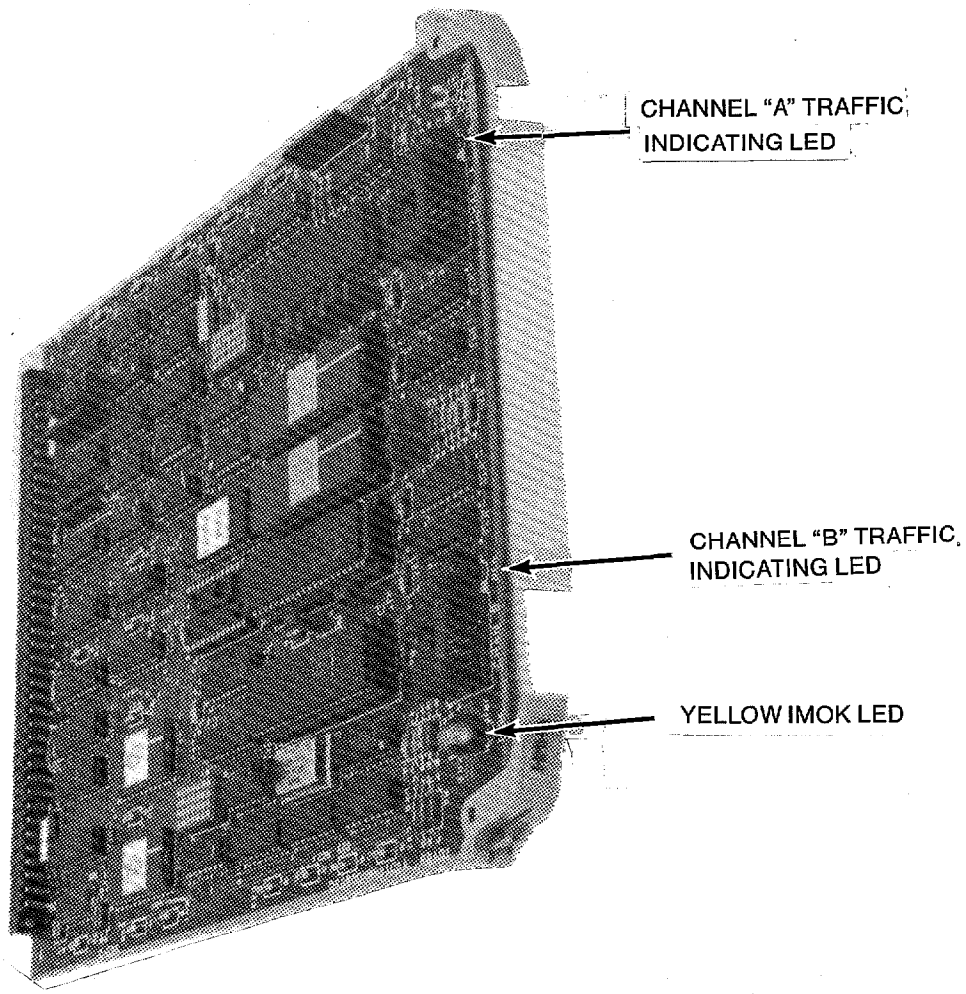
(DSF54140)

Figure 34
DS3800HMPA



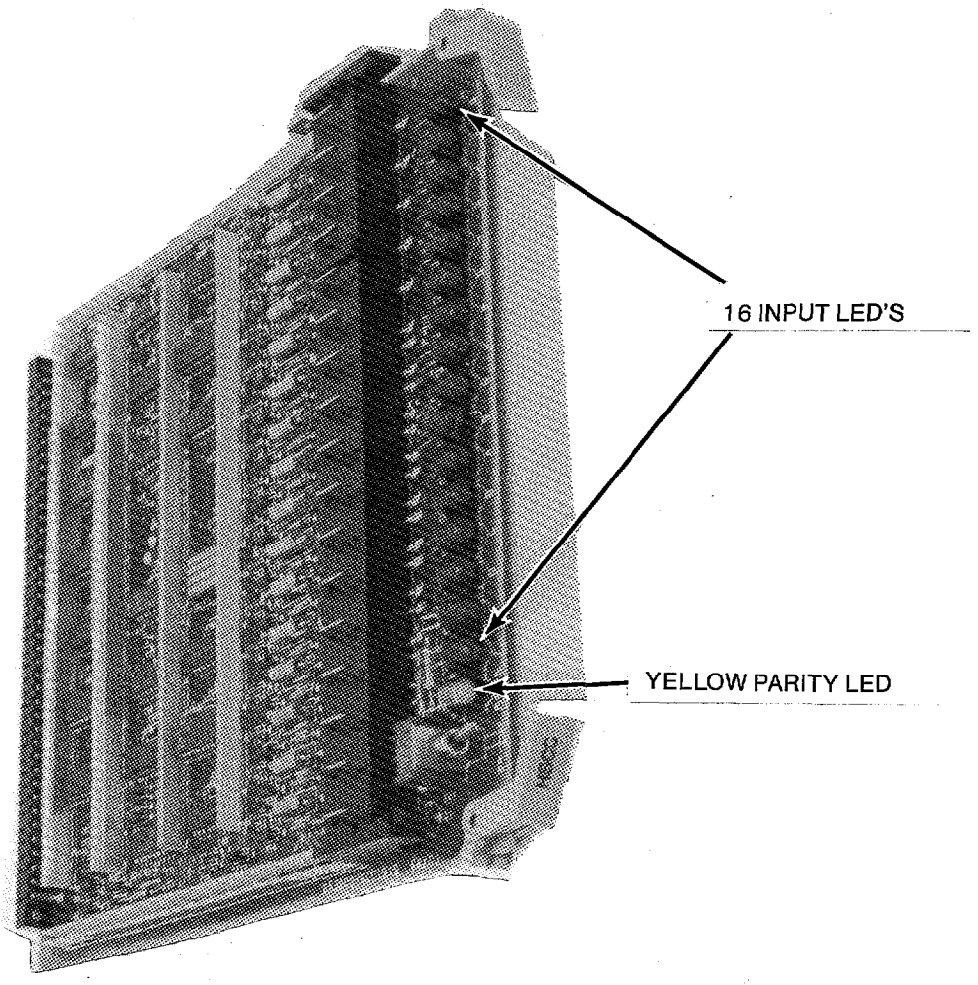
(DSF54138)

Figure 35
DS3800HCMA



(DSF54144)

Figure 36
DS3800HMHA



(DSF54145)

Figure 37
DS3800HSCC

There are 16 INPUT LEDs (all red) on the front of the HSCC to indicate the state of each of the 16 process input lines. Each red LED will be lit when its corresponding process input line is a logic level 1 (28 VDC). Refer to the Station Operator Console and Station Communications Controller elementary for specific inputs to each line.

DS3800HRRB (See Figure 38) The HRRB has a PARITY LED (yellow) which is lit when parity between the HRRB and the HMPA is good.

This card has an OUTPUT LED (red) for each of eight circuits on the HRRB. This LED is lit when the relay is energized.

Self Verification Modules

The Self Verification Module (SVM) (See Figure 39) is a tool that can be used to aid in locating defective hardware within the Data Acquisition System. There are two Self Verification Modules available for Station Communication Controller testing.

WARNING

THE USE OF THIS DIAGNOSTIC SOFTWARE MAY CAUSE OPERATION OF OTHER DEVICES. THE TEST OPERATOR MUST TAKE APPROPRIATE ACTIONS TO ASSURE THAT THE OPERATION OF THESE DEVICES WILL NOT CAUSE INJURY TO PERSONNEL OR DAMAGE TO EQUIPMENT.

Self Verification Module 277A5732G6. This Self Verification Module performs the following tests:

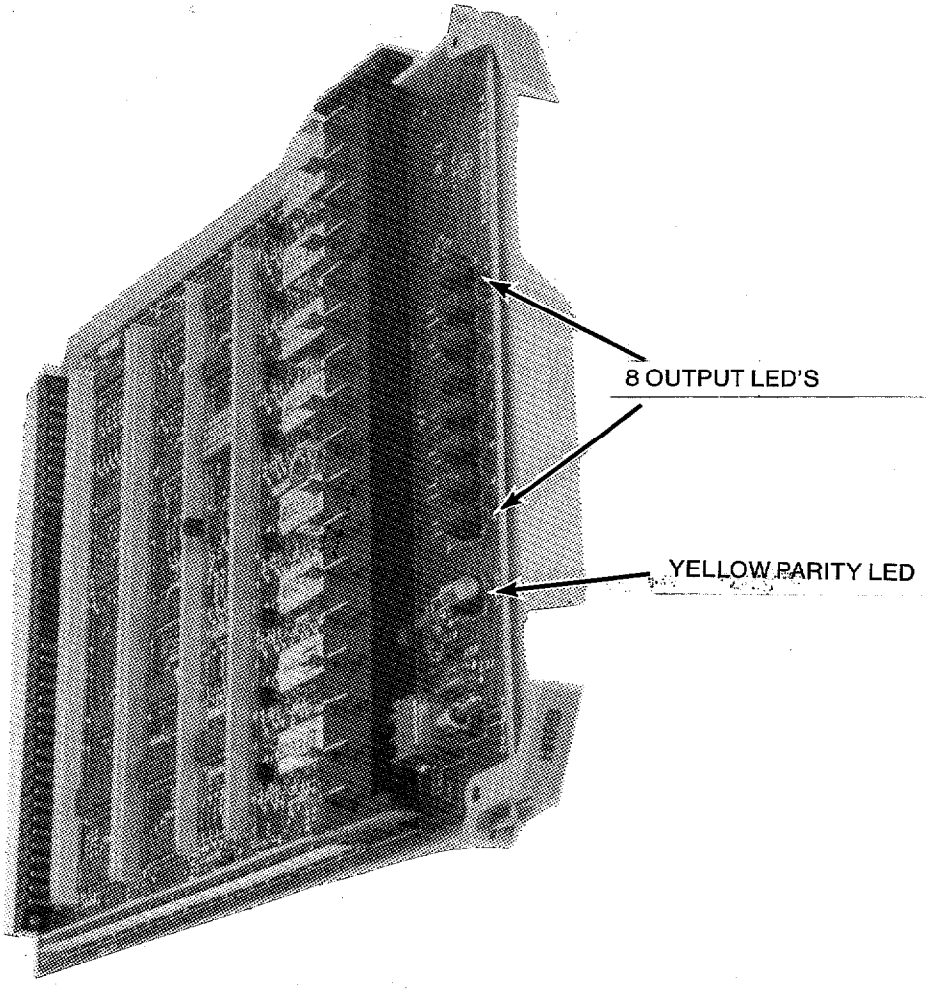
1. CPU check of the HMPA card
2. PROM checksum on HMPA and both HXPA's *
3. RAM test on HMPA and both HXPA's
4. Timer test on HMPA and both HXPA's
5. Stall test
6. Parallel I/O loopback test on HMPA and both HXPA's
7. Interrupt test
8. Analog loopback - runs automatically in ALLTEST but does not apply to the Station Communication Controller Medium Electronics Module. The Station Communication Controller Medium Electronics Module contains no NADC or NDAC cards.
9. HALT test

* Refer to elementary diagram frontsheets to determine the PROM addresses assigned to each PROM.

Follow these steps to set up the Self Verification Module:

1. Turn all Station Communications Controller power off.

2. Load the hardware verification program SVM005A as follows:
   A) Plug diagnostic module containing 68A9172 PROMS programmed with SVM005A into the Medium Electronics Module maintenance test socket JLA (68A9172P0085 and 68A9172P0086) (See Figure 40).
   B) The lock lever must be in the horizontal position for insertion.



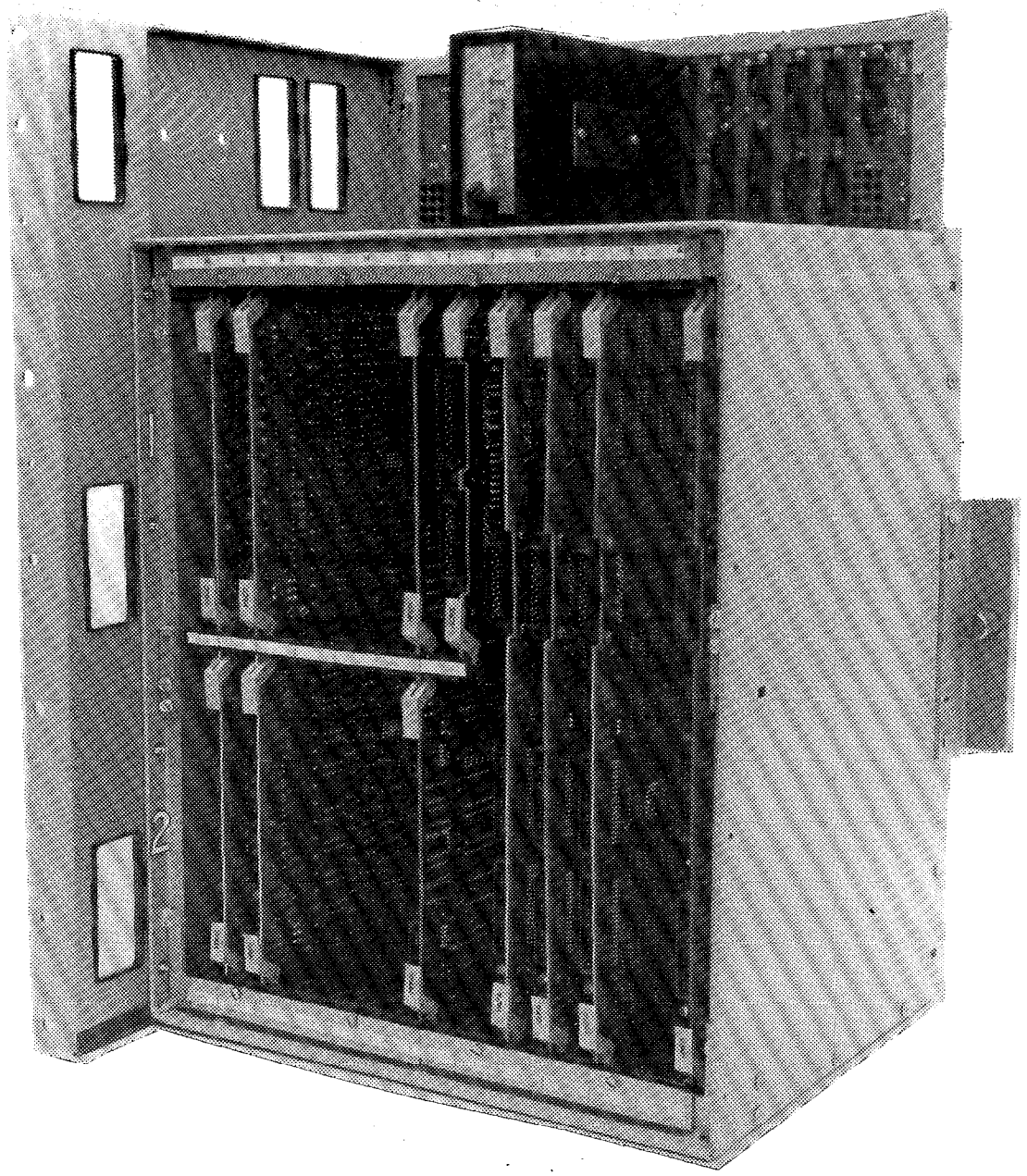
(DSF54143)

Figure 38
DS3800HRRB



(DSF54055)

Figure 39
Self Verification Module



(DSF54135)

Figure 40
Self Verification Module Plugged into MEM

CAUTION

THE SELF VERIFICATION MODULE AND/OR THE MEDIUM ELECTRONICS MODULE CAN BE DAMAGED IF THE PINS IN THE SELF VERIFICATION MODULE PLUG ARE BENT. BEFORE INSERTING THE SELF VERIFICATION MODULE, CAREFULLY INSPECT THE PINS TO ASSURE THAT THEY ARE STRAIGHT. ALL PINS IN A HORIZONTAL ROW SHOULD ALIGN ALONG THE HORIZONTAL ROW WITHIN ± 1/64INCH. PINS WHICH ARE OUT OF ALIGNMENT SHOULD BE REPAIRED BEFORE THE SELF VERIFICATION MODULE IS USED. WHEN INSERTING THE SELF VERIFICATION MODULE, USE MINIMUM FORCE. IF THE SELF VERIFICATION MODULE SEEMS TO "HANG" ABOUT 1/8 INCH FROM FULL INSERTION, DO NOT FORCE SINCE A PIN IS PROBABLY NOT ENTERING A HOLE PROPERLY. WHEN THE SELF VERIFICATION MODULE IS FULLY INSERTED, ROTATE THE LOCK LEVER 1/4 TURN CLOCKWISE TO THE VERTICAL POSITION TO LOCK THE MODULE IN PLACE.

3. Disconnect the cables connected to the HMPA (Station Communication Controller Medium Electronics Module 1A/2A) connectors JA and JB. Connect JA to JB using test cable DS3815RCG1CL020X00.

4. Disconnect the cables connected to the HXPA1 (Station Communication Controller Medium Electronics Module 1C/2C) connectors JA and JB. Connect JA to JB using test cable DS3815RCG1CL020X00.

5. Disconnect the cables connected to the HXPA2 (Station Communication Controller Medium Electronics Module 1D/2D) connectors JA and JB. Connect JA to JB using test cable DS3815RCG1CL020X00.

6. Temporarily connect the Integrated Temperature System display jack JB to the Station Communication Controller Medium Electronics Module jack JKA with a 34-conductor ribbon cable DS3815RCG1BL—-.

7. Monitor the PS3 yellow STATUS light and apply power to the Medium Electronics Module. If PS3 STATUS light does not come on immediately, deenergize immediately! Remove the Self Verification Module and inspect for bent or touching pins.

Follow these steps to run the **ALLTEST Diagnostic Routine**:

1. Make the following settings on the Integrated Temperature System display:
   Set LIST thumbwheel switch = 0
   Set HUNDREDS thumbwheel switch = 1
   Set TENS thumbwheel switch = C
   Set ONES thumbwheel switch = 5
   Set KEYSWITCH = NORMAL

These settings will run the following tests in the following order:

TEST#	TEST	PARAMETERS
1	CPU	None
4	PROM	28K (NOTE:Station Communication Controller ROMS must be installed.)
5	RAM	5K (HMPA and 2-HXPA) (NOTE:HXPA #3 RAM not used.
6	TIMER	HMPA and 2-HXPA *
7	STALL	Fails **
8	PARALLEL I/O LOOPBACK	HMPA and 2-HXPA
9	INTERRUPT	Fails 9-0, 9-1
C	ANALOG LOOPBACK	Fails - Bushog ***

* Due to wiring on the Station Communication Controller Medium Electronics Module backplane, the timer should fail twice in 6-3 (2nd HXPA) and once in 6-1 (HMPA timer 2).

** The behavior of the stall light in the Station Communication Controller Medium Electronics Module is not significant, either in the stall test or normal operation because of hardware differences in the Medium Electronics Module.

*** The Station Communication Controller Medium Electronics Module contains no NADC or NDAC cards. The analog loopback test, which runs automatically in SVM005A ALLTEST, does not apply to the Station communication Controller Medium Electronics Module.

2. ALLTEST is initiated by cycling the HMPA RESET switch to initialize the processor and start execution of the diagnostic program. The INCR will have to be pushed in test 6 (Timer Test) for each of five failures, and the FUNC pushbutton should be pushed on the first failure in test C (NADC/NDAC Loopback) to exit the analog loopback test to the test selector. If desired, each test (1, 4, 5, 6, 8, 9) can be run individually by selecting that test with the LIST thumbwheel switch instead of running ALLTEST. Tests 7 and C, which run automatically in the ALLTEST sequence, have no significance in the Station Communication Controller Medium Electronics Module. Any failures that occur other than those noted above in test 6 (and tests 7 and C, when running ALLTEST), should be investigated and repaired before continuing. Refer to 278A1549 for details on operating the diagnostic program and interpreting failure information.

3. Turn the power off and disconnect the three test cables (DS3815RCG1CL020X00) connected to the HMPA, HXPA1 and HXPA2 cards, and reconnect the system cables to those connectors. Leave the Integrated Temperataure System display connected. Reapply power.

4. Run the remaining diagnostic routines (not part of ALLTEST) one at a time by setting the LIST thumbwheel switch as specified below and cycle ENTER pushbutton to start the test.

   To perform the HALT test, set the LIST thumbwheel switch = 9 (interrupt) and keyswitch = maintenance. Start the test and wait 5 seconds after the message HALT TEST = RESTART appears on the display. Verify the HMPA lights = 93H (1001 0011) and not EEH (110110). Cycle the HMPA RESET switch to resume testing.

To perform the display test, set the LIST thumbwheel switch = 2 and Keyswitch = NORMAL. Cycle ENTER pushbutton to start the display test. Observe the display long enough to assure that each character occurs properly in the following sequence:

CHARACTER	DESCRIPTION
A thru Z	Alphabet
[	Left square bracket
	Tilde ("squiggle")
]	Right square bracket
{	Left braces
}	Right braces
!	Exclamation point
”	Quote (double)
#	Number sign
$	Dollar sign
%	Percent sign
&	Ampersand
’	Apostrophe
(	Left parenthesis
)	Right parenthesis
*	Asterisk
+	Plus sign
,	Comma
-	Hyphen (Minus sign)
.	Period
/	Slash
0 thru 9	Numbers
:	Colon
;	Semicolon
<	Left angle bracket ("Less than")
>	Right angle bracket ("Greater than")
?	Question mark
@	"At" sign

NOTE: Cycle the INCR pushbutton (Hold for 1 second) to cause program to switch to the SINGLE STEP mode where one new character will be displayed each time the INCR pushbutton is cycled, if desired. Cycle the ENTER pushbutton to return to AUTO-INCREMENT mode.

After the display character set has been verified, set Keyswitch = maintenance and cycle FUNCTION pushbutton (hold for at least 1 second) to switch the display test to the echo mode. In this mode the display will show (flashing):

L XHTO Maintenance - where:

L = LIST thumbwheel switch position
X = THOUSANDS thumbwheel switch position
H = HUNDREDS thumbwheel switch position
T = TENS thumbwheel switch position
O = ONES thumbwheel switch position

The lights adjacent to the pushbuttons will light when the corresponding pushbutton is pushed. Check all positions of all thumbwheel switches and verify that the display and the thumbwheel switch setting match. Verify that each pushbutton, singly and in combination with the others, lights the appropriate light. Set Keyswitch = NORMAL to return to the DISPLAY CHARACTER mode.

For a more detailed description of the SVM005 tests and test procedures, refer to drawing 278A1549 (SVM005 Operating Instructions).

Self Verification Module 277A5732G3 (SVM003). This Self Verification Module performs the following tests:

1. Power-up verification
   A) CPU test
   B) HMPA RAM test
   C) PROM checksum test
2. HRMA RAM test
3. HCMA dual ported RAM test
4. HMHA dual ported RAM test
5. 200-Level contact input test (HSCC)
6. 200-Level contact output test (HRRB)

Refer to the SVM003A operating instructions 278A4901 for detailed information on the operation of the diagnostic program.

Follow these steps to run the SVM003A tests using the Integrated Temperature System Display:

1. Turn power off and install the SVM003A diagnostic program by inserting a Self Verification Module with PROMS.

2. The lock lever must be in the horizontal position for insertion.

CAUTION

THE SELF VERIFICATION MODULE AND/OR MEDIUM ELECTRONICS MODULE CAN BE DAMAGED IF THE PINS IN THE SELF VERIFICATION MODULE PLUG ARE BENT. BEFORE INSERTING THE SELF VERIFICATION MODULE, CAREFULLY INSPECT THE PINS TO ASSURE THAT THEY ARE STRAIGHT. ALL PINS IN A HORIZONTAL ROW SHOULD ALIGN ALONG THE HORIZONTAL ROW WITHIN ± 1/64 INCH. PINS WHICH ARE OUT OF ALIGNMENT SHOULD BE REPAIRED BEFORE THE SELF VERIFICATION MODULE IS USED. WHEN INSERTING THE SELF VERIFICATION MODULE, USE MINIMUM FORCE! IF THE SELF VERIFICATION MODULE SEEMS TO HANG ABOUT 1/8 INCH FROM FULL INSERTION, DO NOT FORCE SINCE A PIN IS PROBABLY NOT ENTERING A HOLE PROPERLY.

3. When the Self Verification Module is fully inserted, rotate the lock lever 1/4 turn clockwise to the vertical position to lock the module in place.

4. Monitor the PS3 yellow STATUS light and apply power to the Medium Electronics Module. If PS3 STATUS light does not come on immediately, deenergize immediately! Remove the Self Verification Module and inspect for bent or touching pins.

5. Turn power on and observe the CPU lights for the power-up initialization. When the CPU lights = AAH, push the ENTER pushbutton on the Integrated Temperature System display to get the 'G3 - SEL Test' message.

6. Refer to the following two paragraphs (HRMA 16K RAM Card Test and HCMA Test) to run the following tests from SVM003A:

THUMBWHEEL	TESTS
1	HRMA 16K RAM Card
2	HCMA (Dual-Port RAM Only)
3	HMHA (Dual-Port RAM Only)
5	HRRB Relay Outputs
6	HSCC Contact Inputs

To run the HRMA 16K RAM Card test, set the LIST thumbwheel setting = 1 and cycle the ENTER pushbutton to start the test. The CPU lights will count up from 00H to FFH as the test progresses. The test requires about 7 minutes to run.

To run the HCMA Test (Dual-Port RAM Test), set the LIST thumbwheel switch = 2 and cycle the ENTER pushbutton to start the test. The CPU lights will count up from 00H to FFH as the test progresses. The test requires about 1 minute to run.

To run the HMHA Test (Dual-Port RAM only), set the List thumbwheel switch = 3 and cycle the ENTER pushbutton to start the test. The CPU lights will count up from 00H to FFH as the test progresses. The test requires about 1 minute to run.

To run the HSCC Contact Input test, set the LIST thumbwheel switch = 6 and cycle the ENTER pushbutton to start the test. Use the THOUSANDS and HUNDREDS thumbwheel switch to select the port address to be displayed. If the keyswitch is set to 'MAINT' and the ENTER pushbutton pushed once, the test will switch to the continuous update mode for more convenient testing. Using the table below to identify the port addresses for each HSCC card, check each input of each card by applying a P28V signal through A 15K-ohm resistor to the point and verifying that the displayed bit for that input changes from 0 to 1 when the P28V is applied, and returns to 0 when it is removed.

CARD	LOCATION	LOWER ADDRESS	UPPER ADDRESS
HSCC #1	SCC MEM Slot 1M	88	89
HSCC #2	SCC MEM Slot 1L	8A	8B

To run the HRBB Relay Output test, set the LIST thumbwheel switch = 5 and cycle the ENTER pushbutton to start the test. Use the THOUSANDS and HUNDREDS thumbwheel switches to set the port address, and the TENS and ONES thumbwheel swiches to set the data to be output. Test the #1 HRRB (Station Communications Controller Medium Electronics Module slot 2M) by using port address A0H. Use data setting of 00H, 05H, 0AH, 50H, A0H, and FFH and verify for each that the HRRB output LEDs assume the correct state. Test the #2 HRRB (Station Communications Controller Medium Electronics Module slot 2L) by using port address A1H. Use data setting of 00H, 05H, 0AH, 50H, A0H, and FFH and verify for each that the HRRB output LEDs assume the correct state. When the Station Communication Controller Medium Electronics Module has been verified using the tests above, turn off power and remove the Self Verification Module.

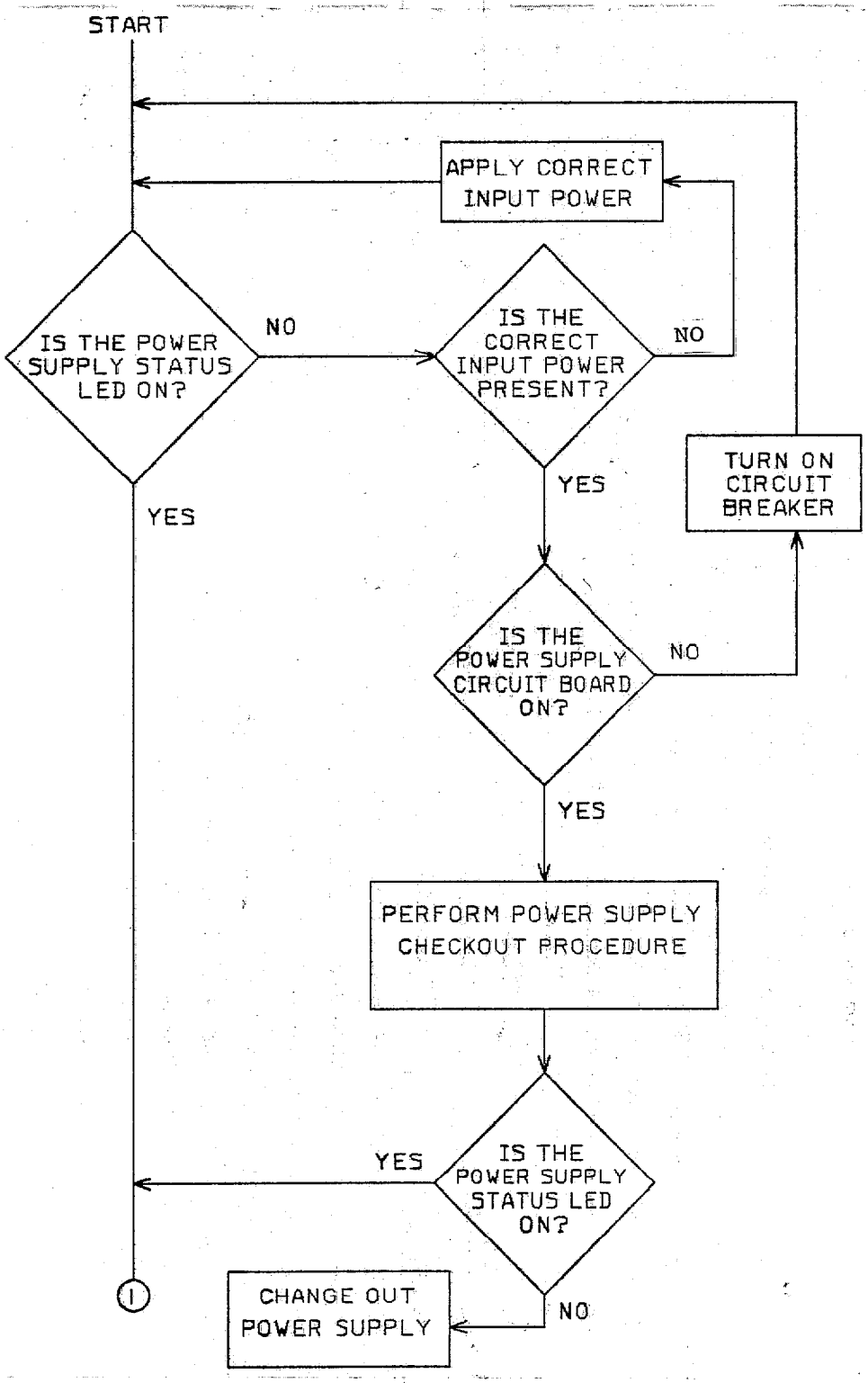
Troubleshooting

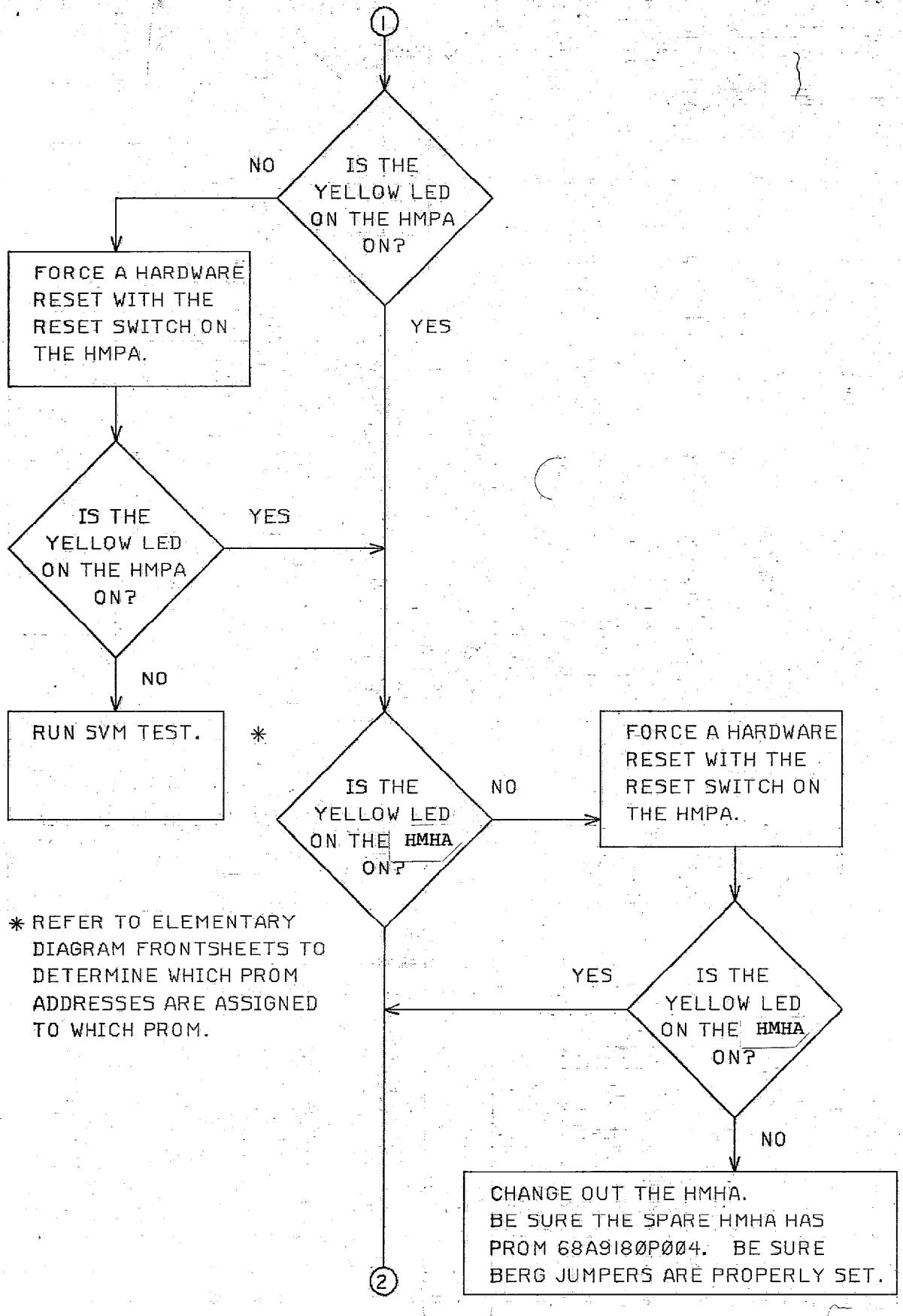
Normal Operation. Before you can begin trouble shooting the Station Communication Controller Medium Electronics Module, you must first have some understanding of its normal operation. The following is a brief description of what you should expect to see when the Station Communication Controller is functioning normally. This is not an attempt to cover all operating contingencies. During normal operation, the following conditions should be present:

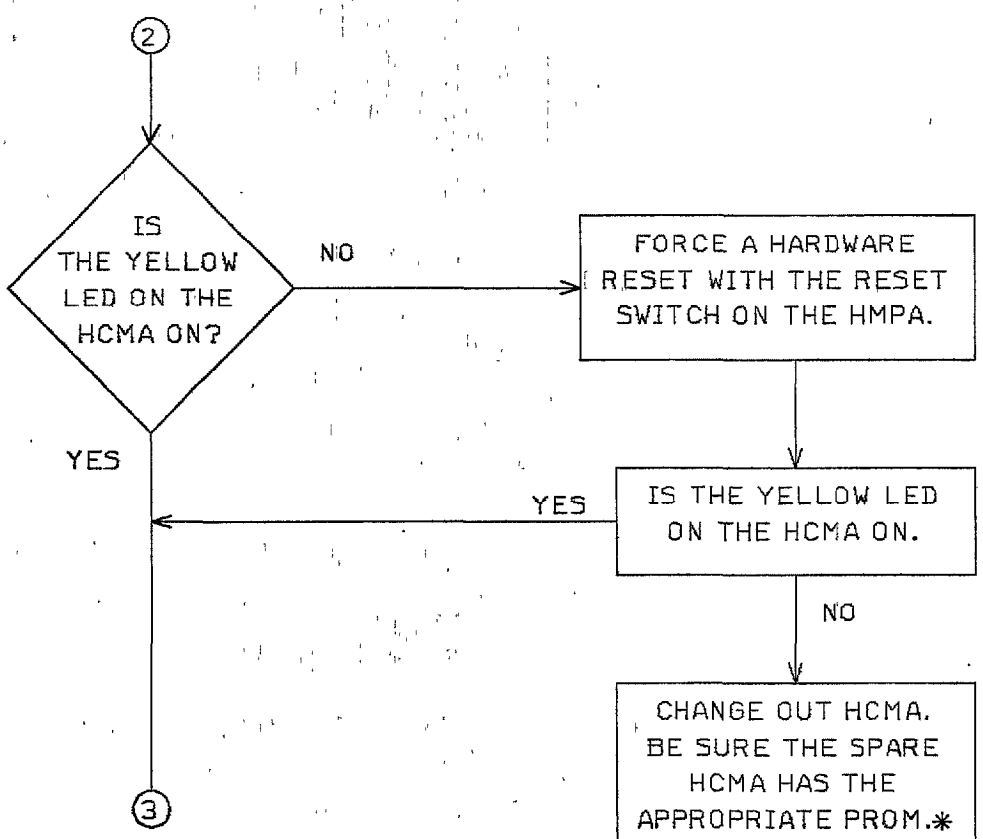
- Yellow STATUS LED on power supply PS3 is on.
- Output voltage of PS-28A is 28 VDC.
- Yellow STATUS LEDs on the HMPA, HXPA, HCMA, HMHA, HSCC AND HRRB are on.
- Yellow STATUS LED on DS3820LIMA module is on.
- The carrier LEDs on the limited distance data set (LDDS) both local set and remote set for the SCADA link and Data Analyzer and Processor link should be on. Be sure the switch on the rear of the Limited Distance Data Sets is in the NORMAL position.
- The TRANSMIT LEDs on the local Limited Distance Data Sets should blink when the Station Communication Controller sends messages to Supervisory Control and Data Acquisition (SCADA) or Data Analyzer and Processor (DAP). The corresponding red LED on the HMPA (i.e., for SCADA, the third LED from bottom, for DAP, the 5th LED from bottom) should also blink. (NOTE: The SCADA link and DAP link each have their own Limited Distance Data Sets (LDDS) which is independent of the other.)
- The remaining red LEDs on the HMPA should blink when there is activity on their respective channels. Refer to the visual diagnostic aids section of the Station Operator's Console Maintenance Instructions for a channel description.
- The red LEDs on the HCMA and HMHA should blink when there is activity on their respective channels.
- The red LEDs on the HSCCs should light when receiving inputs on their respective channels.
- The red LED on the HRRBs should light when sending outputs on their respective channels.
- When a hardware reset is generated with the RESET switch on the HMPA, the following sequence occurs (The entire sequence takes approximately four seconds):

    1. The yellow LEDs on the HMPA, HXPA, HCMA, HMHA, HSCC and HRRB boards go out.
    2. All eight red LEDs on the HMPA and the two red LEDs on the HMHA and HCMA boards come on.
    3. The yellow LEDs on the HMHA and HCMA come on and the two red LEDs on the HMHA and HCMA go out.
    4. The eight red LEDs on the HMPA go out and the yellow LEDs on the HMPA, HXPA, HSCC and HRRB come on.

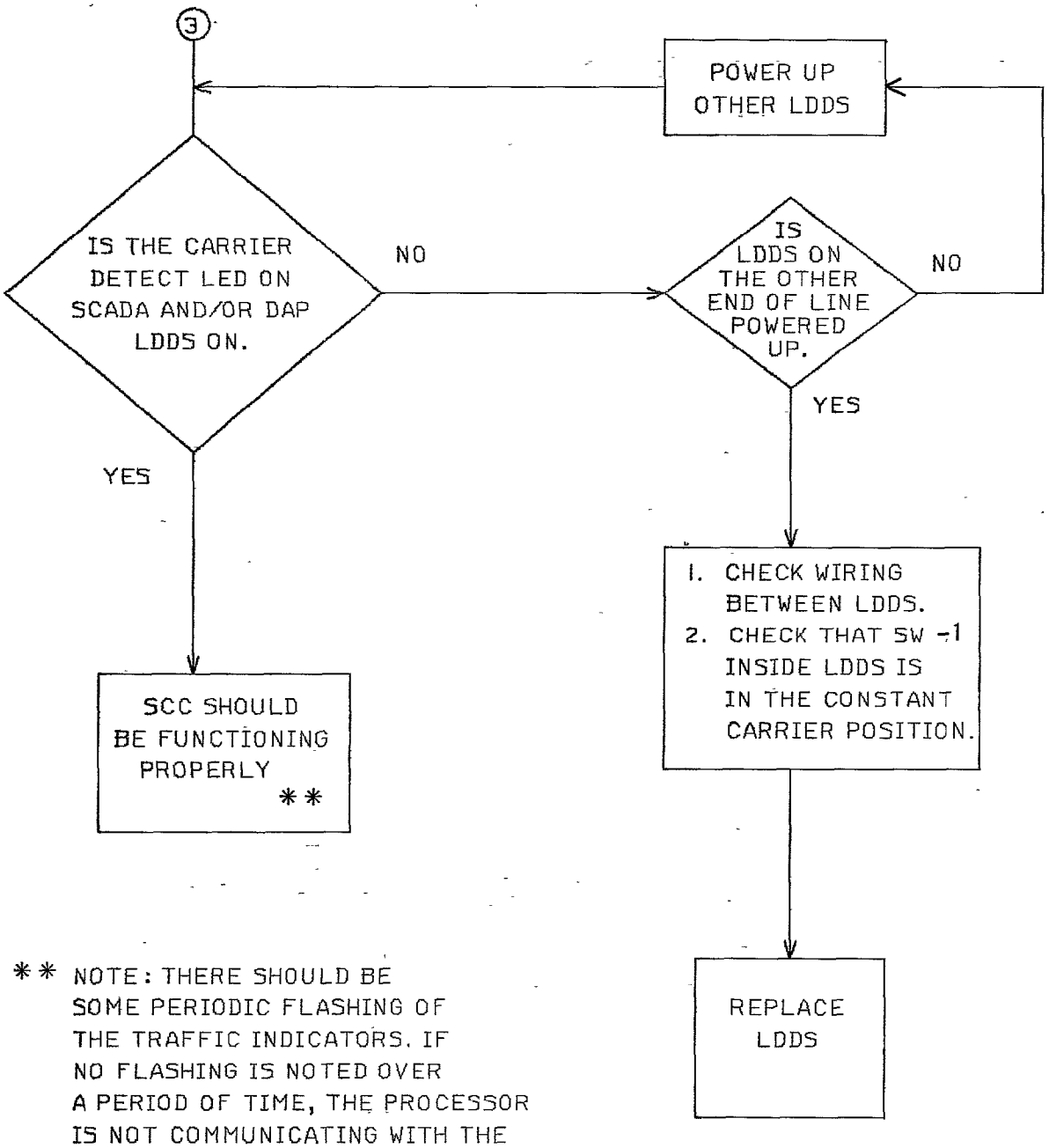
Troubleshooting Guide. The following guide may be used if the Station Communication Controller associated equipment is not operating normally. It is intended as a guide and does not cover every possible contingency.




1
IS THE YELLOW LED ON THE HMPA ON?
NO
YES
FORCE A HARDWARE RESET WITH THE RESET SWITCH ON THE HMPA.
IS THE YELLOW LED ON THE HMPA ON?
YES
NO
RUN SVM TEST.
*
IS THE YELLOW LED ON THE HMHA ON?
NO
FORCE A HARDWARE RESET WITH THE RESET SWITCH ON THE HMPA.
IS THE YELLOW LED ON THE HMHA ON?
YES
NO
CHANGE OUT THE HMHA. BE SURE THE SPARE HMHA HAS PROM 68A9180P004. BE SURE BERG JUMPERS ARE PROPERLY SET.
2
* REFER TO ELEMENTARY DIAGRAM FRONTSHEETS TO DETERMINE WHICH PROM ADDRESSES ARE ASSIGNED TO WHICH PROM.



* FOR THE HCMA (SLOT 1F)
LINK TO THE CRT'S USE
PROM 68A9180P00001.
FOR HCMA (SLOT 2F)
LINK TO THE DATA LOGGER USE
PROM 68A9172P0246



** NOTE: THERE SHOULD BE SOME PERIODIC FLASHING OF THE TRAFFIC INDICATORS. IF NO FLASHING IS NOTED OVER A PERIOD OF TIME, THE PROCESSOR IS NOT COMMUNICATING WITH THE OUTSIDE WORLD. THE CAUSE OF THIS SHOULD BE INVESTIGATED.

GLOSSARY

52G	Generator Breaker. Connects generator to power bus.
200 Level Input/Output	Refers to a family of DIRECTO-MATIC ®Plus I/O boards such as DS3800HSCC, DS3800HRRB, DS3800HSCD, etc.
A/D	Analog-to-digital
AIR	Analog Input Rack. Contains signal conditioning, amplifier circuitry and multiplexing for analog inputs.
ASCII	A commonly used data format usually consisting of seven bits of data per character. A unique ASCII code exists for each alphabetic, numeric, and control character.
Baud Rate	Bits per second
Berg Jumper	Berg jumpers are the small blue rectangular jumpers that are found on some DIRECTO-MATIC Plus circuit boards. These jumpers are used for configuring the board for a particular application.
Bit	Stands for binary digit. (BI)nary digi(T)
Byte	A group of eight bits
Cab	An outdoor enclosure in which the SPEEDTRONIC Control and Data Acquisition System are located. Some installations do not require outdoor enclosures, in which case this equipment is mounted on a skid.
Card	Circuit board
Channel	A communication path. The redundant communication channel option includes a separate Station Communication Controller and a separate party line. These are referred to as channel A and channel B.
Checksum	A method of error checking. The last two bytes of each PROM in the Data Acquisition System and Station Communication Controller are a checksum calculated from the sum of all preceding bytes in that PROM. When the PROM checksum test is run, the processor computes the checksum and then compares it with the last two bytes in that PROM. This is done for each PROM in the system.
CRT	Cathode ray tube
D/A	Digital-to-analog
DAP	Data analyzer and processor. This is the optional maintenance oriented minicomputer.
DAS	Data Acquisition System

DIRECTO-MATIC®II	Refers to a family of General Electric circuit boards. The SPEEDTRONIC Control contains many of these circuit boards.
DIRECTO-MATIC®Plus	Refers to a family of General Electric circuit boards. The Data Acquisition System and Station Communication Controller contain this family of circuit boards.
Dual Ported RAM	RAM memory that may be accessed by two processors. Both the DS3800HMHA and DS3800HCMA contain 2K of dual ported RAM memory.
EROM	Erasable read only memory
Full Duplex	Refers to a communication link that allows simultaneous two way communication. Thus, each station has the capability to send and receive messages at the same time.
Half Duplex	Refers to a communicaton link where a station may be either sending or receiving but not both at the same time.
I'm OK Light	Each processor board has a yellow status LED that is known as an IMOK light. This light is lit when the processor is executing instructions. I/O boards also have yellow LEDs which indicate good parity when lit.
Interrupt	Allows another device to interrupt the processor from what it is currently doing to service a time critical task. After servicing the interrupt, the processor continues where it left off.
ITS	Integrated Temperature System. This system performs the critical task of temperature protection for the turbine.
LDDS	Limited Distance Data sets. These are used in pairs (one at each end of line) to increase drive of RS-232 communication links.
LED	Light emitting diode
MEM	Medium Electronics Module. The Station Communication Controller is housed in a MEM.
Memory	A storage device into which information can be inserted and held for use at a later date.
Microprocessor	One or more chips which can process total system requirements through the use of memories and arithmetic logic units.
MODEM	Short term for (MO)dulator/(DEM)odulator. These are used to increase the distance range of RS-232 communication links. The DATA-TRONIC™ system uses the LDDS for this purpose instead of a MODEM.
Multiplexer	A circuit which carries two or more signals on the same wire or wires on a time shared basis.
Opto-coupler	A solid state version of a relay. A light emitting diode and a photo detector are coupled in a single package. Since this device couples light, it isolates the voltages on each side of the coupler.

Parity	A method of error checking. For an eight bit data field an additional bit would be added to indicate parity. For even parity, the parity bit would be set if there were an even number of ones in the data field. For odd parity, the parity bit would be set if there were an odd number of ones in the data field.
Party Line	A party line is a communication link where there are multiple stations on one common line. Each station has an address to which it will respond.
Party Line Connector	This is a special connector for terminating the party line cable shield.
Party Line Terminator	This referes to the termination impedance which must be present at the ends of the party line. This is inserted with a Berg jumper on the DS3820LIMA module.
PROM	Stands for programmable read only memory. In the DATA-TRONIC system, all PROMs are also EROMs or erasable read only memory. The contents of PROM or EROM is not lost on interruption of power.
RAM	Stands for random access memory. RAM memory may be read from or written to. This is typically used for data storage and other changing parameters. The contents of RAM memory are lost on interruption of power.
Ribbon Cable	Many of the connections between modules in the DATA-TRONIC system are made with ribbon cables. Ribbon cables are flat multiconductor cables.
RS-232	A widely used hardware standard for serial data communications. The standard defines connectors and voltage/curent levels.
RS-422	A widely used hardware standard for serial data communications. Due to the nature of the hardware, an RS-422 link may be operated at a faster baud rate than RS-232.
RTD	Resistance Temperature Detector. A calibrated resistance with a known resistance versus temperature characteristic.
SCADA	Supervisory control and data acquisition. This is the link to the customer's load dispatch computer.
SCC	Station Communications Controller. The SCC handles all data transactions between it and all the DAS's. It also interfaces to the SOC operator devices and CRT's.
SDLC	Synchronous data link control. This is a synchronous data link data format which utilizes a bit stuffing technique. The DATA-TRONIC™ party line is a SDLC link.
SEM	Small Electronic Module. The DAS electronics are housed in a SEM.
Serial Party Line	See party line.
SKID	A lineup of control panels on one common base. The SPEEDTRONIC™ Control, generator control, and DAS are mounted on a skid if an outdoor enclosure is not required.
SOC	Station Operator Console.

SPEEDTRONIC Control	Governs the fuel flow to the gas turbine. It also includes protective circuitry and sequencing functions.
SVM	Self verification module. A maintenance tool that can be used to checkout or troubleshoot hardware. SVM's may be used on the DAS SEM, the ITS SEM, and the SCC MEM. See DAS maintenance or SCC maintenance sections for more details.
Termination Impedance	A resistance that must be inserted at the ends of the party line. See Party Line Terminators.
Unit	Refers to a gas turbine and associated control such as SPEEDTRONIC™ Control and DAS.
USART	Universal Synchronous Asynchronous Receiver Transmitter.

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC